OPTIMAL UNEMPLOYMENT INSURANCE FINANCING: THEORY AND EVIDENCE FROM TWO U.S. STATES*

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Abstract

I study the optimal financing of unemployment benefits, comparing experience rating, where firms' tax rates are proportional to benefit spending from their layoffs, to uniform taxation. I derive a sufficient-statistics formula characterizing the optimal degree of experience rating through the tradeoff between firms' insurance value against tax increases during shocks and two distortions: labor misallocation toward high–unemployment-risk industries and moral hazard from increased layoffs. I estimate the formula for Colorado and South Carolina using unemployment tax filing data and experience-rating reforms. I have three main findings. Firms reduce employment and wages in response to higher taxes. Labor misallocation is the primary cost of uniform taxation. Experience rating is mostly inefficiently low.

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1 Introduction

Unemployment insurance is a fundamental welfare program that helps unemployed workers maintain their consumption levels following job losses. The provision of unemployment benefits, however, comes at a substantial cost, ranging from 0.12% to 2.8% of GDP in Western economies and increasing during downturns (OECD 2023). For perspective, the United States normally spends around 35 billion dollars annually on unemployment benefits, representing 0.2% of its GDP. During the Great Recession and the COVID-19 pandemic, benefit spending reached 125 and 160 billion dollars, or 1.1% and 0.9% of GDP, respectively.[1] Since the effectiveness and sustainability of unemployment insurance depend on the ability of governments to secure resources and promptly allocate them to workers as needed, understanding how best to finance the program is of paramount importance.

Typically, unemployment benefits are funded through payroll taxes levied on firms and workers. In this paper, I focus on firm-specific unemployment tax rates and compare the two prevailing approaches to assign these rates to firms. In the United States, firms are assigned personalized and dynamic tax rates, designed to reflect the costs of the unemployment benefits resulting from their layoffs. This approach, known as *"experience rating"*, holds firms financially responsible for such layoffs. In Europe, Canada, and South America, conversely, firms are assigned uniform rates regardless of their individual contributions to unemployment. This system, known as *"coinsurance"*, limits firms' tax liabilities following negative shocks.

It remains unclear which of these two approaches is preferable. The literature highlights three key factors for this evaluation, two against and one in favor of coinsurance. First, coinsurance generates moral hazard by reducing the private cost of layoffs that firms internalize, which increases the frequency of layoffs and imposes a fiscal externality on government budgets in the form of greater spending on unemployment benefits.[2] Second, coinsurance requires the equal participation of all firms to the financing of the system, even though layoffs are concentrated in specific industries with high unemployment risk. As a result, coinsurance redistributes the cost of the unemployment benefits generated by high-unemployment risk industries to low-risk ones. While cross-subsidization, where premiums paid by all insured parties cover the costs of those experiencing a shock, is a standard feature of insurance systems, concerns arise when the same industries consistently subsidize others over time. The subsidization of high-risk industries reduces their labor costs and increases their labor demand and employment, misallocating productive skills and imposing another fiscal externality in the form of a higher benefit spending as more workers face a high risk of unemployment.[3] Third,

[1]These dynamics are illustrated in Figure C.1.

[2]Experience rating reduces layoffs, stabilizing employment both within the year and over the business cycle (Feldstein 1976, Brechling 1977, Topel 1977, Topel 1983, Topel 1984, Kaiser 1986, Halpin 1979, Burgess et al. 1992, Anderson 1993, Anderson et al. 1994, Card et al. 1994, Katz et al. 1998, Blanchard et al. 2008, Duggan et al. 2023). Fath et al. (2005) formalizes the welfare loss from reducing experience rating as a fiscal externality.

[3]High-unemployment risk industries systematically receive more dollars in unemployment benefits than they pay in unemployment taxes, with low-unemployment risk industries covering the balance (Becker 1972, Munts et al. 1980, Mortensen 1983, Topel 1984, Anderson et al. 1993, Anderson et al. 1993, Laurence 1993, and Leombruni et al. 2003).

critics of experience rating argue that higher unemployment tax rates on firms in economic distress may reduce their labor demand, potentially increasing unemployment. This concern is particularly pronounced during recessions, when layoffs are widespread and higher unemployment taxes may slow down economic recovery and amplify the business cycle.[4] By contrast, uniform tax rates protect firms from large tax increases and the further deterioration of their net worth after negative shocks.

Until now, these three factors—coinsurance reducing the private cost of layoffs, coinsurance redistributing the cost of the unemployment benefits from high-risk to low-risk industries, and the imposition of higher unemployment tax rates on firms in economic distress—have been studied separately, offering policymakers limited guidance when deciding between experience rating and coinsurance.[5] In this study, I bring these factors together within a unified theoretical framework, recognize them as the central forces shaping the optimal design of unemployment insurance financing policies, and empirically investigate their relative importance in order to inform the policy debate.

I develop my analysis in three stages. In the first, I derive a formula for the optimal unemployment insurance financing scheme as a function of estimable sufficient statistics. I present a theoretical framework in which firms hire workers and exert costly effort to avoid negative shocks, the probability of which varies across industries. The government levies taxes on firms to fund unemployment benefits for the workers laid off after these shocks. The key choice of the government is the "degree" of experience rating of the unemployment insurance system that maximizes welfare. The degree of experience rating plays three roles. First, it measures the share of own benefit costs that firms repay in unemployment taxes, capturing the distance of the program from pure coinsurance (in which firms pay taxes equally) and from complete experience rating (in which each firm repays its full benefit costs), and thus determining the distribution of the tax burden between high- and low-unemployment risk industries. Second, it measures the marginal tax cost of each dollar of benefit spending, and hence influences firms' hiring and layoff decisions, generating moral hazard and labor misallocation. Third, it measures the increased cost per dollar of unemployment tax after a shock, reflecting the typical tax rise after layoffs in experience-rated systems, and increasing liquidity needs that firms cannot offset through self-insurance precisely when liquidity is most costly.

I derive the first- and second-best solutions. In the first-best scenario, the government controls firms'

The subsidization of high-risk industries leads to their expansion (Topel et al. 1980, Deere 1991). Topel (1990) and Anderson et al. (1993) measure the associated welfare losses from increased unemployment and resource misallocation.

[4] This argument, developed in Lester et al. (1939), Burdett et al. (1989), and Johnston (2021), is further supported by the literatures on the incidence of payroll taxes and adjustment costs.

[5] The question has long been recognized in the literature but remains theoretically and empirically unresolved. In a review of Becker (1972), McCaffree (1975) comments that *"decision makers are not provided with a clear-cut basis for determining trade-offs and making relevant choices."* After establishing that experience rating reduces layoffs, Topel (1984) observes: *"It is tempting to conclude (...) that subsidies to unemployment should be eliminated via complete experience rating of UI taxes. My analysis does not justify that conclusion, however, since very little is known about the optimal structure of the unemployment insurance financing system."* Guo et al. (2021) stress that *"if the benefits of experience rating are substantial, much of the world would benefit from clear evidence. If its costs outweigh, millions of workers in the U.S. could be spared the consequences"*, and that *"empirical and theoretical work to trace out the implications of these varied costs (. . .) would be helpful for assessing the tradeoffs of greater experience rating."*

hiring and layoff decisions, ensuring they are not distorted. As a result, it is optimal to fully insure firms against tax hikes after shocks through coinsurance. In the second-best scenario, where firms choose their behavior autonomously, full insurance is no longer optimal because it distorts firms' hiring and layoff incentives. In this case, the model delivers a sufficient-statistics formula for the optimal degree of experience rating, which balances the marginal benefit and cost of coinsurance identified by the literature. The marginal benefit is firms' value of insurance against large increases in tax liabilities following a negative shock. The sufficient statistic capturing this insurance value is the degree of experience rating itself, which measures the loss associated with an additional dollar of taxes after a shock. This loss can equivalently be written as the gap in marginal profits between the states of the world with and without a shock, and can therefore be interpreted as firms' willingness to pay to transfer one dollar from the state without shock to the state with the shock. The marginal cost of coinsurance includes two components, which can be interpreted as *ex-ante* and *ex-post* forms of moral hazard. First, coinsurance transfers the financial burden of the unemployment benefits generated by high-unemployment risk industries onto low-risk industries, reducing the labor costs in high-risk industries and increasing their labor demand. The sufficient statistic for this distortion is the elasticity of labor demand with respect to the degree of experience rating: the more elastic labor demand is, the more workers reallocate toward high-risk industries as experience rating declines. This inter-industry labor misallocation is a form of ex-ante moral hazard and is inefficient for two reasons: it leads to output losses due to the misallocation of productive skills, and it generates a fiscal externality by increasing the number of workers exposed to high unemployment risk and, in turn, benefit spending. Second, coinsurance reduces the private cost of layoffs for firms, leading to more frequent layoffs. The sufficient statistic for this distortion is the elasticity of layoffs with respect to the degree of experience rating: the more elastic the layoff rate is, the more workers are laid off as experience rating declines. This is a standard ex-post moral hazard, and is inefficient for two reasons: it reduces output through excessive separations and it imposes a fiscal externality on the government budget by raising spending on benefits. By comparing the marginal benefit and cost of providing insurance to firms, with the benefit given by firms' insurance value and the cost by moral hazard layoff and labor demand elasticities, the formula for the optimal degree of experience rating mirrors the classic formula for the optimal unemployment benefit level, with the benefit given by workers' insurance value and the cost by the moral hazard job-search elasticity (Baily 1978, Chetty 2006). Whether experience rating should be marginally expanded or reduced depends on the comparison between the marginal benefit and cost in each context.

In the second part of the paper, I bring the formula to the data and evaluate the local optimality of the unemployment insurance financing policies in two U.S. states. The U.S. provides a particularly relevant setting because unemployment benefits are primarily financed through taxes on firms, with only three states also taxing workers, and because experience rating rules are frequently reformed, which allows me to evaluate these policies with quasi-experimental methods and creates scope for policy adjustments. I focus on South Carolina and Colorado, which reformed their experience rating systems in 2011 and 2018, respectively. These reforms are identified using a novel database I am assembling, the U.S. Unemployment Insurance Financing Policies Database (UIFP). While the sufficient-statistics formula does not directly evaluate the optimality of non-marginal reforms, comparing policy recommendations before and after these reforms provides suggestive evidence on whether they moved experience rating towards the optimum.

First, I calculate the degree of experience rating in each state in two ways: as the share of firms' own benefit charges repaid through unemployment taxes, calibrated as the share of non-socialized costs following Vroman et al. (2017), and as the marginal tax cost of a dollar of benefit charge, calibrated following Pavosevich (2020) and Duggan et al. (2023) using the slope of the unemployment tax rate schedules from the UIFP. In South Carolina, the share of own benefits repaid in taxes was 34.3% in 2010 and rose to 52.7% after the 2011 reform, and the marginal tax cost increased from 15.4% to 24.9%. In Colorado, the share of own benefits repaid in taxes fell from 73.5% in 2017 to 72.5% after the 2018 reform, and the marginal tax cost decreased from 38.1% to 30.4%.

Second, I evaluate the optimality of these experience rating levels by quantifying the marginal benefit and cost of coinsurance in both states. I estimate the sufficient statistic for labor misallocation, the elasticity of labor demand with respect to the degree of experience rating or, equivalently, to the unemployment tax per worker, using novel restricted unemployment tax filing data provided by the South Carolina Department of Employment and Workforce (SC DEW) and the Colorado Department of Labor and Employment (CO DLE). These data cover the (near-) universe of firms in each state and contain unique information on unemployment tax rates unavailable in commonly used datasets such as the Longitudinal Employer-Household Dynamics (LEHD) data. I leverage quasi-experimental variation in the tax per worker generated by the 2011 and 2018 experience-rating reforms to estimate elasticities within the same state and time as the policies being evaluated.

The identification strategy for South Carolina leverages a 2011 reform that changed the measure of firms' experience with unemployment used to set tax rates from *reserve ratios*, reflecting the firm's long-run balance between taxes paid and benefit charges, to *benefit ratios*, based on benefits charged over a seven-year rolling period. This reform made historical tax payments and benefit charges beyond seven years irrelevant, leaving only *recent* experience to determine current tax rates. Using a differences-in-differences approach, I compare firms with similar post-reform benefit ratios, hence similar *recent* experience over the seven-year lookback period coinciding with the reform pre-period, but different pre-reform reserve ratios, which caused them to experience different changes in their tax rates depending on which mix of taxes paid and benefits charged from the *distant past* the reform "forgot". Because distant past conditions are plausibly uncorrelated with recent experience, this generates quasi-random variation in unemployment tax rates and taxes per worker. The reform increased the unemployment tax per worker by $48 (36%) on average. Employment declined by 0.135 workers (1%), total wages by $7,824 (1.8%), and average wages by $256 (0.7%), driven in part by lower wages among retained employees. In response, unemployment tax payments increased by \$54 per firm less than predicted in the reform-year. These effects are robust across a range of alternative specifications, including scaling outcomes by their pre-reform levels, using finer benefit-ratio classes, including sector-by-time and establishment-year-by-time fixed effects, and discretizing treatment.

The identification strategy for Colorado leverages the removal in 2018 of a surcharge which was proportional to initial tax rates, implying absolute larger reductions for firms with higher rates to begin with. Conditional on firms' benefit ratios, initial tax rates are uncorrelated with current firm conditions, generating quasi-random variation in tax rates and taxes per worker. Using a differences-in-differences approach, I find an average effect of the reform on the tax per worker of \$584 (230%). Employment declined by 0.762 (3.75%), total wages by \$109,357 (9%), and the average wage by \$3,478 (5.8%), partly driven by wage reductions among retained employees. Due to these behavioral responses, unemployment taxes increased by \$508 less than predicted per firm per year.

These findings imply that the incidence of unemployment taxes is shared between firms and workers. Assuming average wages for the missing employees, the share of incidence on firms, calculated as the decline in total wages accounted by the decline in employment, is 64.7% in South Carolina and 41.7% in Colorado. The minimum reduction in total wages due to the decline in workers' wages yields an upper bound on firms' share of incidence of 74.7% in South Carolina and 95.1% in Colorado.

I combine these reduced-form estimates to obtain labor demand elasticities with respect to the unemployment tax per worker of –0.032 in South Carolina and –0.016 in Colorado, serving as sufficient statistics for labor misallocation. As a benchmark, they correspond to elasticities with respect to overall labor costs of –9.1 in South Carolina and –3.9 in Colorado. While larger than conventional estimates, these elasticities are more consistent with studies on unemployment taxes (Anderson et al. 1997, Johnston 2021, Guo 2024) and can be rationalized by three mechanisms: heterogeneity in labor costs that cannot be shifted onto workers' wages in competitive labor markets (Lester 1960, Brechling 1977, Anderson et al. 1997); firms' cash constraints, which lead to employment reductions when wages are rigid; and the head-tax nature of unemployment taxes. I test these mechanisms in the data. Similar employment responses between industries with high and low wage variability suggests that labor market competition is not a key driver. In Colorado in 2018, employment responses are larger for younger and smaller firms, consistent with cash-constraints, while in South Carolina in 2011, the absence of heterogeneity may reflect widespread cash constraints after the Great Recession, explaining the higher elasticity. In Colorado, responses are also larger for lower-wage firms, for which unemployment taxes represent a bigger share of payroll, consistent with the head-tax mechanism. Accounting for these factors reconciles the elasticities with standard estimates.

Last, I complete the calibration of the formula with estimates from the literature and data moments from the ETA Financial Data 394 Handbook, the ETA Experience Rating 204 Report, and the Quarterly Census of Employment and Wages. I calibrate the sufficient statistic for moral hazard, the layoff elasticity with respect to the degree of experience rating, using the estimate of -0.1 from Card et al. (1994), and the value of insurance using the degree of experience rating, as implied by my model. I construct bounds on the marginal cost using alternative layoff and labor demand elasticities from the literature, and on the marginal benefit using alternative assumptions on firms' risk aversion.

Comparing the marginal benefit and cost of coinsurance yields two main findings. First, labor misallocation is the primary cost of incomplete experience rating, accounting for 74-88% of the marginal cost. The biggest distortion arises from output losses due to the reallocation of workers toward less productive, high–unemployment-risk industries as unemployment taxes decline. This finding is important both conceptually, given the limited attention paid to labor misallocation relative to moral hazard in the literature, and for its policy implication that coinsurance remains inefficient even where moral hazard is limited, such as in countries with strong employment protection and in seasonal industries that must downsize due to weather or calendar effects.

Second, policy recommendations depend critically on the measurement of experience rating and the calibration of the marginal benefit and cost. When experience rating is measured as the marginal tax cost of benefit charges, it is consistently inefficiently low. When measured as the share of a firm's own benefit charges repaid through unemployment taxes, experience rating is inefficiently low in South Carolina but close to optimal in Colorado pre-reform. While South Carolina increased experience rating and moved it closer to the optimum, Colorado moved further away. Policy recommendations, however, can reverse when labor misallocation or output losses are ignored, or when lower or upper bound estimates of the marginal cost or benefit are used. This sensitivity underscores the importance of incorporating labor misallocation and output losses, estimating elasticities within the context being evaluated, and tightly bounding firms' insurance value before drawing policy conclusions.

This paper contributes to four strands of literature. First, within the literature on the optimal design of social insurance programs, it complements research on the optimal provision of unemployment benefits (Baily 1978, Kiley 2003, Shimer et al. 2007, Gruber 1997, Chetty 2006, and Schmieder et al. 2016) by providing a framework for the optimal financing of the targeted benefit level. Prior work on optimal unemployment taxes had already emphasizes experience rating as a means of eliminating fiscal externalities and maximizing output by minimizing layoffs (Fath et al. 2005, Blanchard et al. 2008). Blanchard et al. (2008) further acknowledges the existence of a tradeoff between the goals of reducing layoffs and of limiting tax payments following layoffs for "risk-averse" or liquidity-constrained firms. I adapt the worker-level framework for optimal unemployment benefits developed by Baily (1978) and Chetty (2008) to study optimal unemployment taxation from the perspective of firms, and follow the moral hazard literature started by Feldstein (1976) to model firms' behavior. With this setup, I have three main countributions. First, I formalize the notion of firms' insurance value, microfounded with liquidity costs or risk-aversion, and provide its first empirical assessment. Second, whereas prior work on optimal unemployment tax design has focused primarily on moral hazard, I also introduce labor misallocation, and formalize and quantify the fiscal externalities and output losses generated by both inefficiencies. The implications of labor misallocation for social insurance design have been studied in the context of both disability (Michaud et al. 2018) and unemployment (Anderson et al. 1993) insurance. However, my estimates differ from Anderson et al. (1993) because I account for fiscal externalities on top of skill misallocation. Third, I make explicit within a unified sufficient-statistics framework the existence of a tradeoff between providing insurance to firms and limiting behavioral distortions, deriving an applicable tool for policy evaluation.

Second, this paper contributes to the literature on experience rating. Existing studies largely focus on how minimum and maximum tax rates shape experience rating. I highlight a critical but underexplored policy parameter: the measure of firms' "experience with unemployment" used to assign tax rates. Twenty-eight U.S. states employ the reserve ratio, and nineteen employ the benefit ratio, with infrequent transitions between the two. South Carolina's shift from a reserve ratio to a benefit ratio system, combined with newly available data on firms' unemployment accounts, offers a unique opportunity to evaluate this policy and to study how the two systems redistribute the tax burden across firms. This analysis complements existing work on the speed of layoff adjustment (Cook 1997), of tax collection (Lachowska et al. 2020), and on employers' incentives (Miller et al. 2019) implied by the two measures. Additionally, the reforms in South Carolina and Colorado constitute novel sources of variation in labor costs; similar reforms are common across states and readily identifiable in the UIFP data, opening the door to a wide range of empirical applications.

Third, this paper contributes to the literature on the incidence of payroll and other employment taxes. While earlier studies documented substantial pass-through of payroll taxes on workers through reduced wages (Gruber 1997, Anderson et al. 1997, Anderson et al. 2000), recent research supports the notion that the incidence of payroll taxes falls on firms as well, with impacts on employment (Behaghel et al. 2008, Saez et al. 2019, Benzarti et al. 2021, Benzarti et al. 2021, Johnston 2021, Guo 2024) and location decisions (Guo 2021). The same conclusions have been drawn in studies of corporate taxes and depreciation bonuses (Suárez Serrato et al. 2016, Mark et al. 2021). The findings in this paper align with this recent literature, showing that firms can pass on only a portion of payroll tax burdens to workers. I further contribute by testing alternative mechanisms, providing suggestive evidence that cash constraints and the head-tax nature of unemployment taxes help explain the observed employment responses.

Last, this paper contributes to the literature on adjustment costs: while some studies find little effect on hiring (Bentolila et al. 1990), the results here are consistent with other studies showing that adjustment costs can increase long-run unemployment (Hopenhayn et al. 1993, Anderson 1993).

The paper is organized as follows. Section 2 presents the model and derives the formula for the optimal degree of experience rating. Section 3 describes the institutional context, data, identification strategy, and estimation results for the labor demand elasticity with respect to the degree of experience rating in South Carolina and Colorado. Section 4 details the calibration of the remaining parameters of the formula. Section 5 discusses the optimal policy in the two states. Section 6 concludes.

2 Theory

2.1 Model Setup

I develop a model to analyze the tradeoffs and welfare implications of financing unemployment benefits with either experience rating or coinsurance. The model is set in general equilibrium and includes four types of agents: firms, which produce goods; workers, who supply labor and consume goods; capitalists, who supply capital and consume the rent; and the government, which finances unemployment benefits for laid-off workers through taxes on firms. I present each agent in turn.

2.1.1 Firms

Two representative neoclassical firms produce goods with a linear technology, tyipical of the literature on skill transferability (Gomes et al. 2017), where total output y is equal to the sum of workers' heterogeneous productivities $f(i)$, i denotes different workers, and capital is omitted: $y = \int_i f(i)\, di$. Labor costs include wages, set as a fraction $w_i \in (0, 1)$ of workers' productivities $f(i)$, and the unemployment tax paid for each worker τ.[6]

Firms face uncertainty over which of two states of the world will realize. In the *bad* state, occurring with probability $r \in (0, 1)$, a shock leads to the layoff of all the workers. In the *good* state, occurring with probability $1 - r$, there is no shock and firms produce output.[7]

I model firms' insurance value by introducing a profitability wedge between the good and the bad states that cannot be offset through private insurance or risk diversification. To mirror the functioning of experience rating, which increases unemployment taxes after layoffs, I assume that the unemployment tax per worker, τ, paid in both states, increases by e% after a shock, where $e \in (0, 1)$ is the degree of experience rating of the unemployment insurance system. Each dollar of tax costs $1 + e$ in the bad state, making the value of a dollar higher in the bad state than in the good state.

Why would firms value transferring funds to the bad state? The theory of the firm offers three possible microfoundations. First, e may capture bankruptcy costs, or the higher cost of liquidity after a shock, that is, borrowing costs: in presence of asymmetric information, lenders charge higher interest rates to firms whose net worth declines to compensate for increased monitoring costs or lower expected repayment (Bernanke et al. 1994, Holmstrom et al. 1997). Insurance transfers funds from low-cost to high-cost states. Second, e may capture the negative externalities of recessions

[6] As detailed in Section 3.1, unemployment taxes in the United States work as head taxes. Appendix B shows that modeling them as payroll taxes leads to the same solution.

[7] In Abebe et al. (2025), we explore the case in which r represents the the share of laid off workers, introducing a continuum of layoff states. The source of layoffs is irrelevant—they may arise from product demand shocks, as in the moral hazard literature (Feldstein 1976, Topel 1984, Card et al. 1994), productivity, or monetary shocks.

that amplify the initial downturn, such as credit constraints (Bernanke et al. 1994, Holmstrom et al. 1997), which can lead to large employment declines (Chodorow-Reich 2013), or depressed aggregate demand (Huber 2018). Consistent with this interpretation, experience rating has been shown to reduce labor demand and slow recovery by raising labor costs during recessions (Lester et al. 1939, Burdett et al. 1989, Johnston 2021). Insurance mitigates these negative externalities. Third, firms may be risk-averse. Starting from Greenwald et al. (1990), a growing literature has challenged the traditional view of risk neutrality to explain behaviors inconsistent with standard firm theory.[8] With risk aversion, insurance closes the gap in the marginal utility of profits between the bad and good states, just as unemployment insurance closes the gap in the marginal utility of consumption for workers between employment and unemployment (Baily 1978). Due to the complexity of introducing preferences over profits, I model the value of insurance through borrowing costs.

I introduce labor misallocation by assuming that the two firms operate in industries with different risk exposure: the firm in the high-risk industry faces a shock with positive probability $r_H \in (0, 1)$, while the firm in the low-risk industry faces no shock ($r_L = 0$). These differences may reflect underlying risk or predictable seasonality, with the low-risk industry operating year-round and the high-risk industry stopping for a fraction r_H of the year (the low season). Coinsurance lowers the labor costs and changes the hiring decisions of the high risk-firm by transferring the tax burden associated with its benefit spending onto the low-risk firm.

I introduce moral hazard by assuming that the high-risk firm can mitigate risk by exerting costly effort. Firms can manage their risk exposure: structured management practices improve resilience to shocks (Lamorgese et al. 2022), while firms that are "too big to fail" may take on excessive risk. Incurring a layoff risk of r_H requires effort $1 - r_H$ and entails a cost $\psi(1 - r_H)$, where ψ is differentiable and strictly convex in $1 - r_H$, meaning that effort becomes increasingly more costly. More convex cost functions represent stronger shocks, which are harder to avoid. Reducing the layoff probability to zero is prohibitively costly ($\psi(1) = \infty$), implying a strictly positive layoff rate. Coinsurance changes the high-risk firm's layoff decisions by modifying the tax cost of layoffs.

Profits for the two firms are defined in Equations (1) and (2). The high-risk firm's expected profits, Π_H, are the probability-weighted average of profits across states minus the cost of effort to mitigate risk. In the good state, profits equal revenues minus labor costs. In the bad state, all workers are laid off, production stops, and the firm incurs a loss equal to unemployment taxes amplified by experience

[8] Behaviors consistent with risk aversion include suboptimal hiring under uncertainty about worker productivity (Grossman et al. 1981, Frank 1990, Orszag et al. 1996); underinvestment in training amid high turnover (Choudhary et al. 2010); the avoidance of high-return-high-risk investments (Heaton et al. 2000, Moskowitz et al. 2002, Coles et al. 2006, Calvet et al. 2007, Caggese 2012, Michelacci et al. 2013); lower ownership shares when risk is higher (Jensen et al. 1976, Bitler et al. 2005); and other value-destroying actions that allow managers to "play it safe" (Gormley et al. 2016). An increasingly risk-averse culture has been observed among U.S. firms (Casselman 2013). Behavior resembling risk aversion may emerge even when firms are risk-neutral — for instance, when they exercise real options (Bulan 2005); or in response to progressive corporate taxation, which makes post-tax profits concave (Eeckhoudt et al. 1997).

rating. Since the low-risk firm faces no shocks, expected and good state profits Π_L coincide.

$$\Pi_H \equiv \underbrace{(1-r_H)}_{\substack{\text{Prob.}\\ \text{No Shock}}} \left[\underbrace{\int_{i\in H} (\overbrace{f(i)}^{\text{Revenues}} - \overbrace{w_{i,H} f_H(i)}^{\text{Wages}})\, di - \overbrace{\tau_H l_H}^{\substack{\text{Unempl.}\\ \text{Taxes}}}}_{\Pi_H^{good}} \right] + \underbrace{r_H}_{\substack{\text{Prob.}\\ \text{Shock}}} \underbrace{[- \overbrace{\tau_H l_H (1+e)}^{\substack{\text{Unempl. Taxes}\\ \text{with Exp. Rating}}}]}_{\Pi_H^{bad}} - \underbrace{\psi(1-r_H)}_{\text{Effort Cost}} \quad (1)$$

$$\Pi_L \equiv \underbrace{\int_{i\in L} \overbrace{f_L(i)\, di}^{\text{Revenues}} - \overbrace{w_{i,L} f_L(i)}^{\text{Wages}} - \overbrace{\tau_L l_L}^{\substack{\text{Unempl.}\\ \text{Taxes}}}}_{\Pi_L^{good}} \quad (2)$$

2.1.2 Workers

Workers i are uniformly distributed over the unit interval. To model skill-based selection into industries, I follow occupational choice models à la Roy (1951) and assume that workers have known industry specific productivities.[9] Let $(1-r_H)f_H(i)$ and $f_L(i)$ denote worker i's productivities in the high- and low-risk industries. The former is scaled by the probability of production occurring in the good state. A larger gap between the two implies lower skill transferability across industries. I make three assumptions. First, following Gomes et al. (2017), productivities across industries are negatively correlated: the most productive individuals in one industry are also the least productive in the other. Specifically, $(1-r_H)f_H(i)$ is nonincreasing, and $f_L(i)$ is nondecreasing over the unit interval. Second, there exists a threshold $i = l_H^{Eff} \in (0,1)$ such that workers below it are more productive in the high-risk industry, and those above it are more productive in the low-risk one. Third, $f_H(i)$ and $f_L(i)$ are differentiable in $(l_H^{Eff}, 1]$. Figure I shows the specific productivity distribution I employ, where $f_H(i)$ decreases linearly and $f_L(i)$ increases linearly over the unit interval.[10]

I introduce two concepts of efficiency. When the high-risk firm hires workers $i \in [0, l_H^{Eff}]$ and the low-risk firm hires workers $i \in [l_H^{Eff}, 1]$, the economy achieves *production efficiency*, as moving workers between industries would decrease total output, and *occupational choice efficiency*, as workers are employed in the industry where they are most productive. Thus, l_H^{Eff} identifies both the high-risk industry's efficient employment share and the workers efficiently employed in the industry. The position of l_H^{Eff} along the unit interval—determined by the intersection of the two productivity lines—depends on the probability of production in the high-risk industry, $1-r_H$. I fix the efficient employment share at the level attained when $r_H = r_H^{\min}$ and production is maximized.

Workers' utilities are defined in Equation (3). In the low-risk industry, utility $U_{i,L}$, is derived from consuming the wage. In the high-risk industry, expected utility $U_{i,H}$ is the probability-weighted

[9] The model abstracts from the inefficiencies related to on-the-job talent revelation in Terviö (2009).

[10] The model's predictions remain robust to symmetric, non-linear, non-continuous, non-monotonic, and positively correlated productivities, illustrated in Appendix B.1.

average of utility across states. In the good state, utility is derived from consuming the wage; in the bad state, it is a separable function of both pecuniary and non-pecuniary terms, as in Akerlof (1978), and comes from consuming the unemployment benefit b and enjoying leisure L_i. In each state, utility from consumption c is represented by the strictly concave function $u(c)$.

$$U_{i,L} \equiv \underbrace{u(\overbrace{w_{i,L} f_L(i)}^{\text{Wage}})}_{U^{good}_{i,L}} \qquad U_{i,H} \equiv \underbrace{(1-r_H)}_{\substack{\text{Prob. No}\\ \text{Shock}}} \underbrace{u(\overbrace{w_{i,H} f_H(i)}^{\text{Wage}})}_{U^{good}_{i,H}} + \underbrace{r_H}_{\substack{\text{Prob.}\\ \text{Shock}}} \underbrace{[u(\overbrace{\ b\ }^{\substack{\text{Unempl.}\\ \text{Benefit}}}) + \overbrace{L_i}^{\text{Leisure}}]}_{U^{bad}_{i,H}} \tag{3}$$

To focus on firms' inefficiencies, I assume with Equation (4) that workers are indifferent between the high-risk wage and the combination of the benefit and leisure.[11] Since benefits are lower than wages, consumption still declines during unemployment, creating a marginal utility gap between the good and bad states that justifies providing unemployment benefits to risk-averse workers.[12]

$$u(w_{i,H} f_H(i)) = u(b) + L_i \tag{4}$$

2.1.3 Capitalists

Capitalists supply capital for production and consume its rental rate, shown in Equation (5). The rent equals the two firms' total net worth for two reasons. Conceptually, with imperfect information capitalists incur agency costs to monitor firms to whom they lend capital. A higher net worth reduces such costs, increasing returns (Bernanke et al. 1989). Practically, profits must enter the social welfare function to derive the optimal policy.[13]

$$U_C \equiv \Pi_L + \Pi_H \tag{5}$$

2.1.4 Government

The government finances unemployment benefits for laid-off workers with taxes on firms, which are chosen to maximize social welfare and output, subject to a balanced budget constraint. The social welfare function SWF is utilitarian and sums the utilities of the workers and capitalists:

$$SWF = \underbrace{\int_{i \in L} U_{i,L}}_{\substack{\text{Utilities of Workers}\\ \text{Employed in Industry L}}} + \underbrace{\int_{i \in H} U_{i,H}}_{\substack{\text{Utilities of Workers}\\ \text{Employed in Industry H}}} + \underbrace{U_C}_{\substack{\text{Capitalists'}\\ \text{Utilities}}} \tag{6}$$

[11] This condition may explain the employment–unemployment cycles commonly observed among seasonal workers. Appendix B.5 shows that, if workers prefer employment, the second-best degree of experience rating increases to account for workers' utility losses from higher unemployment risk induced by coinsurance.

[12] Workers forgo more consumption to enjoy leisure, increasing insurance value (Chetty 2006).

[13] They could also enter directly or through workers' utilities, reflecting concerns about job security.

Output Y is obtained as the sum of the productivities of all workers in the economy:

$$Y = \underbrace{(1 - r_H)y_H}_{\substack{\text{Output of Workers} \\ \text{in Industry H}}} + \underbrace{y_L}_{\substack{\text{Output of Workers} \\ \text{in Industry L}}} = (1 - r_H) \int_{i \in H} f_H(i)\, di + \int_{i \in L} f_L(i)\, di \tag{7}$$

The budget constraint in Equation (8) requires that the taxes collected from both firms match the expected benefit spending, which depends on the exogenous benefit level b, the layoff rate in the high-risk industry r_H, and the share of workers employed in this industry and exposed to shocks, l_H.

$$\underbrace{\tau_L l_L}_{\substack{\text{Taxes Paid by} \\ \text{Low-Risk Firm}}} + \underbrace{\tau_H l_H}_{\substack{\text{Taxes Paid by} \\ \text{High-Risk Firm}}} = \underbrace{b l_H r_H}_{\substack{\text{Expected Benefit} \\ \text{Expenditure}}} \tag{8}$$

Instead of setting the tax rates τ_H and τ_L, the government chooses how to redistribute the tax burden across firms through the degree of experience rating of the unemployment insurance system, $e \in [0, 1]$. Following Feldstein (1976), e is defined as the share of total benefit spending that the high-risk firm repays in unemployment taxes. The low-risk firm pays the residual share, $1 - e$. The resulting tax bills are shown in Equation (9). When $e = 1$, experience rating is complete: the high-risk firm fully finances the cost of unemployment benefits resulting from its layoffs, while the low-risk firm—having no layoffs—is exempt from paying taxes. When $e < 1$, coinsurance implies that a fraction $1 - e$ of the benefit costs generated by the high-risk firm is shifted to the low-risk firm. Consistent with Topel (1984), e also measures the high-risk firm's marginal tax cost of a dollar of benefit spending, and thus affects their labor demand and layoff decisions. For the low-risk firm, unemployment taxes are exogenous and effectively lump-sum, and thus do not affect its behavior.[14]

$$\underbrace{\tau_H l_H}_{\substack{\text{Unempl. Taxes for} \\ \text{High-Risk Firm}}} = \underbrace{e b l_H r_H}_{\substack{\text{Fraction } e \text{ of Expected} \\ \text{Benefit Expenditure}}} \qquad \underbrace{\tau_L l_L}_{\substack{\text{Unempl. Taxes for} \\ \text{Low-Risk Firm}}} = \underbrace{(1 - e) b l_H r_H}_{\substack{\text{Fraction } 1-e \text{ of Expected} \\ \text{Benefit Expenditure}}} \tag{9}$$

2.2 Model Solution

I characterize the first- and second-best solutions. Before, I characterize workers' and the low-risk firm's optimal choices, which are unaffected by experience rating and thus identical across solutions.

Optimal Labor Supply l_H^S Workers supply labor in the industry providing the highest expected utility: Worker i chooses the high-risk industry if $U_{i,H} \geq U_{i,L}$. The indifferent worker, denoted as i^I

[14] Consistent with the US system, the high-risk firm faces a higher tax per worker when it generates more layoffs. However, in reality tax rates depend on realized—not expected—benefit claims, so any rate increase occurs with a lag. Moreover, as payroll shrinks with employment, the tax burden may initially fall despite higher rates (Brechling 1977). When tax rates adjust sufficiently quickly—as in benefit ratio systems, discussed in Section 3.1—and firms do not heavily discount future taxes, true incentives mirror those implied by the model.

and defined by Equation (10), receives the same utility from the two industries:

$$(1 - r_H)u(w_{i^I,H} f_H(i^I)) + r_H[u(b) + L_{i^I}] = u(w_{i^I,L} f_L(i^I)) \tag{10}$$

For fixed wage rates, workers $i < i^I$ strictly prefer employment in the high-risk industry, where their higher productivity yields higher wages, and workers $i > i^I$ prefer the low-risk industry. This choice generates the labor supply in the two industries presented in Equation (11):

$$l_H^S = \int_0^{i^I} i \, di \qquad\qquad l_L^S = \int_{i^I}^1 i \, di \tag{11}$$

Due to heterogeneity in productivity, workers $i < i^I$ prefer the high-risk industry even at wage rates below $w_{i^I,H}$, while workers $i > i^I$ prefer the low-risk industry even at wage rates below $w_{i^I,L}$, all else equal. Therefore, the wage rate needed to attract a worker to the high-risk industry rises with i, while the wage rate needed to attract a worker to the low-risk industry falls with i.

Low-risk Firm's Optimal Labor Demand l_L^D It is derived in Appendix A.1 by setting to zero the derivative with respect to l_L of low-risk profits Π_L. Because productivity in the low-risk industry increase over the unit interval, the firm hires workers in descending order, starting from $i = 1$. The choice of labor demand can therefore be interpreted as an optimal stopping rule along the unit interval. The first-order condition in Equation (12) characterizes this stopping point as the marginal worker $i = 1 - l_L^D$ whose productivity equals the marginal cost of labor. That is, the firm hires a share l_L^D of workers–including all workers $i \in [1 - l_L^D, 1]$– whose productivity exceeds the marginal cost, including the wage for the marginal worker and the wage increase needed to attract them

$$\overbrace{\underbrace{f_L(1 - l_L^D)}_{\substack{\text{Marginal Worker's} \\ \text{Productivity}}}}^{\text{Marginal Product of Labor}} = \overbrace{\underbrace{w_{1-l_L^D} f_L(1 - l_L^D)}_{\text{Marginal Wage}} + \underbrace{\frac{\partial w_{1-l_L^D}}{\partial l_L^D} f_L(1 - l_L^D)}_{\substack{\text{Wage Increase for} \\ \text{Marginal Worker}}}}^{\text{Marginal Cost of Labor}} \tag{12}$$

2.2.1 First Best Equilibrium

The first-best equilibrium is the solution of the following problem: the government chooses the high-risk firm's layoff rate r_H, labor demand, l_H, and the degree of experience rating e to maximize social welfare (Equation 6) and output (Equation 7), subject to the budget constraint (Equation 9), workers' optimal labor supply (Equation 11), the low-risk firm's optimal labor demand (Equation 12), and labor market clearing in both industries. Appendix A.2 shows the Lagrangian of this problem.

First-Best Layoff Rate r_H^{FB} It is derived in Appendix A.2 by setting the derivative of the Lagrangian with respect to r_H equal to zero. The first-order condition in Equation (13) characterizes the optimal layoff rate as the point where the marginal benefit of reducing r_H equals its marginal cost. Lowering the layoff rate yields five benefits: (i) it increases the probability of earning good-state rather than bad-state profits; (ii) it reduces the expected expenditure in unemployment benefits, hence lowering the unemployment tax bill; (iii) in turn, reducing the tax burden eases borrowing costs after a shock; (iv) it reduces the fiscal externality, i.e, the amount of unemployment taxes passed on to the low-risk firm through the budget constraint; and (v) it raises the likelihood of production in the high-risk industry, mitigating output losses. The marginal cost is the additional effort required to further reduce the layoff rate.

$$\overbrace{\underbrace{\Pi_H^{good} - \Pi_H^{bad}}_{\substack{\Delta \text{ profits} \\ \text{between states}}} + \underbrace{ebl_H}_{\substack{\text{Tax} \\ \text{Decline}}} + \underbrace{r_H^{FB} ebl_H e}_{\substack{\text{Borrowing} \\ \text{Costs}}} + \underbrace{(1-e)bl_H}_{\substack{\text{Fiscal} \\ \text{Externality}}} + \underbrace{\int_0^{l_H} f_H(i)\, di}_{\substack{\text{Output} \\ \text{Loss}}}}^{\text{Marginal Benefit of Effort/Reduced Layoff}} = \overbrace{\underbrace{\psi'(1-r_H^{FB})}_{\substack{\text{Increasing} \\ \text{Effort Cost}}}}^{\substack{\text{Marginal Cost of} \\ \text{Effort/Reduced Layoffs}}} \tag{13}$$

First-Best High-Risk Industry's Employment Share $l_H^{D,FB}$ It is derived in Appendix A.2 by setting the derivative of the Lagrangian with respect to l_H to zero. Since productivity in the high-risk industry declines over the unit interval, the firm hires workers in ascending order, starting from $i = 0$. The choice of labor demand can thus be interpreted as an optimal stopping rule along the unit interval. The first-order condition in Equation (14) identifies the stopping point as the marginal worker $i = l_H^{D,FB}$, whose productivity equals the marginal cost of labor. That is, the firm hires all workers $i \in [0, l_H^{D,FB}]$, whose productivities exceed the marginal cost, which has six components: (i) the wage paid to the marginal worker; (ii) the wage increase needed to attract that worker; (iii) the unemployment tax increase to match the higher expected beneift expenditure; (iv) the additional borrowing costs in the event of a shock due to higher taxes; (v) the fiscal externality, i.e., the higher taxes imposed on the low-risk firm through the budget constraint; and (vi) the output loss from reallocating a worker who would have been more productive in the low-risk industry.

$$\begin{aligned} \overbrace{\underbrace{f_H(l_H^{D,FB})}_{\substack{\text{Marginal Worker's} \\ \text{Productivity}}}}^{\text{Marginal Product of Labor}} = & \overbrace{\underbrace{w_{l_H^{D,FB}} f_H(l_H^{D,FB})}_{\text{Marginal Wage}} + \underbrace{\frac{\partial w_{l_H^{D,FB}}}{\partial l_H} f_H(l_H^{D,FB})}_{\substack{\text{Wage Increase for} \\ \text{Marginal Worker}}} + \underbrace{ebr_H}_{\substack{\text{Tax Per} \\ \text{Worker}}} + \underbrace{\frac{r_H}{1-r_H} e^2 br_H}_{\text{Borrowing Costs}}}^{\text{Marginal Cost of Labor}} \\ & + \underbrace{\frac{(1-e)br_H}{1-r_H}}_{\text{Fiscal Externality}} + \underbrace{\frac{f_L(l_H^{D,FB}) - (1-r_H) f_H(l_H^{D,FB})}{1-r_H}}_{\text{Output Loss}} \end{aligned} \tag{14}$$

Next, wages adjust to ensure that the labor market clears in each industry.

$$l_H^{S,FB} = l_H^{D,FB} \equiv l_H^{FB} \qquad\qquad l_L^{S,FB} = l_L^{D,FB} \equiv l_L^{FB} \tag{15}$$

Wage rates in the high-risk industry are set so that the indifferent worker coincides with the marginal worker, $i^I = l_H^{FB}$. As a result, all workers $i \in [0, l_H^{FB}]$ choose employment in the high-risk industry, demand meets supply, and l_H^{FB} becomes the high-risk industry's employment share. Workers $i > l_H^{FB}$ prefer the low-risk industry, where wage rates adjust to make demand meet supply, making $l_L^{FB} = 1 - l_H^{FB}$ its employment share.

First-Best Experience Rating e^{FB} It is derived in Appendix A.2 by setting the derivative of the Lagrangian with respect to e to zero. The first-order condition in Equation (16) implies that, in the first best, experience rating is equal to zero. In the Appendix, I express e as the gap in marginal profits between the good and bad states. A value of zero indicates that marginal profits are equalized across states, meaning the high-risk firm is fully insured against tax increases following the shock.

$$\left(\frac{\Pi_H'^{bad} - \Pi_H'^{good}}{\Pi_H'^{good}}\right)^{FB} = e^{FB} = 0 \tag{16}$$

Description of First Best Equilibrium It is characterized by (i) a minimized layoff rate, (ii) a minimized employment share in the high-risk industry, and (iii) full insurance for the high-risk firm. Since the government controls the high-risk firm's choices, it can provide full insurance without distorting its incentives. This result mirrors Baily (1978)'s model of optimal unemployment benefits in the first-best, when the government directly controls workers' job search effort and can therefore provide full insurance against unemployment—i.e., close the gap in marginal utility of consumption between employment and unemployment—without discouraging job search. Next section shows that in the second best, when the high-risk employer maximizes profits, both the layoff rate and the high-risk industry's employment share increase with insurance. Moreover, the entire tax burden falls onto the low-risk firm without distorting its behavior, à la Ramsey (1927): $\tau_L l_L = b l_H^{FB} r_H^{FB}$. This equilibrium is illustrated in Figure II, which shows how the low-risk employer's tax burden varies with the degree of experience rating, by the point named "First-Best EQ".

In the absence of reasons for the government to expand the high-risk industry beyond its efficient level, the first-best and efficient allocations coincides: the layoff rate reaches its overall minimum, $r_H^{FB} = r_H^{min}$, and employment in the high-risk industry is efficient, $l_H^{FB} = l_H^{Eff}$.[15] This equilibrium

[15] Risk may deter entrepreneurship (Van Doornik et al. 2022) or induce workers to demand compensating wages (Abowd et al. 1981) that employers may struggle to sustain, potentially making high-risk industries inefficiently small and justifying subsidies to expand them (see Van Doornik et al. (2022) for unemployment insurance and Michaud et al. (2018) for disability insurance). However, industries typically subsidized by unemployment insurance, like restaurants and construction (see Figure H3) typically have low rates of innovation (D'Amico et al. 2024, Goolsbee et al. 2025), making positive externalities unlikely to justify subsidies. Additionally, generous unemployment benefits—with replacement rates often above 50%—reduce the wage premium needed to attract workers (O'Connor 1962). There may be reasons to expand the private (high-risk) sector over the public sector (low-risk), typically seen as oversized. However, sufficient variation in layoff risk exists within the private sector to disregard the public sector. Appendix B.9 shows that introducing positive externalities from the high-risk industry has ambiguous effects on the second-best degree of experience rating: while the expansion of the high-risk sector induced by insurance becomes less distortive,

is illustrated by the point named "First-Best EQ" in Figure I. As workers are efficiently allocated between industries, there is no output loss due to skill misallocation: $(1-r_H^{FB})f_H(l_H^{FB})-f_L(l_H^{FB})=0$. However, due to the positive layoff rate, there is non-zero risk of losing production, captured by the term $\int_0^{l_H^{FB}} f_H(i), di$ in Equation (13) and represented in Figure I by the area below the productivity line, $(1-r_H^{FB})f_H(i)$, for all workers employed in the high-risk industry, $i \in [0, l_H^{FB}]$. This loss is normalized to zero so that any second-best output loss is measured relative to this benchmark.

$$\int_0^{l_H^{FB}} f_H(i) = 0 \tag{17}$$

2.2.2 Second Best Equilibrium

The second-best equilibrium is the solution to two problems: the high-risk firm chooses its layoff rate and labor demand to maximize profits, and the government chooses experience rating to maximize social welfare and output, accounting for firm's responses. I present each problem in turn.

Second-Best Layoff Rate, r_H^{SB} It is derived in Appendix A.3 by setting to zero the derivative with respect to r_H of the high-risk firm's expected profits (Π_H from Equation 1). The first-order condition in Equation (18) characterizes the second-best layoff rate as the point where the marginal benefit of reducing r_H equals its marginal cost. Lowering the layoff rate yields three benefits: (i) it increases the probability of earning the good-state rather than the bad state profits; (ii) it reduces the expected expenditure in unemployment benefits, hence lowering the unemployment tax bill; and (iii) in turn, reducing the tax burden reduces borrowing costs after a shock. The marginal cost is the additional effort required to further reduce the layoff rate.

$$\overbrace{\underbrace{\Pi_H^{good}-\Pi_H^{bad}}_{\substack{\Delta\text{ Profits}\\ \text{Between States}}}+\underbrace{ebl_H}_{\substack{\text{Tax}\\ \text{Decline}}}+\underbrace{r_H^{FB}ebl_He}_{\substack{\text{Borrowing}\\ \text{Costs}}}}^{\text{Marginal Benefit of Effort/Reduced Layoff}} = \overbrace{\underbrace{\psi'(1-r_H^{SB})}_{\substack{\text{Increasing}\\ \text{Effort Cost}}}}^{\text{Marginal Cost of Effort/Reduced Layoff}} \tag{18}$$

Since the high-risk firm ignores the government's objective of maximizing social welfare and output, the formula for the second-best layoff rate excludes the fiscal externality and output loss, terms (iv) and (v) in the formula for the first-best layoff rate (Equation 13). The firm fails to internalize that a higher layoff rate both increases expected benefit expenditures, which increases taxes for the low-risk firm through the budget constraint, and reduces the likelihood of production. Ignoring these costs leads to a privately optimal layoff rate which is higher than the socially optimal one: $r_H^{SB} \geq r_H^{FB}$.

Appendix A.3 shows that the second-best layoff rate declines with the degree of experience rating ($\frac{\partial r_H^{SB}}{\partial e} < 0$), as experience rating raises the firm's marginal benefit of exerting effort through a higher tax cost following layoffs. The sensitivity of the layoff rate to experience rating, captured by the

the higher layoff risk exposes more valuable output to potential loss.

elasticity $\varepsilon_{r_H,e}$, measures this moral hazard.

Second-Best High-Risk Industry's Employment Share, $l_H^{D,SB}$ It is derived in Appendix A.3 by setting to zero the derivative with respect to l_H of the high-risk firm's expected profits (Π_H from Equation 1). Since worker productivity in the high-risk industry declines over the unit interval, the firm hires workers in ascending order, starting from $i = 0$. The choice of labor demand can thus be interpreted as an optimal stopping rule along the unit interval. The first-order condition in Equation (19) identifies the stopping point as the marginal worker $i = l_H^{D,SB}$, whose productivity equals the marginal cost of labor. The firm therefore hires all workers $i \in [0, l_H^{D,SB}]$, whose productivities exceed the marginal cost. The marginal cost has four components: (i) the wage paid to the marginal worker; (ii) the wage increase needed to attract that worker; (iii) the unemployment tax increase to match the higher expected beneift expenditure; and (iv) the additional borrowing costs in case of shock due to higher taxes.

$$\overbrace{\underbrace{f_H(l_H^{D,SB})}_{\substack{\text{Marginal Worker's}\\ \text{Productivity}}}}^{\text{Marginal Product of Labor}} = \overbrace{\underbrace{w_{l_H^{D,SB}} f_H(l_H^{D,SD})}_{\text{Marginal Wage}} + \underbrace{\frac{\partial w_{l_H^{D,SB}}}{\partial l_H^{D,SB}} f_H(l_H^{D,SB})}_{\substack{\text{Wage Increase for}\\ \text{Marginal Worker}}} + \underbrace{ebr_H}_{\substack{\text{Tax Per}\\ \text{Worker}}} + \underbrace{\frac{r_H}{1-r_H} e^2 br_H}_{\text{Borrowing Costs}}}^{\text{Marginal Cost of Labor}} \tag{19}$$

Since the high-risk firm ignores the government's objective of maximizing social welfare and output, the formula for the second-best labor demand excludes the fiscal externality and output loss, terms (v) and (vi) in the formula for the first-best labor demand (Equation 14). The firm fails to internalize that increasing labor demand raises expected benefit expenditures, which increases taxes for the low-risk firm through the budget constraint, and induces output losses from hiring workers who would be more productive in the low-risk industry. Ignoring these costs leads to a privately optimal employment share in the high-risk industry which is larger than the socially optimal one: $l_H^{D,SB} \geq l_H^{D,FB}$.

Appendix A.3 shows that the high-risk firm's labor demand declines with the degree of experience rating ($\frac{\partial l_H^{D,SB}}{\partial e} < 0$): experience rating raises the marginal cost of labor, prompting the firm to stop hiring at a higher marginal productivities, which is found at a lower l_H due to the declining productivity curve.[16] Because in equilibrium increasing the tax burden on the high-risk firm reduces its labor demand and shifts employment toward the low-risk firm, the sensitivity of the high-risk firm's labor demand to experience rating, captured by the elasticity $\varepsilon_{l_H,e}$, measures the subsidization of the high-risk firm and the misallocation of labor across industries.

Next, wages adjust to ensure that the labor market clears in each industry.

$$l_H^{S,SB} = l_H^{D,SB} \equiv l_H^{SB} \qquad\qquad l_L^{S,SB} = l_L^{D,SB} \equiv l_L^{SB} \tag{20}$$

[16] Labor demand would similarly fall with labor costs in a Cobb-Douglas production function with decreasing returns.

Wage rates in the high-risk industry are set so that the indifferent worker coincides with the marginal worker, $i^I = l_H^{SB}$. As a result, all workers $i \in [0, l_H^{SB}]$ choose employment in the high-risk industry, making l_H^{SB} the high-risk industry's employment share. Workers $i > l_H^{SB}$ prefer the low-risk industry, where the wage adjusts so that demand meets supply. Consequently, the employment share in the low-risk industry is $l_L^{SB} = 1 - l_H^{SB}$.

Second-Best Experience Rating, e^{SB} It is the solution of the following constrained optimization problem: the government chooses the degree of experience rating e to maximize social welfare (Equation 6) and output (Equation 7), subject to the budget constraint (Equation 9), workers' optimal labor supply (Equation 11), the low-risk firm's optimal labor demand (Equation 12), labor market clearing in both industries, (Equation 20), and the high-risk firm's optimal layoff rate (Equation 18) and labor demand (Equation 19). Appendix A.3 shows the Lagrangian associated with this problem and its derivative with respect to e, capturing the welfare effects induced by a marginal change in the degree of experience rating. By an envelope condition, because the high-risk firm optimally chooses its layoff rate and labor demand, these welfare effects emerge only from the budget constraint (fiscal externalities) or output. To ease interpretation, I scale this derivative to obtain Equation (21), which equalizes the marginal benefit and cost of a one dollar decrease in the unemployment tax paid by the high-risk firm:

$$e^{SB} \equiv \overbrace{\underbrace{\left(\frac{\Pi_H^{\prime bad} - \Pi_H^{\prime good}}{\Pi_H^{\prime good}}\right)^{SB}}_{\text{Insurance Value}}}^{\text{Marginal Benefit of Coinsurance}} = \overbrace{\underbrace{(\lambda_f + \lambda_o)\varepsilon_{l_H,e}}_{\text{Labor Misallocation}} + \underbrace{(\rho_f + \rho_o)\varepsilon_{r_H,e}}_{\text{Moral Hazard}}}^{\text{Marginal Cost of Coinsurance}} \tag{21}$$

The left-hand side of Equation (21) represents the marginal benefit of reducing the unemployment tax paid by the high-risk firm by one dollar. The benefit, equal to e, correspods to the firm's loss per dollar of unemployment tax in the bad state and captures its insurance value. As shown in Appendix A.2, e can be equivalently expressed as the marginal profit gap between the good and bad states. A positive e thus implies a positive gap, indicating incomplete insurance against losses in the bad state. The equation also shows that experience rating, e^{SB} should equal the marginal cost of reducing unemployment taxes, captured by the right-hand side of the equation.

The right-hand side of Equation (21) represents the marginal cost of reducing the unemployment tax paid by the high-risk firm by one dollar. The total cost can be decomposed into the portions due to interindustry labor misallocation and moral hazard.

The elasticity of the high-risk firm's labor demand with respect to the degree of experience rating, $\varepsilon_{l_H,e} < 0$, serves as a sufficient statistic for the cost of labor misallocation. A more elastic response implies larger expansion in labor demand, and thus greater misallocation, when unemployment taxes decline. The two scaling factors, λ_f and λ_o, defined in Equation (22), contain the two social costs that the firm fails to internalize when choosing labor demand. λ_f measures the fiscal externality, which arises because greater labor demand increases expected benefit expenditures by br_H, of which a share $1-e$ is shifted to the low-risk firm through the budget constraint. λ_o captures the output loss from the misallocation of productive skills, which emerges when the marginal worker employed in the high-risk industry, l_H^{SB}, would have be more productive if employed in the low-risk industry, resulting in a productivity loss equal to their skills' transferability, $(1-r_H^{SB})f_H(l_H^{SB}) - f_L(l_H^{SB})$.

$$\overbrace{\lambda_f}^{\substack{\text{Fiscal Externality}\\\text{from Labor}\\\text{Misallocation}}} = -\frac{\overbrace{(1-e)br_H^{SB}}^{\substack{\text{Higher Benefit}\\\text{Expenditure}}}}{2eb(r_H^{SB})^2} \qquad \overbrace{\lambda_o}^{\substack{\text{Output Loss}\\\text{from Labor}\\\text{Misallocation}}} = \frac{\overbrace{\frac{1}{2}\left[(1-r_H^{SB})f_H(l_H^{SB}) - f_L(l_H^{SB})\right]}^{\text{Productivity Loss}}}{eb(r_H^{SB})^2} \tag{22}$$

The elasticity of the high-risk firm's layoff rate with respect to the degree of experience rating, $\varepsilon_{r_H,e} < 0$, serves as a sufficient statistic for moral hazard. A more elastic response implies larger increases in layoffs when unemployment taxes decline. The two scaling factors, ρ_f and ρ_o defined in Equation (23), represent the two social costs that the high-risk firm fails to internalize when choosing the layoff rate. ρ_f measures the fiscal externality, which arises because a higher layoff rate increases expected benefit expenditures by bl_H, of which a share $1-e$ is shifted to the low-risk firm through the budget constraint. Notice that $\rho_f = \lambda_f$. ρ_o measures the output loss due to the greater probability that the high-risk firm stops producing in the bad state: $\int_0^{l_H^{D,FB}} f_H(i), di$.

$$\overbrace{\rho_f}^{\substack{\text{Fiscal Externality}\\\text{from Moral}\\\text{Hazard}}} = -\frac{\overbrace{(1-e)bl_H^{SB}}^{\substack{\text{Higher Benefit}\\\text{Expenditure}}}}{2ebr_H^{SB}l_H^{SB}} \qquad \overbrace{\rho_o}^{\substack{\text{Output Loss}\\\text{from Moral}\\\text{Hazard}}} = \frac{\overbrace{\left(-\int_0^{l_H^{SB}} f_H(i)\,di\right)}^{\text{Production Loss in H}}}{2ebl_H^{SB}r_H^{SB}} \tag{23}$$

Description of Second-Best Equilibrium The second-best equilibrium is characterized by: (i) an inefficiently high layoff rate: $r_H^{SB} \geq r_H^{FB} = r_H^{min}$; (ii) an inefficiently high employment share in the high-risk industry: $l_H^{SB} \geq l_H^{FB} = l_H^{Eff}$; and (iii) incomplete insurance for the high-risk firm. The government cannot offer full insurance as in the first best, because lowering e raises the firm's layoff rate and labor demand, inducing fiscal externalities and output losses from moral hazard and labor misallocation. Instead, it must choose e to balance the insurance value, favoring lower e, against these distortions, favoring higher e. So long as the distortions on the right-hand side of Equation (21) are positive, the second-best experience rating is also positive, implying a positive marginal profit gap and thus incomplete insurance. The same result is obtained in Baily (1978)'s model of optimal unemployment benefits, where the government must balance workers' value of insurance, captured by the gap in marginal utility of consumption between employment and unemployment, against the job search distortions induced by insurance. The classic Baily formula captures this tradeoff by defining the optimal benefit level as the one equating the marginal benefit and cost of insuring *workers*. The firm-side formula in Equation (21) defines the optimal degree of experience rating as the one that equates the marginal benefit and cost of insuring *firms*.

The trade-off between insurance and incentives is clearest under complete experience rating ($e = 1$), which completely eliminates insurance for the high-risk firm but allows the government to implement the first-best, efficient allocation. At $e = 1$, the high-risk firm receives no insurance, as it bears the full tax burden its benefit charges, and faces the maximum profit gap between states. This equilibrium, marked by complete experience rating and a zero tax burden for the low-risk firm, is illustrated by the point named "Exp. Rat. EQ" in Figure II. Moreover, both the layoff rate and labor demand of the high-risk firm fall to their first-best levels. When $e = 1$, the fiscal externality terms in Equations (13) and (14) for the first-best layoff rate and labor demand vanish, while the output losses are already equal to zero. Consequently, these first-best first-order conditions coincide with the second-best ones in Equations (18) and (19). Therefore, complete experience rating recovers the first-best, efficient equilibrium, denoted by the point named "Exp. Rat. EQ" in Figure I, where $r_H^{ER} = r_H^{min}$ and $l_H^{ER} = l_H^{Eff}$.

With coinsurance ($e < 1$), the gap in marginal profits across states narrows, but two distortions arise. Figure I illustrates the output losses from incomplete experience rating. As shown in Figure A.1, reducing the degree of experience rating leads to a new equilibrium in the labor market, characterized by a higher (lower) employment share and wage rate in the high (low)-risk industry. Accordingly, the coinsurance equilibrium is represented by two points in Figure I: "Coinsurance EQ H", with higher employment share and lower marginal productivity in the high-risk industry, and hence a higher marginal wage rate, and "Coinsurance EQ L", with lower employment share and higher marginal productivity in the low-risk industry, implying a lower marginal wage rate than in the first best. Because the layoff rate increases with coinsurance, "Coinsurance EQ H" lies on a lower expected productivity line, $(1 - r_H^{CO})f_H(i)$. The blue-shaded triangle represents the output loss from labor misallocation. All the workers between the efficient employment, $L_H^{Eff/ER}$, and the new employment in the high-risk industry, l_H^{CO} would have been more productive in the low-risk industry. The area is thus calculated as the product of the height of the triangle, the vertical distance between "Coinsurance EQ L" and "Coinsurance EQ H" which measures the productivity loss for the marginal worker in the high-risk industry under coinsurance, $(1 - r_H)f_H(l_H^{CO}) - f_L(l_H^{CO})$, and the base, which is the change in the employment share in the high-risk industry induced by the reduction in experience rating, $l_H^{CO} - l_H^{Eff/ER} \approx \varepsilon_{l_H,e}$, divided by a half. Inspecting Equations (21) and (22) reveals that this area is proportional to $\lambda_o \varepsilon_{l_H,e}$. The gray and green parallelograms represent the output loss from moral hazard. With coinsurance, a higher layoff rate increases the risk of production loss in the high-risk industry. The area is calculated as the product of the height of the parallelogram, which measures the increase in the layoff rate induced by the lower degree of experience rating, $r_H^{CO} - r_H^{ER} \approx \varepsilon_{r_H,e}$, and the base, equal to the productivity at risk, i.e., the output of all workers in the high-risk sector, $\int_0^{l_H^{CO}} f_H(i)\, di$. Inspecting Equations (21) and (23) shows that this area is proportional to $\rho_o \varepsilon_{r_H,e}$. The gray portion represents a loss that would occur even in the first-best, normalized to zero in Equation (17). The green and blue areas overlap as these two inefficiencies reinforce each other.

Figure II illustrates the fiscal externalities from incomplete experience rating. Reducing the degree of experience rating from e' to e increases the unemployment tax burden on the low-risk firm from $T(e')$ to $T(e)$. This increase can be decomposed into the four components shown in Equation (24). First, there is a mechanical effect: holding benefit spending constant, a lower e shifts a larger share of the tax burden onto the low-risk firm. Second, as e declines, the high-risk industry's employment share rises by $\varepsilon_{l_H,e} = l_H - l'_H$. The resulting tax increase for the low-risk firm equals the fraction $(1-e)$ of the benefit b that these additional high-risk workers would claim in the event of a shock, occurring with probability r_H. This term is proportional to $\lambda_f \varepsilon_{l_H,e}$ in Equations (21) and (22), capturing the fiscal externality from labor misallocation. Third, the lower e induces a higher layoff rate in the high-risk sector, $\varepsilon_{r_H,e} = r_H - r'_H$. The associated tax increase for the low-risk firm equals the fraction $(1-e)$ of the benefit b that the l_H workers would claim if laid off. This term is proportional to $\rho_f \varepsilon_{r_H,e}$ in Equations (21) and (23), capturing the fiscal externality from moral hazard. Last, a second-order term accounts for their interaction. The higher labor demand and layoff rate explain why the coinsurance equilibrium with experience rating equal to zero, "Coinsurance EQ" in the figure, implies larger taxes for the low-risk firm relative to the first-best equilibrium "First-Best EQ".

$$T_L(e) - T_L(e') = \underbrace{b l_H r_H (e'-e)}_{\text{Mechanical effect}} + \underbrace{\overbrace{b r_H (1-e)}^{\propto \lambda_f} \overbrace{(l_H - l'_H)}^{\varepsilon_{l_H,e}}}_{\text{Labor misallocation}} + \underbrace{\overbrace{b l_H (1-e)}^{\propto \rho_f} \overbrace{(r_H - r'_H)}^{\varepsilon_{r_H,e}}}_{\text{Moral hazard}} + \underbrace{b(1-e)(r_H - r'_H)(l_H - l'_H)}_{\text{Second-order term}} \tag{24}$$

2.3 Model Discussion

Optimal Policy The model delivers a Baily (1978)-style sufficient-statistic formula (Equation 21) for the optimal degree of experience rating, which can be empirically implemented to evaluate the optimality of unemployment insurance financing policies. To ensure meaningful comparisons, all its parameters should be estimated from the same context and time of the policies being evaluated. Marginal costs, captured by firms' elasticities in response to experience rating, can be estimated using quasi-experimental variation from policy reforms, while the marginal benefit depends on a deeper primitive, firms' value of insurance, conceptualized here for the first time. The formula also depends on the degree of experience rating itself, e, the employment share of the high-risk industry, l_H, the layoff rate in the high-risk industry, r_H, the benefit level b, and industry-specific marginal productivities, which are potentially endogenous to the degree of experience rating. As a result, the formula cannot be extrapolated to assess how the marginal benefit and cost vary with experience rating without imposing assumptions on how these parameters change with e. However, the formula is well suited to evaluating the optimality of the degree of experience rating in place. Once both sides of the formula are estimated, the direction of the optimal reform is: reduce experience rating if the marginal benefit exceeds the marginal cost, and increase it otherwise. Non-marginal reforms, such as moving from a uniform tax system to experience rating, require richer structural models.

(Chetty 2009). By contrast, the minor, frequent adjustments to experience rating in the US and the small changes in unemployment tax rates for specific firms or workers in Europe or South America may fall within the scope of the sufficient statistics approach.

External Validity Parameters estimated in one context may not apply elsewhere and, even within the same context, may vary over the business cycle. Although the model is static, tracking how the two sides of the formula evolve over time can inform policy adjustments. For example, if the value of insurance rises in recessions due to higher liquidity costs while moral hazard declines as layoffs become unavoidable, experience rating should be reduced. Similar patterns have been established for unemployment benefits: workers' insurance value rises in recessions (East et al. 2015) and job-search distortions decline (Schmieder et al. 2012, Kroft et al. 2016), supporting more generous benefits in downturns. In many countries, firms already receive support to preserve jobs during recessions through short-time work programs. If the optimal degree of experience rating declines in downturns, the U.S. practice of raising experience rating to finance increased benefit claims precisely during recessions, described in Section 3.1, would be inefficient.

Industries and Firms They coincide in the model, but the distinction is empirically important. Labor misallocation occurs within industries as well (Anderson et al. 1993), although it induces smaller inefficiencies because firms within an industry tend to face similar unemployment risks and require similar skills. Moral hazard is generally considered at the firm-level, but may be minimal in seasonal industries, where layoffs reflect external constraints, such as the weather, rather than tax incentives. In such industries, where unemployment risk is fully predictable, the value of insurance may be limited, whereas insuring industries subject to large, unpredictable shocks may be more valuable than targeting potentially inefficient individual firms (Giupponi et al. 2022).

Extensions The model relies on a set of simplifying assumptions that increase its tractability but can be relaxed without altering its core insights. Several model extensions are discussed in Appendix B. The solution is unchanged when modeling unemployment taxes are as payroll rather than head taxes (Appendix B.2) when eligibility and benefit take-up are incomplete (Appendix B.7), and when incorporating adjustment costs, career ladders, mobility costs, and other factors modifying the high-risk firm's profits or worker's utilities (Appendices B.3 and B.4). This robustness arises because these changes either leave the distribution of the tax burden across firms unchanged, or enter firms' and workers' optimal decisions, shifting both the first- and second-best solutions without affecting the gap between them. The optimal degree of experience rating increases relative to baseline when workers prefer employment to unemployment (Appendix B.5); when the government places a higher welfare weight on workers (Appendix B.6); and when benefit take-up declines with experience rating—for example, because firms appeal claims more aggressively as their tax liability rises (Lachowska et al. 2020), or higher taxes discourage formal hiring (Bertrand et al. 2021) (Appendix B.7). The optimal degree of experience rating decreases relative to baseline when firm profits must be protected because corporate taxes finance other public goods (Appendix B.8). The implication is ambiguous when expanding the high-risk industry has social value—for example, if it represents the private sector and the low-risk/public sector is inefficiently large, if it offers high returns, or if it is inefficiently small due to risk deterring entrepreneurship or raising the wages demanded by workers, because coinsurance also increases the risk of losing more valuable output.

Next Step: Implementation In the rest of the paper, I calibrate the parameters of Equation (21) for South Carolina and Colorado to evaluate the local optimality of their degrees of experience rating. Section 3 presents the estimation of the elasticity of labor demand with respect to the degree of experience rating, $\varepsilon_{l_H,e}$. Section 4 details the calibration of the remaining parameters. Section 5 evaluates the optimal policy.

3 Estimation

I estimate the elasticity of labor demand with respect to the degree of experience rating, $\varepsilon_{l,e}$ for South Carolina and Colorado. Identification is challenging due to difficulties in measuring the degree of experience rating and in finding suitable quasi-experimental variation. However, because the unemployment tax per worker increases linearly with the degree of experience rating ($\tau = ebr_H$) the elasticity with respect to experience rating is equivalent to the elasticity with respect to the unemployment tax per worker. The latter is easier to estimate because of the availability of quasi-experimental variation in the tax per worker from state-level experience-rating reforms.

$$\varepsilon_{l_H,e} = \frac{\partial l_H}{\partial \tau}\frac{\partial \tau}{\partial e}\frac{e}{l_H} = \frac{\partial l_H}{\partial \tau}\frac{br_H e}{l_H} = \frac{\partial l_H}{\partial \tau}\frac{\tau}{l_H} = \varepsilon_{l_H,\tau} \tag{25}$$

I describe the institutional framework, the data, the empirical strategy and the results for each state.

3.1 Unemployment Insurance Financing in the United States

The Unemployment Compensation Program, established in response to the Great Depression with the 1935 Social Security Act, provides temporary and partial wage replacement to workers who become unemployed after involuntary layoffs. Benefits are primarily funded through payroll taxes levied on firms. Vroman et al. (2017) provides a comprehensive overview the program. Here, I highlight two distinctive features that create quasi-experimental variation in unemployment taxes which I leverage for identification.

First, unemployment taxes vary countercyclically to cover higher benefit spending during recessions. Each state maintains a separate Unemployment Trust Fund, into which its unemployment taxes are deposited and from which benefits are paid. When fund balances fall below specified thresholds due to higher benefit demand, states raise unemployment taxes to restore fund solvency and guarantee benefit payments; once benefit demand declines and balances recover, taxes are reduced. States adjust tax revenue by changing the unemployment tax rate and/or the taxable wage base, which is the portion of workers' wages subject to taxation.

Second, unemployment taxes vary across firms and over time due to experience rating: firms that lay off more workers, and thus generate higher benefit spending, are assigned higher tax rates. The assignment occurs in two steps. First, states calculate an "experience factor" for each firm, based on benefits claimed by its laid-off workers and updated each year to reflect new layoffs. Second, firms are assigned tax rates according to a schedule (a table) that maps experience factors to tax rates. The schedule may change annually: states use higher-rate schedules restore fund solvency, and adopt lower-rate schedules once funds are replenished.

Despite the presence of these policies designed to make firms internalize the costs of their layoffs, experience rating in the United States remains far from complete. This holds under both measures used in the model. First, following Vroman et al. (2017), I calibrate the average share of a firm's own benefit charges repaid through unemployment taxes and find an average of 78% across states between 2000 and today ($e_{share} = 0.78$). Second, Pavosevich (2020) estimates that the average marginal cost of an additional dollar of benefit charges in 2018 was only 29 cents ($e_{mtc} = 0.29$). Consequently, the distortions associated with incomplete experience rating remain a relevant concern. At the same time, the frequent adjustment of these policies leaves scope for optimal reforms.

Importantly for my empirical strategy, solvency and experience rating policies generate substantial variation in unemployment taxes across firms and over time, which I use for identification. Figure C.2 shows that unemployment taxes range between 0.4 and 1% of total wages over the business cycle, rising during and after recessions. These increases reflect both higher layoffs under experience rating and state-level adjustments to restore fund solvency. The identification strategies for South Carolina and Colorado, presented next, leverage quasi-experimental variation in unemployment taxes arising from the interaction of state-level reforms with underlying experience rating policies, while accounting for tax changes driven by firms' evolving layoff histories.

3.2 Reduced Form Effects of Unemployment Taxes in South Carolina

Data and Sample The data provided by the South Carolina Department of Employment and Workforce (SC DEW) cover 358,758 firms operating in South Carolina between 2002 and 2022. To prevent identification, the top 1% largest firms were excluded from the data. The remaining firms account for half of state employment (Figure D1), and their sectoral distribution mirrors that of the state economy (Table D1). For each firm, the dataset includes quarterly employment, total, minimum, average, and maximum wages, taxable wages and unemployment taxes; annual reserve ratios, benefit ratios, and unemployment tax rates; the NAICS four-digit sector code and the date of establishment. I restrict the sample to 40,272 private-sector experience-rated firms continuously observed between 2007 and 2014, an eight-year window around the 2011 tax reform used for identification. In 2010, firms in the study sample had on average 11 employees, 17 years of activity, paid average annual wages of $39,340, and operated in services (57%), trade and transport (21%), construction (12.5%), manufacturing (5.5%), finance, insurance, and real estate (2.5%) and the primary sector (1.5%). Consistent with the exclusion of new (non-experience rated) firms and firms going out of business during the study period, the sample is positively selected: included firms are larger, pay higher wages, have been established for longer, and have fewer benefit charges per worker than excluded firms (Table D2). Appendix D.1 provides details on data cleaning, the construction of key variables used in the analysis, and sample restrictions.

Identification Strategy Identification relies on quasi-experimental variation in firms' unemployment taxes per worker from South Carolina's 2011 reform of its experience rating policies. The goal of the reform was to raise tax revenue after the state became insolvent in 2008, when trust fund reserves were insufficient to cover the increase in benefit claims during the Great Recession and the state had to borrow $1 billion in federal funds (see panel [c] of Figure C.2). Initiated in 2010, the reform took effect in 2011, and by 2014 the loan had been repaid.

At the core of the reform was South Carolina's modification of the experience factor used to assign unemployment tax rates to firms. Equation (26) shows the experience factors in place before and after the reform and reveals two key differences between them. Until 2010, the state used the *reserve ratio*, which is calculated by dividing the net reserves in a firm's individual account, equal to the difference between the sum of all of the unemployment benefits ever claimed by the workers laid off by a firm and the sum of all of the unemployment taxes ever paid by the firm since its establishment, by the previous year's taxable wages. The reserve ratio could be positive or negative depending on whether benefit charges exceed tax payments. Layoffs that generate new benefit claims increase a firm's reserve ratio.[17] Since 2011, South Carolina employs the *benefit ratio*, which is calculated as the ratio of the benefits charged to a firm to the sum of the taxable wages paid by the firm during the most recent seven years.[18] The benefit ratio only assumes non-negative values. Layoffs that generate new benefit claims increase a firm's benefit ratio. There are two key differences between these two experience factors. First, they differ in the length of the lookback period, j, used to assess firms' layoff histories. While the reserve ratio is calculated using data from the firm's establishment date to the calculation date, the benefit ratio relies on a rolling seven-year lookback period, discarding any earlier data. Second, while unemployment taxes factor into the calculation of the reserve ratio to determine the firm's net position within the program, they are irrelevant to the calculation of the

[17] I have inverted the sign of the reserve ratio so that higher values of all experience factors consistently correspond to higher unemployment tax rates. Net reserves are in fact calculated as the difference between firms' total tax payments and total benefit charges.

[18] The lookback period of the benefit ratio was updated to three years in 2014.

benefit ratio, which only depends on benefit charges.

$$\text{RR}_{it} = \frac{\sum_{j=-\infty}^{t-1} \text{Benefits}_{ij} - \sum_{j=-\infty}^{t-1} \text{Taxes}_{ij}}{\text{Taxable Wages}_{i,t-1}} \longrightarrow \text{BR}_{it} = \frac{\sum_{j=-7}^{t-1} \text{Benefits}_{ij}}{\sum_{j=-7}^{t-1} \text{Taxable Wages}_{ij}} \tag{26}$$

With this transition, South Carolina aimed to accelerate the replenishment of the state unemployment trust fund. Since most firms had built up substantial reserves through years of unemployment tax payments, the benefit costs charged during the Great Recession did not significantly increase their reserve ratios, and, consequently, their unemployment tax rates remained largely unaffected. Indeed, the reserve ratio tends to be a "sticky" metric of experience with unemployment, predominantly reflecting firms' historical conditions rather than their present conditions, and responding slowly to labor market changes (Vroman et al. 2017, Miller et al. 2019). By employing the benefit ratio, the government could ignore tax reserves and impose elevated tax rates on those firms that faced substantial layoffs during the Great Recession. Consistent with this goal, Lachowska et al. (2020) finds that benefit ratio-states increased their tax revenue and restored fund solvency at double the rate of reserve ratio-states after COVID-19.

The transition from the reserve ratio to the benefit ratio caused a sudden change in firms' experience factors and, consequently, in their unemployment tax rates, leading to a redistribution of the unemployment tax burden among the firms in the state. Intuitively, the reform penalized the firms with cumulated tax payments by disregarding unemployment taxes in the experience calculation and placing greater emphasis on any benefit costs incurred during the Great Recession. As a result, these firms experienced a sharp increase in their unemployment tax rates just as the economy was beginning to bounce back.[19] By contrast, the firms that had laid off workers in the distant past but not during the benefit ratio's lookback period benefitted from the neglect of their past cumulated benefit charges and saw their unemployment tax rates decline.

My identification strategy leverages the heterogeneous impact of the transition from reserve to benefit ratios on firms with different layoff histories. Figure III presents the variation employed and motivates the specification required for identification. Panel (a) displays firms' benefit ratios (on the y-axis) against their predicted reserve ratios (on the x-axis) in 2011. Predicted reserve ratios, calculated following Equation (26), measure the experience factors firms would have had in 2011 had the transition to benefit ratios not occurred. This measure is preferable to actual 2010 reserve ratios, as it accounts for changes in firms' experience factors between 2010 and 2011 that would have occurred even without the reform, while still being derived entirely from pre-reform data. Nonetheless, the correlation between true reserve ratios in 2010 and predicted ones in 2011 is 0.81

[19] In April 2011, the Greenville Business Magazine reported the concerns of a long-standing firm who was severely impacted during the Great Recession: *"Two of those years, 2008 and 2009, are what I call the 'Katrina' years as far as the economy is concerned. They were devastating. I've been in business here for thirty years. Do the other twenty-three years not count for anything?"* The magazine also emphasized widespread concerns that *"the new rates will discourage companies from hiring new employees as the economy begins its uptick."*

(shown in panel [a] of Figure E1), confirming the persistence of reserve ratios over time. While a positive correlation between benefit and predicted reserve ratios exists, as both measures reflect firms' experience with unemployment, substantial variation remains. As a result, firms that were similarly classified and taxed under the old system were reassigned to different categories and tax rates under the new one.

Using firms' predicted reserve ratios, I construct a measure of their predicted unemployment tax rate in 2011, which is the rate they would have faced had the reform not occurred and had the 2010 unemployment tax schedule (shown in panel [c] of Figure E1) remained in place in 2011 as well. I then compare these predicted rates with the true unemployment tax rates assigned in 2011 based on true benefit ratios and the new schedule, which also introduced new lower and higher unemployment tax rates (shown in panel [d] of Figure E1). Panel (b) of Figure III presents all possible combinations of true (y-axis) and predicted (x-axis) tax rates in 2011, weighted by their frequency in the study sample. The broad representation of tax-rate combinations in the data confirms substantial heterogeneity in the reform's impact on firms' unemployment tax rates; using predicted reserve ratios isolates this impact from changes that would have occurred even without the reform due to evolving layoff histories and experience factors over time.

I construct a firm-level continuous measure of treatment as the difference between true and predicted unemployment tax rates in 2011. This difference is then multiplied by the taxable wage base in 2011 ($10,000) to obtain the firm's predicted change in unemployment tax per worker.[20] The difference between true and predicted taxes per worker therefore captures the variation in firms' tax burdens generated by the reform, relative to what their taxes would have been had the transition from reserve to benefit ratios not occurred. Panel (c) of Figure III displays the distribution of the predicted change in the tax per worker, which ranges between -$600 and $1,000. 43% of firms experience a predicted change of -$114; these are firms assigned the minimum tax rate under both systems.

My identification strategy consists in comparing firms with the same benefit ratios post-reform but different reserve ratios pre-reform. Because of their similar benefit ratios, these firms had similar trends during the seven-year lookback period used to calculate the benefit ratio, which coincides with the reform pre-period (July 2003-July 2010, or the *"recent past,"*). The different predicted reserve ratios, shaped by the different compositions of their *"distant past"* reserves, generate variation in predicted tax rates in 2011 and, consequently, in the predicted change in tax per worker. Panel (d) of Figure III provides a simple example to illustrate my approach. The figure shows the average unemployment tax rates for two groups of firms, both with a benefit ratio equal to zero post-reform, one with a predicted reserve ratio corresponding to the minimum pre-reform tax rate (1.24%) and the other to the maximum (6.1%). Because their benefit ratios are zero, neither group incurred benefit charges during the reform's lookback period, and should thus display parallel trends prior to

[20]The taxable wage base increased from $7,000 to $12,000 between 2011 and 2014 (panel [b] of Figure E1). This common change across firms is absorbed by year fixed effects and does not affect identification.

the reform, in the recent past. However, due to differing predicted reserve ratios, their unemployment tax rates and taxes per worker differed substantially before the reform. Importantly, these differences must be entirely driven by layoff behavior occurring before the seven-year window captured under the reserve ratio system, in the distant past. When in 2011 benefit ratios replaced reserve ratios, these historical differences became irrelevant: the tax rates of the two groups were equalized, leading to a larger reduction in the unemployment tax per worker for firms with higher initial rates.

A key identification step highlighted by this example is controlling for firms' benefit ratios, which measure firms' layoffs in the recent past.[21] Without this control, predicted tax changes would be correlated with pre-reform behavior, since high reserve ratios predict high benefit ratios (panel [a] of Figure III), violating the identifying assumption that treatment is quasi-randomly assigned. Because firms in 2011 are ranked by their benefit ratios and assigned to one of twenty tax-rate classes, I use these classes as a natural grouping of firms into different "risk-sets". Panel (e) of Figure III shows that class-one firms have a benefit ratio of zero, which means no benefit charges during the lookback period, while higher classes include firms with progressively larger pre-reform average benefit charges per worker. Consistently, panel (f) shows that class-one firms maintained stable employment pre-reform, whereas higher-class firms experienced increasingly severe workforce reductions. Panel (d) confirms that, after controlling on benefit-ratio classes, substantial variation remains in the predicted change in unemployment taxes per worker.

Equation (27) shows my preferred specification. $Y_{i,t}$ is the outcome for firmr i at time t, which is quarter or year depending on the outcome, α_i are firm fixed effects, Pred Δ Unemp. Tax Per Worker$_i$ is the continuous treatment measuring the predicted impact of the reform on a firm's unemployment tax per worker, b_i are the twenty benefit ratio classes, $\alpha_{b(i),t}$ are benefit-ratio class-by-time fixed effects; and $\epsilon_{i,t}$ is an error term. I cluster the standard errors at the firm level to account for correlated outcomes within firms over time. In regressions where the outcome is measured yearly, β_{2010} is normalized to zero. When the outcome is measured quarterly, β_{2010q4} is normalized to zero. I multiply benefit ratio classes by the time fixed effects to control for different layoff and employment trends over the pre-period. Following common practice (Lachowska et al. 2018, Johnston 2021), I winsorize outcomes at the 99^{th} or 99.9^{th} percentiles to remove the influence of outliers and errors, which would otherwise introduce substantial noise in the estimates. Appendix D.1 describes the cleaning steps in place and winsorization details.

$$Y_{i,t} = \alpha_i + \sum_{y=2007}^{2014} \beta_y 1_{y=t} \text{Pred } \Delta \text{ Unemp. Tax Per Worker}_i + \alpha_{b(i),t} + \epsilon_{i,t} \tag{27}$$

For the β coefficients to identify the average treatment effect on treated firms (ATTs), two assumptions need to be satisfied: parallel trends and no-anticipation. The parallel trend assumption requires

[21] An equivalent approach was proposed by Johnston (2021) (footnote 32).

that average evolution of outcomes that units with any predicted tax change ("dose") d would haved experienced without treatment is the same as the evolution of outcomes actually experienced by groups with $d = 0$. To establish the credibility of this assumption, I directly examine parallel trends in firms' outcomes. Additionally, I show that firms' predicted reserve ratios, which determine treatment intensity, mainly reflect behavior in the distant past, making it unlikely that treatment is correlated with current outcomes, especially after accounting for recent trends through benefit ratio fixed effects. To this end, Equation (28) decomposes the numerator of the reserve ratio, the firm's *total reserves*, into three components: (i) the *recent benefits*, which correspond to the numerator of the benefit ratio, (ii) the *distant past reserves*, which are calculated as the difference between the benefits charged to the firm and the unemployment taxes paid by the firm from the establishment to seven years before the calculation date, and (iii) the *recent taxes* paid by the firm during the seven-year lookback period of the benefit ratio. Variation in distant past reserves and recent taxes generates variation in the reserve ratios of firms with the same benefit ratio. Consistent with treatment being primarily determined by distant past factors, Figure E2 shows that, conditional on benefit ratio classes, most variation comes from distant past reserves compared to recent taxes.

$$\underbrace{\sum_{-\infty}^{0}\text{Benefits}_i - \sum_{-\infty}^{0}\text{Taxes}_i}_{\text{RR num= Tot. reserves}} = \underbrace{\sum_{-7}^{0}\text{Benefits}_i}_{\substack{\text{BR num.=}\\ \text{recent benefits}}} + \underbrace{\left(\sum_{\infty}^{-7}\text{Benefits}_i - \sum_{\infty}^{-7}\text{Taxes}_i\right)}_{\text{Distant past reserves}} - \underbrace{\sum_{-7}^{0}\text{Taxes}_i}_{\text{Recent Taxes}} \tag{28}$$

Second, identification hinges on the non-anticipation assumption. Several factors make it unlikely that the reform influenced firms before its implementation in 2011. The depletion of the unemployment trust fund drew little attention until South Carolina became insolvent in December 2008. Yet, legislative action was delayed until 2009, and the reform was approved only in spring 2010.[22] Firms could not predict how they would have been individually impacted until late November 2010, when they were notified their tax rates for 2011. Anticipating these rates was unfeasible because the calculation of the benefit ratios required data on workers' benefit charges which firms do not have access to, and because rates are assigned based on firms' benefit ratio ranking rather than level. Additionally, the law stipulated a ten-year lookback period for the calculation of the benefit ratio, but SC DEW only had seven years of data available in 2011. When the tax rates were announced, *"after most companies began their fiscal years with budgets already in place"*, firms remained *"blindsided"* with *"tens of thousands of dollars in unplanned expenses"* (The Greenville Business Magazine, April 2011). Since both the benefit ratio and the predicted reserve ratio, the two determinants of treatment intensity, are based on firms' behavior up to July 2010, firms had no opportunity to adjust their behavior to influence their new tax rates and sort into different treatment intensities.

[22] On March 19, 2010, The Sun News attributed the delay to legislators seeking re-election, noting that *"none publicly told his colleagues what he had heard (about the coming insolvency). Not one alerted the media nor, as far as I can tell, anyone else. Another option would have been simply to tell someone - the press, the colleagues, anyone, and begin working on a solution in April 2008"*.

Results Figure IV presents the estimated β_y coefficients from Equation (27) for firms in the SC DEW study sample with fewer than 50 employees in 2010, the 95$^{\text{th}}$ percentile of the employment distribution, to limit noise from outliers. Results with large firms are presented among the robustness tests. Table E1 shows year-level estimates. Table D3 describes the construction of the outcomes.

Panel (a) shows the first stage: the effect of the predicted unemployment tax per worker on the actual tax per worker. The tax is flat before the reform and increases by 74 cents in 2011. The estimate declines over time but remains positive and statistically significant, averaging 35.7 cents post-reform. Given the average pre-reform tax per worker of \$134.9, the estimate implies an average increase in the tax per worker of \$48 (0.357×134.9), or 36% of the pre-reform mean.

Panel (b) shows the corresponding changes in quarterly employment. Employment is stable before the reform and begins to gradually decline in 2011. The average post-reform reduction of 0.001 implies an average decline of 0.135 (0.001×134.9) workers, equal to 1% of pre-reform average employment. The decline unlikely results from firms manipulating their data to lower their tax bills: since 2004, the SUTA Dumping Prevention Act has prohibited such practices, and SC DEW enforces it by conducting routine annual audits of 1,500 firms and imposing high "delinquent" tax rates on violators. More importantly, estimating a labor demand elasticity requires that employment changes reflect reduced hiring rather than increased separations. Although the data do not distinguish hirings from separations, several factors point towards reduced hiring as the main driver. First, the reform occurred after the Great Recession, when most separations had already happened and firms' primary decision margin was hiring.[23] Second, Guo (2024) shows, using LEHD data that directly observe hiring and separations, that higher unemployment taxes reduced employment mainly by discouraging hiring in several states, including South Carolina, after the Great Recession. Finally, I provide indirect evidence that the effect is driven by hiring by examining taxable wages, discussed below.

Panel (c) shows that total annual wages were stable before the reform and declined gradually starting in 2011, by between \$17 and \$90 per year. The average post-reform estimate of \$58 implies an average reduction in total wages of \$7,824 (58×134.9), or 1.8% of the pre-reform mean.

Panel (d) shows that average annual wages were flat before the reform, remained unchanged in 2011, and declined starting in 2012.[24] The average post-reform estimate of -\$1.9 implies a reduction in average wages of \$256 per year (1.9×134.9), or 0.7% of the pre-reform mean. Examining taxable wages, discussed next, helps distinguish whether the decline in average wages reflects changes in workforce composition or actual wage reductions among workers who remain employed.

Panel (e) shows that taxable wages were stable prior to the reform, dropped by \$13 in 2011, and

[23] This sentiment is echoed by the Greenville Business Magazine (April 2011), suggesting that *"the new rates will discourage companies from hiring new employees as the economy begins its uptick."* and that *"no employer should feel safe. This structure will keep South Carolina in a recession and make sure (employers) will not recover."*

[24] Figure E3 shows the effects on total, minimum, average, and maximum quarterly wages. Total wages decline gradually throughout the post-reform period, while the average, minimum, and maximum are largely unaffected except for significant declines in the fourth quarters of 2012, 2013, and 2014, which drive the annual patterns.

continued to decline gradually by up to \$34 per year. The average post-reform estimate is \$26, implying an average reduction in taxable wages of \$3,507 (26×134.9), or 3.7% of the pre-reform mean. The reduction in taxable wages provides indirect evidence that the decline in employment in panel (b) is due to reduced hiring. If separations were the main drivers, taxable wages would not fall, because firms pay taxes on wages paid to workers whom they lay off during the year, unless these workers were hired and then laid off before earning a wage equal to the taxable wage base (\$7,000–12,000 per year). Moreover, the magnitude of the decline clarifies what drives the reduction in average wages in panel (c). If the missing workers in panel (b) earned more than the taxable wage base, taxable wages would have declined by the average employment reduction, 0.135, multiplied by the average post-reform taxable wage base of \$11,500, yielding \$1,552.5. If the missing workers earned less than the taxable wage base, the implied decline in taxable wages would be even smaller. The excess reduction in taxable wages documented here, \$3,507 minus \$1,552.5, must therefore reflect lower wages among the remaining workers, with some workers' earnings falling below the taxable wage base.[25]

Finally, panel (f) shows the effect on unemployment taxes, which were flat pre-reform and increased by \$4.23 in 2011, implying an increase of \$571 (4.23×134.9), or 29%. The estimate declines to \$0.44 in 2012, implying an average increase of \$59, and becomes statistically insignificant thereafter, consistent with firms reducing their taxable wages to bring their tax bills down to pre-reform levels. I compare observed unemployment taxes with simulated taxes, calcluated under the assumption that taxable wages remained at their 2010 level—to assess the implications of these behavioral responses for tax revenue. Absent employment and wage adjustments, unemployment taxes would have been \$0.40 higher in 2011 and unchanged thereafter, implying an average revenue gap of \$53.6 (0.40×134.9) per firm in 2011. Over time, the gap between simulated and observed taxes is attenuated by the structure of the benefit-ratio system: although higher simulated taxable wages mechanically raise simulated taxes, they also reduce the benefit ratio (see Equation 26), which lowers the simulated tax rate and offsets part of the increase.

Robustness I test the stability of my findings in several ways. Figures E4 and E5 report estimates from a specification with large firms. Because their inclusion introduces substantial variability, all outcomes are scaled by their 2010 levels. To limit noise arising from scaling by small denominators, I exclude firms with fewer than one employee in 2010. Results are consistent with baseline.

Figure E6 shows that the results are robust to several modifications of the baseline specification: (i) adding sector-by-time fixed effects to account for heterogeneous industry trends; (ii) adding

[25] Panel (b) of Figure E1 shows the evolution of the taxable wage base in South Carolina. Figure C.3 shows that, under any assumption on the wages of the missing workers, reductions in the wages of retained workers are necessary to explain the larger-than-predicted decline in taxable wages and the decline in average wages. Consistently, Panel (e) of Figure E3 shows a post-reform increase of 0.004 percentage points in the share of workers earning below the taxable wage base in a quarter. This evidence is only suggestive, because individual workers' annual wages may still exceed the taxable wage base, in which case reductions in quarterly wages would not translate into lower taxable wages.

year-of-establishment-by-time fixed effects to control for differential trends across firms with different number of years entering the calculation of their reserve ratios; (iii) replacing the twenty benefit-ratio classes with 330 bins created by rounding the benefit ratio to the third decimal place, to capture variation in layoff behavior within broad classes; (iv) restricting the sample to firms with a benefit ratio in 2011 equal to zero, to check the results for the 47% of the sample with no benefit charges pre-reform; and (v) removing manufacturing firms, repeatedly hit across recessions.

Figure E7 compares the true average estimated effects, $\overline{\beta_{y,post}}$ with the distribution of average effects from a permutation exercise. I randomly assign each firm a predicted reserve ratio in 2011 drawn from a uniform distribution between −1.5 and 1.5, effectively generating random predicted tax rate changes among those available. The true employment effect is much larger in absolute value than any of the one hundred placebo estimates, which are an order of magnitude smaller and centered around zero. The estimated effect on average wages is also larger than 97% of the placebo effects. The finding indicates that the estimated impacts are not the result of a lucky draw.

Finally, Table E2 shows consistent results from a specification where I regress the change in outcomes between the pre-reform (2007–2010) and post-reform (2011–2014) periods on an indicator for having a non-zero predicted change in the unemployment tax per worker, controlling for benefit-ratio class fixed effects. This specification identifies ATTs if the β_y coefficients from Equation (27) are biased by treatment heterogeneity across units receiving different "treatment doses" (Callaway et al. 2024). To avoid pooling firms with positive and negative treatment, I take the negative of all outcomes for positive-dose units, and compare untreated units with negative-treatment ones. Because no employer has a predicted change exactly equal to zero, I define untreated units as those with an absolute predicted change below 10 (2.36% of the sample) or below 100 (25% of the sample).

Overall, these tests confirm the stability of the main findings across alternative specifications and under stricter requirements on the identifying variation.

3.3 Reduced Form Effects of Unemployment Taxes in Colorado

Data and Sample The data provided by the Colorado Department of Labor and Employment (CO DLE) cover the universe of 366,125 firms operating in the state between 2013 and 2022. For each firm the dataset includes monthly employment; quarterly total wages, taxable wages, unemployment taxes, and benefit charges; annual unemployment tax rates and reserve ratios; the NAICS six-digit sector code and date of estalishment. I restrict the sample to 69,776 private-sector experience-rated firms observed continuously between 2014 and 2021, for eight years around the 2018 unemployment tax reform used for identification. In 2017, firms in the study sample had on average 21 employees, 15 years of activity, paid average annual wages of $66,000, and operated in services (54%), trade and transport (17%), finance, insurance, and real estate (12%), construction (11%), manufacturing (4%), and the primary sector (2%). Consistent with the exclusion of new (non-experience rated) firms and firms going out of business during the study period, the sample is positively selected: included firms are larger, pay higher wages, have been established for longer, and have cumulated fewer benefit charges per worker than excluded firms (Table D4). Appendix D.2 provides details on data cleaning, the construction of key variables used in the analysis, and sample restrictions.

Identification Strategy Identification relies on quasi-experimental variation in firms' unemployment taxes per worker generated by the removal in 2018 of a surcharge that had previously increased unemployment tax rates. The surcharge was introduced in 2013 to raise unemployment tax revenue after the state became insolvent in 2009, when trust-fund reserves were insufficient to cover the increase in benefit claims during the Great Recession (see panel [d] of Figure C.2). To pay benefits, Colorado secured a federal loan in 2010 and repaid it in 2012 through the issuance of \$630 million in bonds. The surcharge, denoted by s, was implemented in 2013 to repay the bond principal and increased all the rates in the schedule from τ to $\tau(1+s)$.[26] The bond principal was repaid in May 2017 and the surcharge was eliminated in 2018.

The removal of the surcharge had different effects on firms depending on their original tax rates. I show this heterogeneity in Panel (a) of Figure V, displaying the unemployment tax rate schedules in place between 2014 and 2021. The schedules remained virtually unchanged until 2017: the base schedule stayed the same, and the only year-to-year variation came from small changes in the surcharge. In 2018, the base schedule remained unchanged, but the surcharge of 23.94% was eliminated. As a result, the schedule shifted downward by the same proportion for all the rates, implying larger absolute reductions at higher rates. The minimum tax rate declined by 0.0015 percentage points; the maximum by 0.0195. Panel (b) presents the corresponding changes in unemployment taxes per worker (product of tax rate and taxable wage base), which in 2018 declined by between \$18 at the lowest rate and \$245 at the highest rate.

I leverage the heterogeneous impact of the surcharge removal to construct my identification strategy, illustrated in panel (c) of Figure V. For each firm, I define the *predicted* unemployment tax rate in 2018 as the rate that the firm would have been assigned if their reserve ratio remained unchanged. In the illustrative example, an employer with a reserve ratio of 0.5 and a tax rate of 10.1% in 2017 has a predicted rate of 8.15% in 2018: this is the rate corresponding to a reserve ratio of 0.5 under the 2018 schedule without surcharge. Formally, predicted rates for 2018 can be calculated as:

$$\text{Predicted } \tau_{2018} = \frac{\tau_{2017}}{1+s} \tag{29}$$

Next, I define the predicted change in the unemployment tax rate as the difference between the predicted rate in 2018 and the actual rate in 2017. Graphically, it corresponds to the vertical distance

[26]Table F1 reports the surcharge applied in each year. Moreover, between 2012 and 2020, Colorado gradually raised the taxable wage base from \$10,000 to \$13,600 (Figure F1); in 2013, it replaced its original twelve tax rate schedules (with rates from 0 to 5.4%) with seven new schedules with higher rates (0.51–10.39%). These changes do not affect identification, as they either predate the study period or apply uniformly across firms.

between the two schedules evaluated at the 2017 reserve ratio (the red arrow in the figure). Formally,

$$\text{Predicted } \Delta\tau_{2018} = \frac{\tau_{2017}}{1+s} - \tau_{2017} = -\tau_{2017}\frac{s}{1+s} \tag{30}$$

The *actual* change in the unemployment tax rate combines two components (the two arrows in panel [c]): the vertical shift caused by the removal of the surcharge, the *predicted* rate change, and the change due to employers' updated reserve ratios between 2017 and 2018 (the blue arrow), a change that would have occurred even had the surcharge not been removed (although along the 2017 schedule rather than the 2018 schedule). The part of the change due to the surcharge removal is always negative; the part due to the updated reserve ratio is weakly negative for employers whose reserve ratios decreased between 2017 and 2018 (their tax rates would have declined even without the surcharge removal), and weakly positive for employers whose reserve ratios increased (their tax rates would have risen in the absence of the surcharge removal and instead either fall or rise by a smaller amount because of its removal). Formally:

$$\Delta\tau_{2018-2017} = \underbrace{\left[\tau_{2018} - \frac{\tau_{2017}}{1+s}\right]}_{\text{Updated Reserve Ratio (+/-)}} + \underbrace{\left[\frac{\tau_{2017}}{1+s} - \tau_{2017}\right]}_{\text{Surcharge Removal = Pred. } \Delta\tau_{2018} \text{ (-)}} \tag{31}$$

Given the strong persistence of reserve ratios over time (94% correlation between any two years in the data), the predicted rate change is expected to closely track the actual change, while remaining clean from changes due to firms' layoff histories and reserve ratios evolving at the time of the reform.

Finally, I multiply the predicted rate change by the 2018 taxable wage base ($12,600) to obtain the predicted change in unemployment taxes per worker, which I use as a continuous firm-level treatment. Panel (d) of Figure V shows its distribution. While the treatment is small for most firms, 10% are predicted to experience tax reductions exceeding $100 per worker, with effects as large as $245. This heterogeneity is driven by the underlying distribution of reserve ratios in 2017, shown in panel (e): 90% of firms have negative reserve ratios and therefore face low unemployment tax rates, resulting in limited predicted tax changes following the removal of the surcharge.

Equation (32) shows my preferred specification. $Y_{i,t}$ is the outcome of interest for firm i at time t, which is quarter or year depending on the outcome; α_i are firm fixed effects; Pred Δ UTPW$_i$ is the firm's predicted unemployment tax per worker in 2018; and $\epsilon_{i,t}$ is the error term. Standard errors are clustered at the firm level. β_{2017} or β_{2017q4} normalized to zero. I winsorize outcomes at the 99^{th} or 99.9^{th} percentiles to remove the influence of outliers and errors, which would otherwise introduce substantial noise in the estimates. Appendix D.2 describes cleaning and winsorization details.

$$Y_{it} = \alpha_i + \sum_{y=2014}^{y=2021} \beta_y 1_{y=t} \text{Pred } \Delta \text{ UTPW} + \epsilon_{i,t} \tag{32}$$

The β coefficients identify ATTs if parallel trends and no-anticipation hold. I assess parallel trends by direct inspection of firms' outcomes pre-reform. In addition, I show that, after excluding firms with substantial layoff activity in the pre-period, treatment intensity is plausibly as good as random. The treatment, equal to the vertical shift between tax schedules, depends only on firms' positions on the 2017 schedule. Firms with larger treatment doses have higher reserve ratios and, as shown in panel (f) of Figure V, also higher pre-reform benefit charges. Excluding employers in the top 5% of the 2018 benefit-ratio distribution, with the largest benefit charges in the pre-period, yields a sample in which benefit charges are essentially flat across the reserve-ratio distribution. In this restricted sample, treatment intensity is uncorrelated with recent layoff behavior, implying that firms' positions along the schedule, and hence their treatment exposure, reflect only distant past behavior and can be considered as good as random. Accordingly, I estimate Equation (32) only for the bottom 95% of the 2018 benefit-ratio distribution.

I then discuss no-anticipation. Although the bond was repaid in May 2017, firms were notified of their 2018 unemployment tax rates only in December 2017, leaving little room for pre-reform adjustments. Identification is not threatened if firms anticipated tax cuts but such expectations did not vary by treatment status, or if their expectations were primarily driven by the endogenous component of the tax change, such as anticipating rate declines due to declining reserve ratios, which should average out for each treatment dose.

Results Figure VI presents the estimated β_y coefficients from Equation (32). The sample is restricted to firms in the bottom 95% of the benefit ratio distribution in 2018, in support of exogeneity, and with fewer than twelve employees in 2017, the 75^{th} percentile of the distribution, to limit noise from outliers. Results with large employers are presented among the robustness tests. Table F2 shows year-level estimates. Table D5 describes the construction of the outcomes.

Panel (a) shows the first stage: the effect of the predicted increase in the unemployment tax per worker on the actual increase. The tax is flat pre-reform but it rises sharply in 2018 and remains persistently high. The average increase between 2018 and 2021 is \$2.3. Given the average tax per worker pre-reform of \$254, the estimates imply an average increase in the tax per worker of \$584 or 230% of the pre-reform mean.

Panel (b) shows the corresponding changes in quarterly employment. Employment is stable prior to the reform, but begins a gradual decline in the first quarter of 2018. The average post-reform estimate is of 0.003, implying an average employment decline of 0.762 workers (0.003×254), or 3.75% of the pre-reform mean (0.003×254).

Panel (c) reports the coefficients for total wages, which are flat pre-reform and decline gradually from 2018. The average post-reform estimate is \$431, implying an average reduction in total wages of \$109,357 (431×254) or 9%.

Panel (d) presents the coefficients for average wages. The pre-reform coefficients are jointly insignificant, though slightly downward-sloping. In 2018, average wages decline sharply, by an average of \$13.7. This estimate translates into an average causal effect of \$3,478 (13.7×254), or 5.8%.

Panel (e) shows that taxable wages were flat pre-reform and declined by an average of \$60.71. This estimate translates into an average effect of \$15,420 (60.71×254) or 5.5%. The reduction in taxable wages provides indirect evidence that the employment decline is driven by reduced hiring rather than increased layoffs, which is crucial for interpreting the reduced-form estimates as labor demand elasticities. If the decline had been driven by layoffs, taxable wages would not have fallen, as the wages of workers laid off during the year are still taxable at the time of hire. Moreover, the magnitude of the decline clarifies what drives the reduction in the average wage in panel (c). If the missing workers in panel (b) earned above-average wages, then the total decline in taxable wages would be equal to the product of the average employment decline, 0.762, and the average post-reform taxable wage base, \$13,225, which gives \$10,077. Figure C.3 discusses that the larger observed reduction in taxable wages reflects a decline in the wages of retained workers below the taxable wage base.

Finally, panel (f) shows the coefficients for unemployment taxes, which increase by \$8.6 on average after the reform. This estimate implies an average annual increase of \$2,184 per year, or 35% of the pre-reform mean. I compare this observed increase with the change in simulated unemployment taxes, calculated assuming that firms kept their 2017 taxable wages. In the absence of behavioral responses, the government would have collected \$2 more, corresponding to \$508 per firm-year.

Robustness I test the stability of my findings in several ways. Figure F2 reports estimates from a specification that includes large employers. Because their inclusion introduces substantial variability, all outcome variables are scaled by their 2017 levels. The results remain consistent with the baseline, with the exception of the effect on the average wage: although the coefficient is negative in 2020, the other coefficients are smaller and statistically insignificant in the other years and on average.

Figure F3 shows that the results are robust to several modifications of the baseline specification: (i) adding sector-by-time fixed effects to account for heterogeneous industry trends; (ii) adding year-of-establishment-by-time fixed effects to control for differential trends across firms with different number of years entering the calculation of their reserve ratios; (iii) including benefit-ratio-by-time fixed effects instead of excluding employers in the top 5% of the 2018 benefit-ratio distribution; and (iv) excluding firms with a 2017 reserve ratio between –0.2 and 0.2, that in panel (f) of Figure V show positive pre-period benefit charges that rise and then decline over this interval.

Figure F4 compares the true effects with the distribution of average effects on employment and average wages obtained from a permutation exercise. In each of one hundred iterations, treatment is calculated using a randomly drawn 2017 tax rate from the set of available rates. The simulated average effects cluster tightly around zero and are one order of magnitude smaller than the true effects. The finding indicates that the estimated impacts are not the result of a lucky draw.

Finally, figure F5 presents results from a specification where I regress the change in outcomes between the pre-reform (2014–2017) and post-reform (2018–2021) periods on a set of indicators for predicted tax changes in 2018. With discrete multivalued treatment, each indicator identifies a separate ATT for a given treatment dose if the β_y coefficients in Equation (32) are biased by treatment-effect heterogeneity across doses (Callaway et al. 2024). Because no firms have a predicted change in the tax per worker equal to zero, I normalize to zero the coefficient for the smallest treatment dose ($\beta_{d=-19}$). All other coefficients are therefore interpreted relative to this baseline. Consistent with the main results, larger predicted reductions in the tax per worker are associated with progressively larger declines in the actual tax per worker (ranging between 0 and -\$600) and with corresponding increases in employment (0-2) and average wages (0-\$10,000).

Overall, these tests confirm the stability of the main findings across alternative specifications and under stricter requirements on the identifying variation.

3.4 Incidence and Implied Elasticities

Incidence For each state, I calculate the share of the incidence of unemployment taxes falling on firms in Equation (33) as the fraction of the average decline in total wages, $\overline{\beta_{l \cdot w}}$, that can be attributed to the average decline in employment, $\overline{\beta_l}$, under alternative assumptions about the wages of missing employees, w^M.

$$\text{Firms' Incidence Share } |_{w^M} = \frac{\overline{\beta_l} \cdot w^M}{\overline{\beta_{l \cdot w}}} \tag{33}$$

Equation (34) provides an upper bound on firms' incidence share. The gap between the total decline in taxable wages, $\overline{\beta_{tw}}$, and the portion explained by employment losses alone, $\overline{\beta_l} \cdot \overline{TWB_{Post}}$, measures the minimum reduction in total wages due to lower wages among retained workers. The residual reduction in taxable wages determines the maximum share attributable to firms.

$$\text{Firms' Maximum Incidence Share} = 1 - \frac{\overline{\beta_{tw}} - \overline{\beta_l} \cdot \overline{TWB_{Post}}}{\overline{\beta_{l \cdot w}}} \tag{34}$$

The lower bound on firms' incidence share is zero, occurring when missing workers have zero wages, $w^M = 0$. Importantly, these calculations do not account for potential responses in hours worked, profits, or consumer prices.

Table I reports the results. Panel A shows the average reduced-form effects, Panel B the sample averages, and Panel C the implied incidence shares.

For South Carolina, total wages decline by $\overline{\beta_{l \cdot w}} = -58$. By Equation (33), if $\overline{\beta_l} = 0.001$ fewer workers earned the pre-reform average wage of $\overline{w_{pre}} = 37{,}732$, the employment decline would account for 64.7% of the reduction in total wages. The average reduction in taxable wages is $\overline{\beta_{tw}} = -26$, of which the employment decline explains only $\overline{\beta_l} \cdot \overline{TWB_{post}} = 11.5$. By Equation (34), the difference of 14.5 represents the minimum decline in total wages attributable to changes in workers' average wages, implying a maximum incidence share on firms of 74.7%.

For Colorado, total wages decline by $\overline{\beta_{l \cdot w}} = -430.543$. By Equation (33), if $\overline{\beta_l} = 0.003$ fewer workers earned the pre-reform average wage of $\overline{w_{pre}} = 59{,}916$, the employment decline would account for 41.7% of the reduction in total wages. The average reduction in taxable wages is $\overline{\beta_{tw}} = -60.71$, of which the employment decline explains only $\overline{\beta_l} \cdot \overline{TWB_{post}} = 39.68$. By Equation (34), the difference of 21.03 represents the minimum decline in total wages attributable to changes in workers' average wages, implying a maximum incidence share on firms of 95.1%.

These results align with a growing empirical literature showing that payroll tax incidence falls partly on firms, despite the theoretical prediction that firms can fully shift such taxes onto workers' wages. Evidence of substantial employment responses comes from studies of payroll tax changes over time and across locations (Saez et al. 2012; Ku et al. 2020; Benzarti et al. 2021; Benzarti et al. 2021), hiring subsidies targeted to specific worker groups (Cahuc et al. 2019; Saez et al. 2019), and unemployment tax changes (Behaghel et al. 2008; Johnston 2021; Guo 2024). This pattern has been characterized as a recurring anomaly in the literature (Benzarti 2025). One proposed mechanism is fairness norms, which can prevent payroll tax cuts targeted to specific groups from generating wage differentials relative to equally productive workers (Saez et al. 2012). In the context of unemployment insurance taxes that vary across firms, earlier work argues that firms facing unusually high tax rates cannot fully shift these costs onto wages or consumer prices in competitive labor and output markets, leading to employment adjustments (Lester 1965; Hamermesh 1977; Anderson et al. 1997). Across these studies, the share of unemployment tax incidence borne by firms ranges from 50% to 100%. While this range is wide, it is consistent with the estimates reported in Table I.

Implied Elasticities For each state, I use the reduced-form estimates to calculate the three elasticities in Equation (35): (i) the elasticity of labor demand with respect to the unemployment tax per worker, which is the sufficient statistic for labor misallocation in the formula for the optimal degree of experience rating (Equation 21), $\varepsilon_{l,\tau}$; (ii) the implied elasticity of labor demand with respect to labor costs, to compare my estimates with the literature, $\varepsilon_{l,(w+\tau)}$; and (iii) the elasticity of average wages with respect to the unemployment tax per worker, which tests whether the labor market clears as predicted by the model in Figure A.1, $\varepsilon_{w,\tau}$.

$$\varepsilon_{l,\tau} = \frac{\overline{\beta_l}}{\overline{\beta_\tau}} \cdot \frac{\overline{\tau_{Pre}}}{\overline{l_{Pre}}} \qquad \varepsilon_{l,(\tau+w)} = \frac{\overline{\beta_l}}{\overline{\beta_\tau}} \cdot \frac{\overline{\tau_{Pre}} + \overline{w_{Pre}}}{\overline{l_{Pre}}} \qquad \varepsilon_{w,\tau} = \frac{\overline{\beta_w}}{\overline{\beta_\tau}} \cdot \frac{\overline{\tau_{Pre}}}{\overline{w_{Pre}}} \tag{35}$$

Table I shows the results. Panel A reports the average reduced-form effects, Panel B the full-sample pre-reform averages, and Panel C the implied elasticities.

For South Carolina, the reduced-form effects on the tax per worker, employment, and the average wage are $\overline{\beta_\tau} = 0.357$, $\overline{\beta_l} = -0.001$, and $\overline{\beta_w} = -1.9$. I scale these estimates by the pre-reform full-sample averages in the SC DEW data: $\overline{\tau_{pre}} = 134.9$, $\overline{l_{pre}} = 11.714$, and $\overline{w_{pre}} = 37{,}732$. The resulting labor demand elasticity with respect to the unemployment tax per worker is $\varepsilon_{l,\tau} = -0.032$, which implies a labor demand elasticity with respect to labor costs of $\varepsilon_{l,(\tau+w)} = -9.1$. The wage elasticity with respect to the tax per worker is $\varepsilon_{w,\tau} = -0.019$.

For Colorado, the reduced-form effects on employment, the average wage, and the tax per worker are $\overline{\beta_l} = -0.003$, $\overline{\beta_w} = -11.5$, and $\overline{\beta_\tau} = 2.039$. I scale these estimates by the pre-reform full-sample averages in the CO DLE data: $\overline{l_{pre}} = 20.3$, $\overline{w_{pre}} = 59{,}916$, and $\overline{\tau_{pre}} = 254$. The resulting labor demand elasticity with respect to the unemployment tax per worker is $\varepsilon_{l,\tau} = -0.016$, which implies a labor demand elasticity with respect to labor costs of $\varepsilon_{l,(\tau+w)} = -3.9$. The wage elasticity with respect to the tax per worker is $\varepsilon_{w,\tau} = -0.025$.

For both states, the elasticities are consistent with the model predictions on labor market clearing shown in Figure A.1: when the unemployment tax per worker rises, both labor demand and the average wages of affected firms decline. Interpreted through the model and Figure I, these results suggest that as labor costs increase, firms facing higher unemployment taxes stop hiring earlier along the worker productivity distribution. This shift leads them to employ higher-productivity workers who, due to skill-based sorting across industries, are also willing to accept lower wages.

The magnitudes of the labor demand elasticities with respect to labor costs, -9.1 for South Carolina and -3.9 for Colorado, exceed conventional estimates of $[-1, 0]$ (Lichter et al. 2015, Johnston 2021), but are in line with the unemployment tax literature: Anderson et al. (1997) report a value of -2.3, Guo (2024) estimate a lower bound of -2.24, and Johnston (2021) finds an elasticity of -4, rising to -8 and -12 in alternative specifications.[27] More broadly, labor demand has become more elastic over time (Lichter et al. 2015), and large elasticities have also been documented for standard payroll taxes: -3.6 in Ku et al. (2020); $[-2.9, -4.16]$ in Benzarti et al. (2021).

Johnston (2021) proposes three explanations for the large employment responses to experience-rated unemployment taxes, which I test in my setting. First, when taxes are heterogeneous across firms, employers cannot coordinate to shift the tax onto wages or prices in competitive labor and output markets, resulting in high incidence on firms. To test this hypothesis, I examine whether employment responses in the two states are larger in industries with smaller variability in average wages. Results are shown in panels (a) of Figures E8 and F6. The observed lack of heterogeneity suggests that this channel does not play a major role in driving the large labor demand elasticities.

Second, cash-constrained firms that need to maintain positive profits may disproportionately reduce employment when unemployment taxes rise and wages are rigid in the short run (Schoefer 2021). When the unemployment tax increases, reducing employment lowers costs faster than revenues, as marginal productivity rises while marginal costs decline through both lower wage payments and lower

[27] The author notes in footnote 24 that an employment response of -0.013 percentage points corresponds to a 1.2% change; dividing 1.2% by the 0.3% change in the wage bill yields an elasticity of 4. Larger effects of -0.027 percentage points reported in Table 3 (2.5%) and -0.039 in Online Appendix Figure 8 (3.6%) imply elasticities of -8.3 and -12.

unemployment taxes due to reduced benefit expenditures, as illustrated in Figure I and Equation (19). Inspired by Johnston (2021), Appendix G presents a simple application of my model showing that tax increases that push profits below zero generate larger employment declines, with elasticities up to twice as large than in the absence of cash constraints. Two pieces of evidence support this mechanism. First, the labor demand elasticity is larger in South Carolina in the aftermath of the Great Recession (2010) than in Colorado during a relatively stable economic period (2017). Consistently, Johnston (2021) (Online Appendix Figure A8) reports estimates consistent with elasticities between -8 and -12 in post-recession years (2010–2012), which are nearly four times larger than those estimated in earlier years. Second, panels (b) and (c) of Figure F6 show that employment responses in the CO DLE sample are larger for small and young firms, which are typically considered more cash-constrained (Saez et al. 2019). This heterogeneity does not emerge in the SC DEW sample (panels (c) and (d) of Figure E8), potentially because during recessions even larger and older firms face cash constraints.

Third, unemployment insurance in the United States effectively operates as a head tax, since earnings above the taxable wage base are exempt from taxation. As a result, increases in unemployment taxes raise the cost of employment rather than payroll, amplifying hiring responses. Johnston (2021) compares revenue-equivalent payroll and head taxes and show that, when it is costly for firms to increase hours of work, labor demand elasticities are four times larger under a head tax. Consistent with this mechanism, employment responses in Colorado are concentrated among firms with below-median average wages, which face a higher tax burden relative to payroll (panel [d] of Figure F6). This pattern does not emerge in South Carolina (panel [d] of Figure E8).

To evaluate the quantitative importance of cash constraints and head taxes, suppose that all firms were cash-constrained in South Carolina, while 81% of firms were cash-constrained in Colorado, corresponding to the share of firms that are either small or young. Under this assumption, the implied elasticities absent cash constraints would be $-9.1/2 = -4.55$ in South Carolina and $-3.9/(0.19 + 2 \times 0.81) = -2.15$ in Colorado. Accounting additionally for the head-tax nature of the tax and dividing these elasticities by four yields $-4.55/4 = -1.14$ in South Carolina and $-2.15/4 = -0.54$ in Colorado, which fall within the conventional elasticity range.

4 Calibration

I calibrate the remaining parameters in the formula for the optimal degree of experience rating, Equation (21), using data moments and estimates from the literature. Table II summarizes the calibration strategy and reports the corresponding parameter values for South Carolina and Colorado in the years surrounding the unemployment tax reforms used for identification: 2010–2011 for South Carolina and 2017–2018 for Colorado. Appendix H details the procedure and data sources.

Degree of experience rating, e It is calibrated using two alternative approaches, corresponding to its two interpretations within the model. First, e measures the share of a firm's own benefit expenditures that the firm repays in unemployment taxes. Following Van Doornik et al. (2022), I calibrate e as the fraction of non-socialized benefit charges in a given state-year from the ETA 204 Report, that is, charges that are not repaid by the firms who generated them: benefits not assigned to any firm, benefits assigned to firms that have gone out of business, and benefits assigned to firms already at the maximum tax rate. This approach yields e_{share} equal to 0.343 for South Carolina in 2010, 0.527 in 2011, 0.735 for Colorado in 2017, and 0.725 in 2018, indicating that 34.3-52.7% and 73.5-72.5% of benefit charges were repaid by the firms who generated them. The exceptionally high socialized costs after the Great Recession explain the unusually low degree of experience rating in South Carolina in 2010 relative to both 2011 and Colorado (Figure H1).

Second, e measures the marginal tax cost of a dollar of benefit charges. Following Pavosevich (2020) and Duggan et al. (2023), I calibrate it as the average increase in the tax bill for a firm of average size. A one-dollar benefit charge induces a change in the firm's experience factor, which in turn generates a change in its tax rate. I calculate the change in the experience factor using the formula specific to each state and approximate the change in the tax rate with the average slope of the unemployment tax rate schedule in each state-year calculated from the Unemployment Insurance Financing Policies Database (Figure H2). I then convert the resulting change in the tax rate into a change in total tax liability by multiplying the rate change by the taxable wage base and average firm size. This approach yields e_{mtc} equal to 0.154 in South Carolina in 2010, 0.249 in 2011, 0.381 in Colorado in 2017, and 0.304 in 2018, implying that employers could expect taxes to increase by 15.4–24.9 cents and 38.1–30.4 cents, respectively, for each dollar of benefit charges. The increase in South Carolina reflects the tenfold increase in the average slope of the schedule after the introduction of the benefit ratio. The difference between South Carolina pre-reform and Colorado reflects Colorado's steeper schedule, due to a higher maximum tax rate and shorter flat segments (Figure H2).

Appendix H.1 explains the calibration in detail. Across states and consistent with prior studies, the degree of experience rating is systematically higher when measured as the share of benefits repaid than as the marginal tax cost of benefits.

Unemployment Benefit, b It is defined as the product of the average weekly benefit amount and potential duration in weeks in the ETA 394 Handbook. I obtain \$5,201 for South Carolina in 2010, \$4,662 in 2011, \$10,066 for Colorado in 2017, and \$10,285 in 2018.

Labor Market I classify industries, identified by their six-digit NAICS codes, into low- and high-unemployment risk based on the average within-year standard deviation of quarterly employment between 1990 and 2020 in the QCEW. Most industries have stable employment within the year, but a subset displays substantial seasonal fluctuations (Figure H3). To capture these pronounced variations, I define high-risk industries as those in the top 5% of the standard deviation distribution within each state. Examples include ski facilities, restaurants, hotels, tax preparation services, landscaping services, and highway, street, and bridge construction. Appendix H.2 provides additional details on this classification and the calibration of industries' characteristics.

I calibrate l_H as the fraction of state employment in high-risk industries in the QCEW, obtaining 40.4% for South Carolina in 2010, 40.3% in 2011, 32.7% for Colorado in 2017, and 32.6% in 2018. Next, I calibrate the layoff rate in high-risk industries, r_H, as the excess percentage difference between the minimum and maximum quarterly employment within the year for high- versus low-unemployment-risk industries in the QCEW. This yields 4.4% for South Carolina in 2010, 11.3% in 2011, 12.4% for Colorado in 2017, and 8.6% in 2018. Then, I calibrate the marginal productivities in the two industries, $f_H(l_H)$ from Equation (19) and $f_L(l_H)$ from Equation (12), using industries' average annual wages from the QCEW. Productivity is consistently higher in low-risk industries relative to high-risk ones, by ∼\$4,500 per worker per year in South Carolina and ∼\$8,000 in Colorado. Finally, the output loss from moral hazard, $\int_0^{l_H} f_H(i)\,di$, is calculated as the area of the green and gray parallelograms in Figure I, given by the product of the base, the employment share in high-risk industries, l_H, and the height, the increased potential output loss between the first- and second-best $(r_H - r_H^{FB})f_H(l_H)$. The first-best layoff rate, r_H^{FB}, is calibrated as the lowest nonzero annual layoff rate in the QCEW. I obtain a loss of ∼515 in South Carolina in 2010, ∼1,600 in 2011, ∼770 in Colorado in 2017 and ∼210 in 2018. Changes in layoff rates over time explain these differences.

Excess Output Losses Since the scaling factors measuring output losses, λ_o and ρ_o in Equations (22) and (23), do not converge to zero as the degree of experience rating approaches one, I remove the output losses that would arise even under complete experience rating by subtracting the values obtained at $e = 1$ from the calibrated values.

Elasticity of Labor Demand with Respect to Experience Rating, $\varepsilon_{l_H,e}$ By Equation (25), it is equal to the elasticity of labor demand with respect to the unemployment tax per worker, $\varepsilon_{l_H,\tau}$. In Section 3, I obtained -0.032 for South Carolina and -0.016 for Colorado.

Elasticity of the Layoff Rate with Respect to Experience Rating, $\varepsilon_{r_H,e}$ It is calibrated with the elasticity of layoffs with respect to experience rating from Card et al. (1994), equal to -0.1.

Insurance Value By Equation (21), it is equal to the degree of experience rating, e_{share} or e_{mtc}, in a state-year: 0.343 and 0.154 for South Carolina in 2010, 0.527 and 0.249 in 2011, 0.735 and 0.381 in Colorado in 2017, and 0.725 and 0.304 in 2018.

5 Optimal Unemployment Insurance Financing Policy

I plug the parameters calibrated in Table II into the formula for the optimal degree of experience rating (Equation 21) to calculate the marginal benefit and marginal cost of incomplete experience rating in South Carolina and Colorado and evaluate the optimality of the states' financing policies. The implied policy recommendation is to increase experience rating when the marginal cost exceeds the marginal benefit, and to reduce it otherwise. For each state–year, I evaluate the optimality of the two measures of experience rating from the model, the share of own benefits repaid in taxes, e_{share}, and the marginal tax cost, e_{mtc}, both before and after the reforms used to estimate the sufficient statistic for labor misallocation. While this comparison does not directly establish the optimality of these non-marginal reforms, changes in the policy recommendation around each reform provide suggestive evidence on whether experience rating moved toward the optimum.

I calculate bounds for the marginal benefit and cost. The bounds for the marginal benefit rely on the alternative interpretation of the insurance value as the gap in the marginal utility of profits between the good and bad states. The lower bound is 0 when firms are risk-neutral, while the upper bound is the highest estimate of workers' insurance value, 3.13 from Landais et al. (2021), assuming that firms are less risk averse than workers. The bounds for the marginal cost are based on extreme elasticity estimates. Table H3 summarizes labor demand elasticities from unemployment tax studies. Guo (2024) reports a lower-bound elasticity of -2.24, which I convert to an elasticity with respect to the tax per worker (Equation 66) and use as lower bound. The estimate of –0.032 from South Carolina serves as the upper bound. Table H4 reports layoff elasticities; I use 0 from Johnston (2021) as lower bound and -0.43 from Card et al. (1994) as upper bound.

Marginal benefits and costs by state, year, and experience rating method are reported in Table III and Figure VII. The table also presents the implied policy recommendations under the main calibration and twelve alternative scenarios: considering only fiscal externalities to evaluate the role of output losses, considering only moral hazard to evaluate the role of labor misallocation, and comparing upper and lower bounds, to test the robustness of the main recommendation to firms' responsiveness and risk preferences.

Figure VIII illustrates how the gap between marginal benefit and cost varies with the degree of experience rating for each state-year. The degree at which this gap equals zero identifies the optimal level of experience rating, which is then compared to the prevailing degree under both measurement approaches. Importantly, the figure assumes that all other variables in the formula remain constant as experience rating changes, even though they may themselves be endogenous to the degree of experience rating in place.

South Carolina In 2010, the share of own benefit charges repaid by firms in unemployment taxes was $e_{share} = 34.3\%$. Panel (a) of Figure VII and column (1) of Table III show that the marginal benefit of reducing experience rating is 0.343, while the marginal cost is 22.42. These values imply that a one-dollar reduction in unemployment taxes on high-risk industries generates a welfare gain of 34.3 cents and a welfare loss of \$22.42. The fiscal externality from moral hazard accounts for \$2.20, the output loss from moral hazard for \$0.54, the fiscal externality from labor misallocation for \$0.70, and the output loss from labor misallocation for \$18.98. Since the marginal cost exceeds the benefit, increasing the degree of experience rating would have increased welfare. This conclusion holds in 83% of the alternative scenarios, and reverses only when firms' maximum risk aversion is compared with either fiscal externalities only or moral hazard only.

In 2011, South Carolina introduced a steeper tax rate schedule. Consistently, the share of own benefits repaid in taxes increased to $e_{share} = 52.7\%$. Panel (a) of Figure VII and column (2) of Table III show that the marginal benefit of reducing experience rating is 0.527, while the marginal cost is 2.78: 40 cents due to the fiscal externality from moral hazard, 34 cents to the output loss from moral hazard, 13 cents to the fiscal externality from labor misallocation, and \$1.92 to the output loss from labor misallocation. The gap between the marginal benefit and cost declines substantially, but the policy recommendation remains to increase the degree of experience rating. This conclusion is robust in 50% of the scenarios and reverses when firms' maximum risk aversion is compared with the calibrated marginal cost, fiscal externalities only, moral hazard only, and the lower bound on the marginal cost; the degree of experience rating is optimal when the calibrated insurance value is compared with fiscal externalities only and with the lower bound on the marginal cost.

Stronger conclusions emerge from evaluating South Carolina's marginal tax cost of benefit charges, equal to $e_{mtc} = 15.4\%$ in 2010. Panel (b) of Figure VII and column (3) of Table III show that the marginal benefit of reducing experience rating is 0.154 and the marginal cost is 64.14: \$6.29 due to the fiscal externality from moral hazard, \$1.54 to the output loss from moral hazard, \$2.01 to the fiscal externality from labor misallocation, and \$54.30 to the output loss from labor misallocation. Since the fiscal externality from moral hazard alone is larger than the upper bound on insurance value, increasing experience rating is optimal across all scenarios. In 2011, the marginal tax cost increased to $e_{mtc} = 29.4\%$. Panel (b) of Figure VII and column (4) of Table III show that the marginal benefit of reducing experience rating increased to 0.249 and the marginal cost decreased to 9.33. The policy recommendation remains to increase experience rating. The conclusion holds in 75% of the alternative scenarios and only reverses when firms' maximum risk-aversion is compared with either fiscal externalities, or moral hazard, or the marginal cost's lower bound.

Figure VIII identifies the optimal degree of experience rating as the level at which the marginal benefit equals the marginal cost. In 2010, the optimal degree is $e^* = 92.7\%$, substantially higher than both the prevailing share of benefits repaid through taxes and the marginal tax cost. In 2011, the optimal degree declines to $e^* = 79.6\%$, which remains above, but is closer to, the prevailing levels.

Colorado In 2017, the share of own benefit charges repaid by firms in unemployment taxes was $e_{share} = 73.5\%$. Panel (c) of Figure VII and column (5) of Table III show that the marginal benefit of reducing experience rating is 0.735 and the marginal cost is 0.73: 13 cents due to the fiscal externality from moral hazard, 3 cents to the output loss from moral hazard, 4 cents to the fiscal externality from labor misallocation, and 53 cents to the output loss from labor misallocation. Because the marginal benefit equals the marginal cost, the degree of experience rating was optimal. The policy recommendation becomes to increase the degree of experience rating in 33% of cases, those based on risk-neutrality, and to reduce the degree of experience rating in the remaining 67% of cases in which either firm's maximum risk aversion is employed, or the marginal cost is reduced to fiscal externalities, moral hazard, or to the lower bound.

In 2018, Colorado reduced unemployment tax rates by removing a surcharge. Consistently, the share of own benefits repaid in taxes decreased to $e_{share} = 72.5\%$. Panel (c) of Figure VII and column (6) of Table III show that the marginal benefit declines to 0.725, while the marginal cost rises to 1.23. The policy recommendation is to increase experience rating, and is robust in 33% of the scenarios, the ones assuming firm risk neutrality, and reversing when either firm's maximum risk aversion is employed, or the marginal cost is reduced to fiscal externalities, moral hazard, or to the lower bound.

Stronger conclusions emerge when evaluating Colorado's marginal tax cost of a dollar of benefit charges, equal to $e_{mtc} = 38.1\%$ in 2017. Panel (d) of Figure VII and column (7) of Table III show that the marginal benefit is 0.381 and the marginal cost is 3.61: 66 cents reflect the fiscal externality from moral hazard, 15 cents the output loss from moral hazard, 21 cents the fiscal externality from labor misallocation, and \$2.59 the output loss from labor misallocation. The policy recommendation to increase experience rating holds in 67% of the cases, and reverses when firms' maximum risk aversion is compared with the calibrated marginal cost, fiscal externalities, moral hazard, and the lower bound on the marginal cost. Panel (d) of Figure VII and column (8) of Table III show that, when in 2018 the marginal tax cost decreased to $e_{mtc} = 30.4\%$, the marginal benefit declined to 0.304, and the marginal cost increased to 8.53, strengthening the recommendation to increase experience rating. The result holds in 75% of the cases, and only reverses when firms' maximum risk aversion is compared with either fiscal externalities, moral hazard, or the lower bound on the marginal cost.

Figure VIII identifies the optimal degree of experience rating as the level at which the marginal benefit and cost are equal. In 2017, the optimal degree is $e^* = 74.9\%$, close to the prevailing share of own benefits repaid in taxes, but higher than the prevailing marginal tax cost. In 2018, the optimal degree rises to $e^* = 82\%$, exceeding the prevailing levels and moving further from the optimum.

Two conclusions First, in both South Carolina and Colorado, labor misallocation is the primary source of inefficiency from incomplete experience rating, accounting for 74–88% of the marginal cost, despite receiving limited attention in the existing literature compared to moral hazard. The main distortion arises from output losses due to the reallocation of workers toward less productive, high–unemployment-risk industries as unemployment taxes decline.

Second, policy recommendations depend critically on the measurement of experience rating and the calibration of the marginal benefit and cost. When experience rating is measured as the marginal tax cost of benefit charges, it is consistently inefficiently low. When measured as the share of firms' own benefit charges repaid through unemployment taxes, it is inefficiently low in South Carolina but optimal in Colorado pre-reform. South Carolina increased experience rating and moved it closer to the optimum, while Colorado moved it further away. Policy recommendations, however, can reverse when ignoring labor misallocation or output losses and using lower or upper bound estimates of the marginal cost or benefit. This sensitivity underscores the importance of accounting for labor misallocation and output losses, estimating elasticities within the relevant context, and tightly bounding firms' insurance value before drawing policy conclusions. Importantly, these results ignore workers' disutility from unemployment. Appendix B.5 shows that accounting for them increases the marginal cost and the optimal degree of experience rating.

6 Conclusion

The cost of providing unemployment benefits to laid-off workers is borne largely by firms, yet little is known about how firms respond to unemployment taxes or how different tax designs affect welfare. In the United States, firms are assigned individualized unemployment tax rates based on the benefit cost of their layoffs (experience rating), while all the other countries assign uniform rates, irrespective of firms' individual contributions to unemployment (coinsurance). Each approach generates distinct distortions, but the literature lacks a comprehensive framework to compare them.

In this paper, I investigate theoretically and empirically the optimal design of unemployment insurance financing policies. I derive an employer-Baily-type sufficient statistic formula specifying that the optimal degree of experience rating balances the marginal benefit and cost of insuring firms through coinsurance. Coinsurance insures firms against sharp increases in unemployment taxes during economic shocks, when liquidity is particularly costly, but has two costs. First, it increases layoffs by reducing their cost for firms. This moral hazard generates a fiscal externality through higher benefit spending and reduces output through excessive layoffs. Second, coinsurance subsidizes the expansion of high-unemployment risk industries. This labor misallocation creates an additional fiscal externality, as more workers are exposed to unemployment risk, and leads to the misallocation of productive skills. I empirically implement the formula to evaluate the optimality of South Carolina and Colorado's degrees of experience rating. I use firm-level unemployment tax filing data and quasi-experimental variation from state-level experience-rating reforms to estimate the sufficient statistic for labor misallocation: the elasticity of labor demand with respect to the the unemployment tax per worker. I then calibrate moral hazard and insurance value using data moments and estimates from the literature.

The paper has three main findings. First, firms respond to higher unemployment taxes by reducing employment and lowering incumbent workers' wages, leading to an approximately equal division of tax incidence between firms and workers when evaluated at the average wage. The analysis also shows that alternative experience-rating systems redistribute the tax burden across firms and workers, and that higher taxes may erode the tax base, reducing revenue below predictions and limiting the scope for financing more generous benefits through tax increases, a policy option currently under debate (Wandner 2023).

Second, despite having received limited attention in the literature, labor misallocation emerges as the primary source of inefficiency from incomplete experience rating in both states. These inefficiencies are driven by sizable output losses resulting from the concentration of employment in low-productivity, high–unemployment risk industries. This finding implies that coinsurance can generate substantial welfare costs even in settings where employer moral hazard is limited, such as many European countries, where strict employment protection constrains layoffs. The findings motivate further research in these contexts, particularly as several countries introduce layoff taxes and consider experience rating within the European Unemployment Benefit Scheme to limit systematic cross-country subsidization (Fuest et al. 2005; Simonetta 2017).

Third, policy recommendations depend critically on how experience rating is measured and on how the marginal benefit and cost are calibrated. While recent work has improved the estimation of experience rating and its marginal cost, far less is known about the marginal benefit: firms' value of insurance. Ongoing work (Abebe et al. 2025) aims to disentangle the roles of borrowing constraints, which would call for policies expanding access to liquidity, from risk aversion, for which coinsurance would be appropriate. It is also crucial to assess whether firms' insurance value varies over the business cycle. If it rises during recessions, the optimal policy may involve higher experience rating during booms and lower experience rating during downturns, in contrast to current U.S. practice.

The theoretical framework and empirical application developed in this paper provide a foundation for future work evaluating experience rating in the United States and for studying the introduction of experience rating in other countries. Future work should examine the joint design of all the policy parameters involved. Jointly analyzing firm and worker tax rates is essential for a complete incidence analysis, and benefit generosity cannot be studied independently from tax design when higher taxes induce more aggressive appeals and lower take-up (Lachowska et al. 2025) and generous benefits raise unemployment both by reducing workers' search effort and by discouraging hiring through higher tax burdens (Johnston 2021). These interdependencies call for a unified framework that accounts for behavioral responses and risk preferences on both sides of the labor market.

Tables and Figures

Table I: Quasi-Experimental Responses to and Incidence of Unemployment Taxation

	South Carolina	**Colorado**
Panel A. Estimated Effects		
Unemp. Tax per Worker $\overline{\beta_\tau}$	0.357***	2.093***
Employment $\overline{\beta_l}$	-0.001***	-0.003***
Total Wages $\overline{\beta_{l\cdot w}}$	-58.32***	-430.543***
Average Wage $\overline{\beta_w}$	-1.899**	-13.696***
Taxable Wages $\overline{\beta_{tw}}$	-26.260***	-60.71***
Panel B. Full-Sample Pre-/Post-Reform Averages		
Unemp. Tax per Worker $\overline{\tau_{Pre}}$	134.942	254.199
Employment $\overline{l_{Pre}}$	11.714	20.300
Average Wage $\overline{w_{Pre}}$	37,732	59,916
Taxable Wage Base $\overline{TWB_{Post}}$	11,500	13,225
Panel C. Incidence		
Firm's Incidence Share $\|_{w^M=\overline{w_{Pre}}}$	64.7%	41.7%
Firm's Maximum Incidence Share	74.7%	95.1%
Panel D. Implied Elasticities		
Labor Demand w.r.t. Unemployment Tax Per Worker: $\epsilon_{l,\tau}$	-0.032	-0.016
Labor Demand w.r.t. Labor Costs: $\epsilon_{l,(\tau+w)}$	-9.055	-3.889
Wage w.r.t. Unemployment Tax Per Worker: $\epsilon_{w,\tau}$	-0.019	-0.025

Notes: Panel (A) reports the average of the estimated post-reform β_y coefficients from Equations (27) for South Carolina and (32) for Colorado. The corresponding year/quarter-level estimates are shown in Figure IV and Table E1 for South Carolina and VI and Table F2 for Colorado. Panel (B) reports pre- and post-reform averages of key variables for the full South Carolina and Colorado study samples. For South Carolina, the pre-reform period is 2007–2010 and the post-reform period is 2011–2014. For Colorado, they are 2014–2017 and 2018–2021. Appendix D details the steps to clean and winsorize the data and the construction of key outcomes for each state. Panel (C) reports the share of incidence borne by firms under the assumption that missing employees earned the average wage reported in Panel (B), calculated using Equation (33), and the maximum share of incidence on firms, calculated using Equation (34). Panel (D) reports elasticities implied by the reduced-form estimates in Panel (A) and the full-sample averages in Panel (B), calculated using Equation (35). *** p<0.01, ** p<0.05, * p<0.1.

Table II: Calibration of Parameters for Optimal Experience Rating

Parameter	**Description**	**Calibration / Source**	***South Carolina***	***Colorado***
e_{share}	Degree of experience rating: share of own benefits repaid in taxes	Share of non-socialized benefit charges from ETA 204 (Eq. 63)	0.343 (2010) 0.527 (2011)	0.735 (2017) 0.725 (2018)
e_{mtc}	Degree of experience rating: marginal tax cost of $1 of benefit charge	Avg. marginal tax cost of $1 of benefit charges (Eq. 64) from QCEW, ETA394, UIFP	0.154 (2010) 0.249 (2011)	0.381 (2017) 0.304 (2018)
b	Unemployment benefit level	Avg. weekly benefit × potential benefit duration from ETA 394	$5,201 (2010) $4,662 (2011)	$10,066 (2017) $10,285 (2018)
l_H	Employment share in high-risk industry	Employment share in industries in the top 5% of distribution of within-year std. dev. of quarterly employment, averaged 1990–2020 from QCEW.	0.404 (2010) 0.403 (2011)	0.327 (2017) 0.326 (2018)
r_H	Layoff rate in high-risk industry	Differential % change in employment over year between low- and high-risk industries (Eq. 65) from QCEW	0.044 (2010) 0.113 (2011)	0.124 (2017) 0.086 (2018)
$f_L(l_H)$	Marginal productivity in low-risk industry	Average annual wage in low-risk industries from QCEW (Eq. 12)	$39,362 (2010) $41,114 (2011)	$58,393 (2017) $61,047 (2018)
$f_H(l_H)$	Marginal productivity in high-risk industry	Sum of average annual wage in high-risk industries from QCEW +calibrated unempl. taxes and borrowing costs (Eq. 19)	$34,839 ($e_{share}$, 2010) $34,795 ($e_{mtc}$, 2010) $37,649 ($e_{share}$, 2011) $37,487 ($e_{mtc}$, 2011)	$50,209 ($e_{share}$, 2017) $49,700 ($e_{mtc}$, 2018) $52,324 ($e_{share}$, 2017) $51,915 ($e_{mtc}$, 2018)
$\int_0^{l_H} f_H(i)\,di$	Output in high-risk industry	Area of green and gray parallelograms in Fig. I	$516 ($e_{share}$, 2010) $515 ($e_{mtc}$, 2010) $1,614 ($e_{share}$, 2011) $1,607 ($e_{mtc}$, 2011)	$780 ($e_{share}$, 2017) $772 ($e_{mtc}$, 2017) $218 ($e_{share}$, 2018) $214 ($e_{mtc}$, 2018)
$\epsilon_{l_H,e}$	Elasticity of labor demand w.r.t. experience rating	Elasticity of employment w.r.t. unemployment tax per worker (Eq. 25); SC DEW, CO DLE	-0.032	-0.016
$\epsilon_{r_H,e}$	Elasticity of layoff rate w.r.t. experience rating	Elasticity of permanent layoffs w.r.t. experience rating; Card et al. (1994)	-0.1	-0.1
$\frac{\Pi_H'^{good}-\Pi_H'^{bad}}{\Pi_H'^{good}}$	Insurance value	Degree of experience rating (Eq. 21)	0.343 (e_{share}, 2010) 0.154 (e_{mtc}, 2010) 0.527 e_{share}, 2010) 0.249 (e_{mtc}, 2010)	0.735 (e_{share}, 2017) 0.381 (e_{mtc}, 2017) 0.725 (e_{share}, 2018) 0.304 (e_{mtc}, 2018)

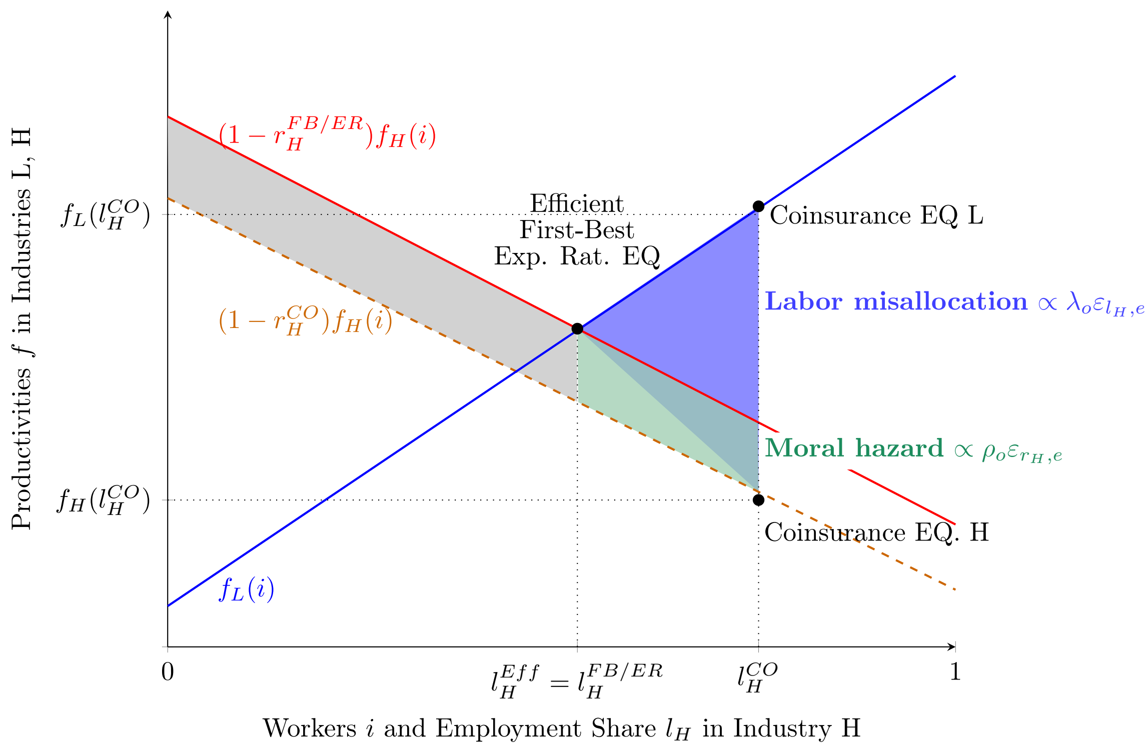
Notes: The table reports the calibration approach, data sources, and values for each parameter in the formula for the optimal degree of experience rating, Equation (21).

Table III: Optimal Degree of Experience Rating

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		South Carolina				**Colorado**			
		e_{share}		e_{mtc}		e_{share}		e_{mtc}	
		2010	*2011*	*2010*	*2011*	*2017*	*2018*	*2017*	*2018*
Current Degree of Experience Rating e:		0.343	0.527	0.154	0.249	0.735	0.725	0.381	0.304
Marginal Cost:									
$\rho_f \varepsilon_{r_H,e}$	Fiscal Externality from Moral Hazard	2.20	0.40	6.29	1.33	0.13	0.19	0.66	1.34
$\rho_o \varepsilon_{r_H,e}$	Output Loss from Moral Hazard	0.54	0.34	1.54	1.13	0.03	0.01	0.15	0.07
$\lambda_f \varepsilon_{l_H,e}$	Fiscal Externality from Labor Misallocation	0.70	0.13	2.01	0.43	0.04	0.06	0.21	0.43
$\lambda_o \varepsilon_{l_H,e}$	Output Loss from Labor Misallocation	18.98	1.92	54.30	6.45	0.53	0.97	2.59	6.71
	Total	22.42	2.78	64.14	9.33	0.73	1.23	3.61	8.53
	% Labor Misallocation	0.88	0.74	0.88	0.74	0.78	0.84	0.78	0.84
	Lower Bound	4.98	0.52	14.25	1.74	0.14	0.26	0.71	1.81
	Upper Bound	33.10	5.64	94.70	18.94	1.37	2.01	6.76	14.00
Marginal Benefit:									
$\frac{\Pi_H^{\prime good} - \Pi_H^{\prime bad}}{\Pi_H^{\prime good}}$	Insurance Value	0.343	0.527	0.154	0.249	0.735	0.725	0.381	0.304
	Lower Bound	0	0	0	0	0	0	0	0
	Upper Bound	3.13	3.13	3.13	3.13	3.13	3.13	3.13	3.13
Implied Policy Recommendation:									
Main Result	MC: Total; MB: Insurance Value	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\approx$	$e\uparrow$	$e\uparrow$	$e\uparrow$
Alternative Scenarios									
(1)	MC: Fiscal Only; MB: Insurance Value	$e\uparrow$	$e\approx$	$e\uparrow$	$e\uparrow$	$e\downarrow$	$e\downarrow$	$e\uparrow$	$e\uparrow$
(2)	MC: Moral Hazard Only; MB: Insurance Value	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\downarrow$	$e\downarrow$	$e\uparrow$	$e\uparrow$
(3)	MC: Lower Bound; MB: Insurance Value	$e\uparrow$	$e\approx$	$e\uparrow$	$e\uparrow$	$e\downarrow$	$e\downarrow$	$e\uparrow$	$e\uparrow$
(4)	MC: Lower Bound; MB: Upper Bound	$e\uparrow$	$e\downarrow$	$e\uparrow$	$e\downarrow$	$e\downarrow$	$e\downarrow$	$e\downarrow$	$e\downarrow$
(5)	MC: Upper Bound; MB: Upper Bound	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\downarrow$	$e\downarrow$	$e\uparrow$	$e\uparrow$
(6)	MC: Lower Bound; MB: Lower Bound	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$
(7)	MC: Fiscal Only; MB: Upper Bound	$e\downarrow$	$e\downarrow$	$e\uparrow$	$e\downarrow$	$e\downarrow$	$e\downarrow$	$e\downarrow$	$e\downarrow$
(8)	MC: Moral Hazard Only; MB: Upper Bound	$e\downarrow$	$e\downarrow$	$e\uparrow$	$e\downarrow$	$e\downarrow$	$e\downarrow$	$e\downarrow$	$e\downarrow$
(9)	MC: Fiscal Only; MB: Lower Bound	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$
(10)	MC: Moral Hazard Only; MB: Lower Bound	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$
(11)	MC: Total; MB: Lower Bound	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$	$e\uparrow$
(12)	MC: Total; MB: Upper Bound	$e\uparrow$	$e\downarrow$	$e\uparrow$	$e\uparrow$	$e\downarrow$	$e\downarrow$	$e\downarrow$	$e\uparrow$
Optimal Degree of Experience Rating: e^*		0.927	0.796	0.927	0.796	0.749	0.820	0.749	0.820

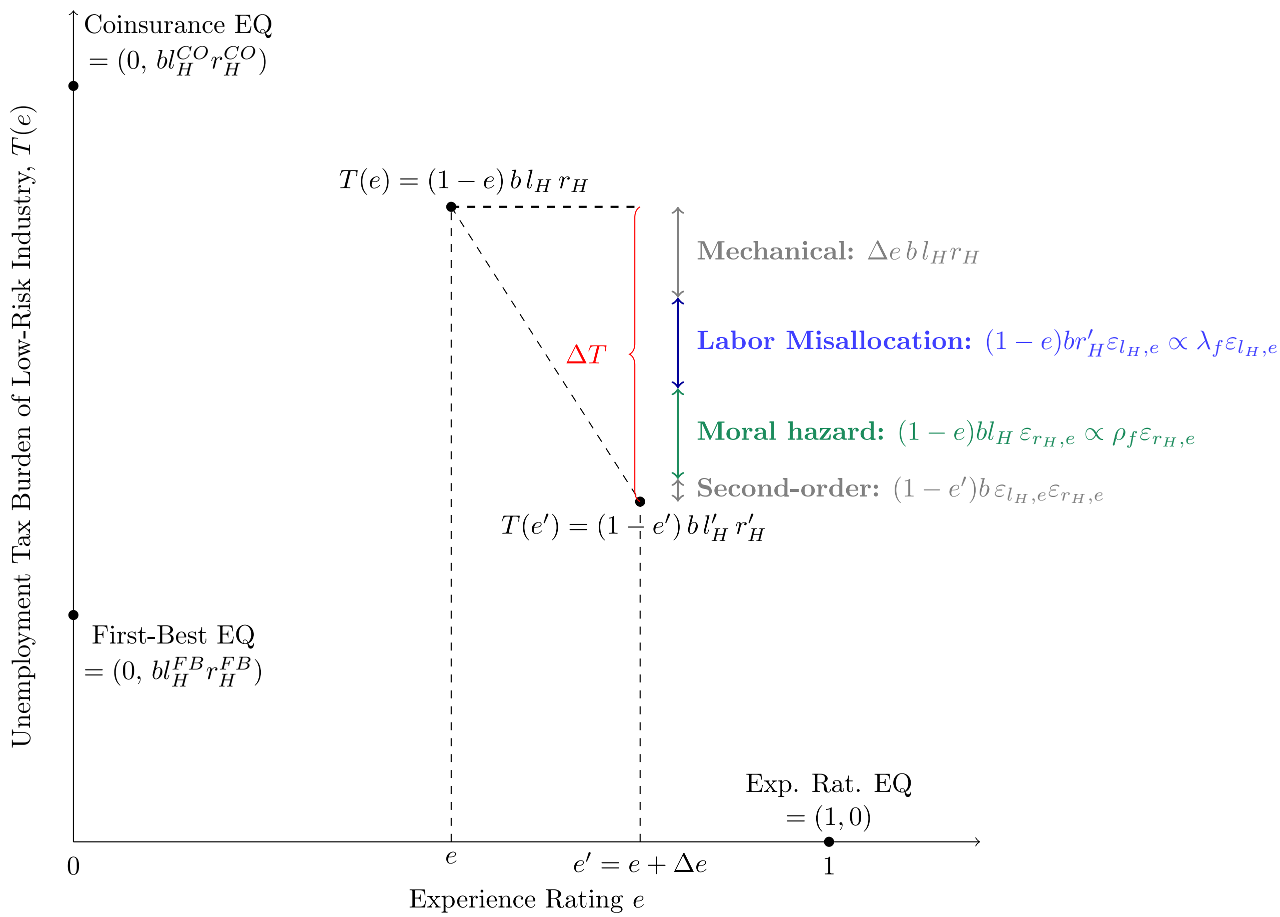
Notes: The table shows the calibration of the formula for the optimal degree of experience rating in Equation (21) for South Carolina in 2010–2011 and Colorado in 2017–2018. Experience rating is measured either as the share of own benefit charges repaid by firms in unemployment taxes, e_{share}, or as the marginal tax cost of one dollar of benefit charges, e_{mtc}. For each state-year, the table reports the prevailing degree of experience rating, the marginal cost and its components, the share attributable to labor misallocation, and the marginal benefit. It also reports upper and lower bounds for the marginal benefit and cost. Bounds on the marginal cost are based on alternative elasticity estimates from the literature, while bounds on the marginal benefit reflect different assumptions about firms' risk aversion. Details are provided in Section 5. The table compares marginal benefits and costs under alternative scenarios and reports the implied policy recommendation: increase experience rating when the marginal cost exceeds the marginal benefit ($e\uparrow$), decrease it when the marginal benefit exceeds the marginal cost ($e\downarrow$), or leave it unchanged when the two are equal ($e\approx$). Scenarios include combinations of considering fiscal externalities only, moral hazard only, and upper and lower bounds on the marginal benefit and cost. Finally, the table shows the optimal degree of experience rating, obtained as the level that equalizes the marginal benefit and cost of reducing experience rating in each state-year under the assumption of constant formula parameters.

Figure I: Output Losses from Coinsurance



Notes: The figure shows the distribution of workers' productivities f in the low- and high-risk industries (L/H) over the unit interval of worker types i. Productivity in the high-risk industry is scaled by the probability of no shock, $1-r$ in the first-best (FB), second-best complete experience rating (ER), and second-best coinsurance (CO) scenarios. l_H^{Eff} is the efficient employment share in the high-risk industry and coincides with the first-best and second-best complete experience rating employment share, $l_H^{FB/ER}$. l_H^{CO} is the second-best coinsurance employment share in the high-risk inudstry. The light-blue area is proportional to the term $\lambda_o \varepsilon_{l_H,e}$ in Equation (21) and represents the output loss from labor misallocation. The green and gray areas are proportional to $\rho_o \varepsilon_{r_H,e}$ in Equation (21) and represent the output loss from moral hazard. The gray area represents the output at risk of loss even in the first-best equilibrium given the nonzero layoff rate in the high-risk industry.

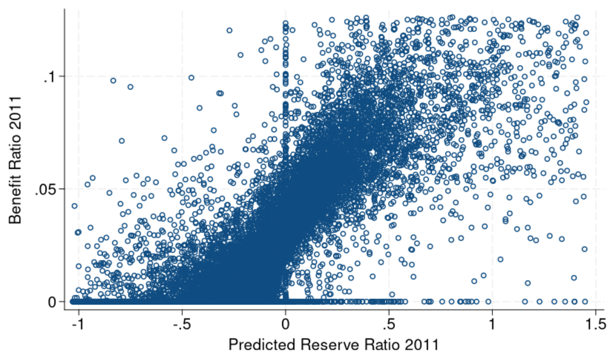
Figure II: Fiscal Externalities from Coinsurance



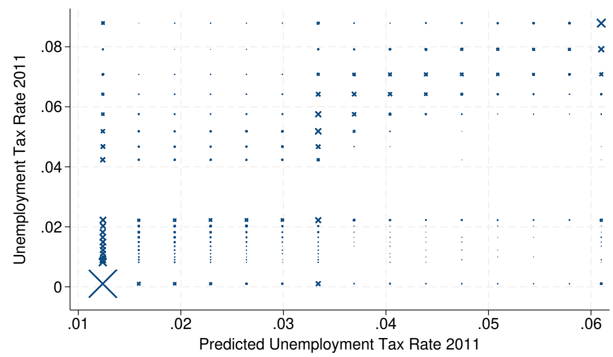
Notes: The figure shows the unemployment tax burden on the low-risk industry, $\tau_L l_L = T(e) = (1-e)bl_H r_H$, as a function of the degree of experience rating e. The first-best (FB), second-best complete experience-rating (ER) and second-best coinsurance (CO) equilibria are highlighted in the graph. Moreover, the figure shows how the unemployment tax burden of the low-risk industry increases when experience rating declines from e' to e, ΔT, and its decomposition into four components: a mechanical increase due to the decline in e for fixed benefit expenditure; an increase from the expansion of the high-risk industry's employment share, proportional to $\lambda_f \varepsilon_{l_H,e}$ in Equation (21) and capturing labor misallocation; an increase from the rise in the layoff rate, proportional to $\rho_f \varepsilon_{r_H,e}$ in Equation (21) and capturing moral hazard; and a second-order interaction term.

Figure III: South Carolina's Unemployment Tax Reform and Identifying Variation

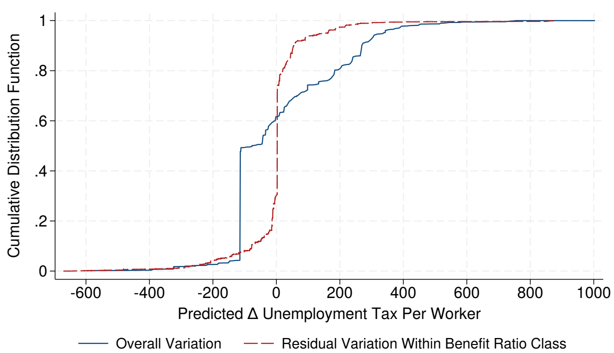
(a) Variation across Experience Factors



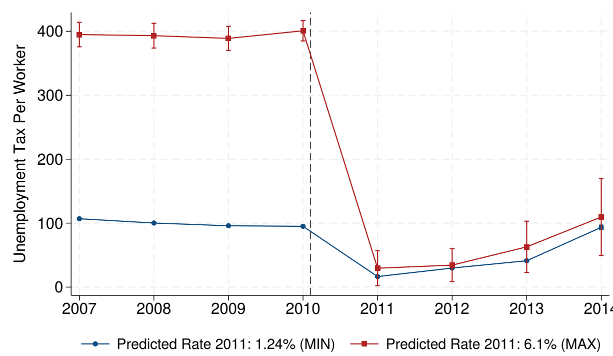
(b) Unemployment Tax Rates' Combinations



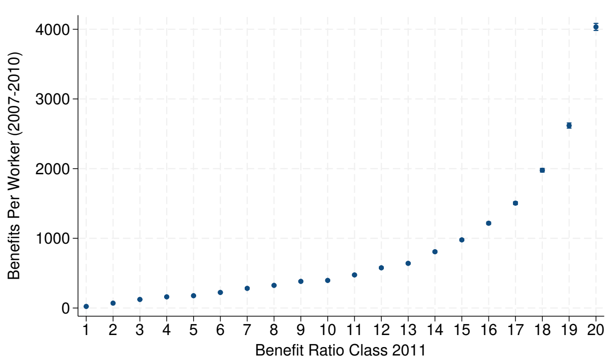
(c) Distribution of Predicted Change in Tax per Worker



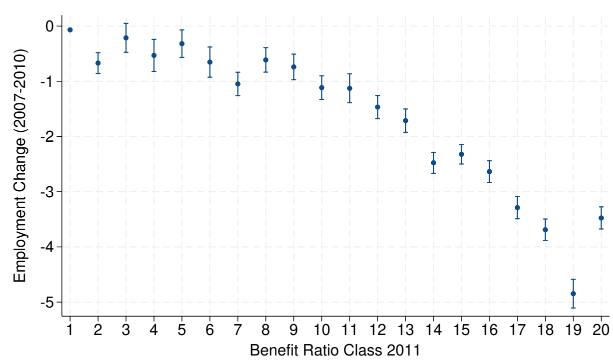
(d) Tax Per Worker for Same-Benefit Ratio Employers



(e) Pre-Reform Benefit Charges by Benefit Ratio Class

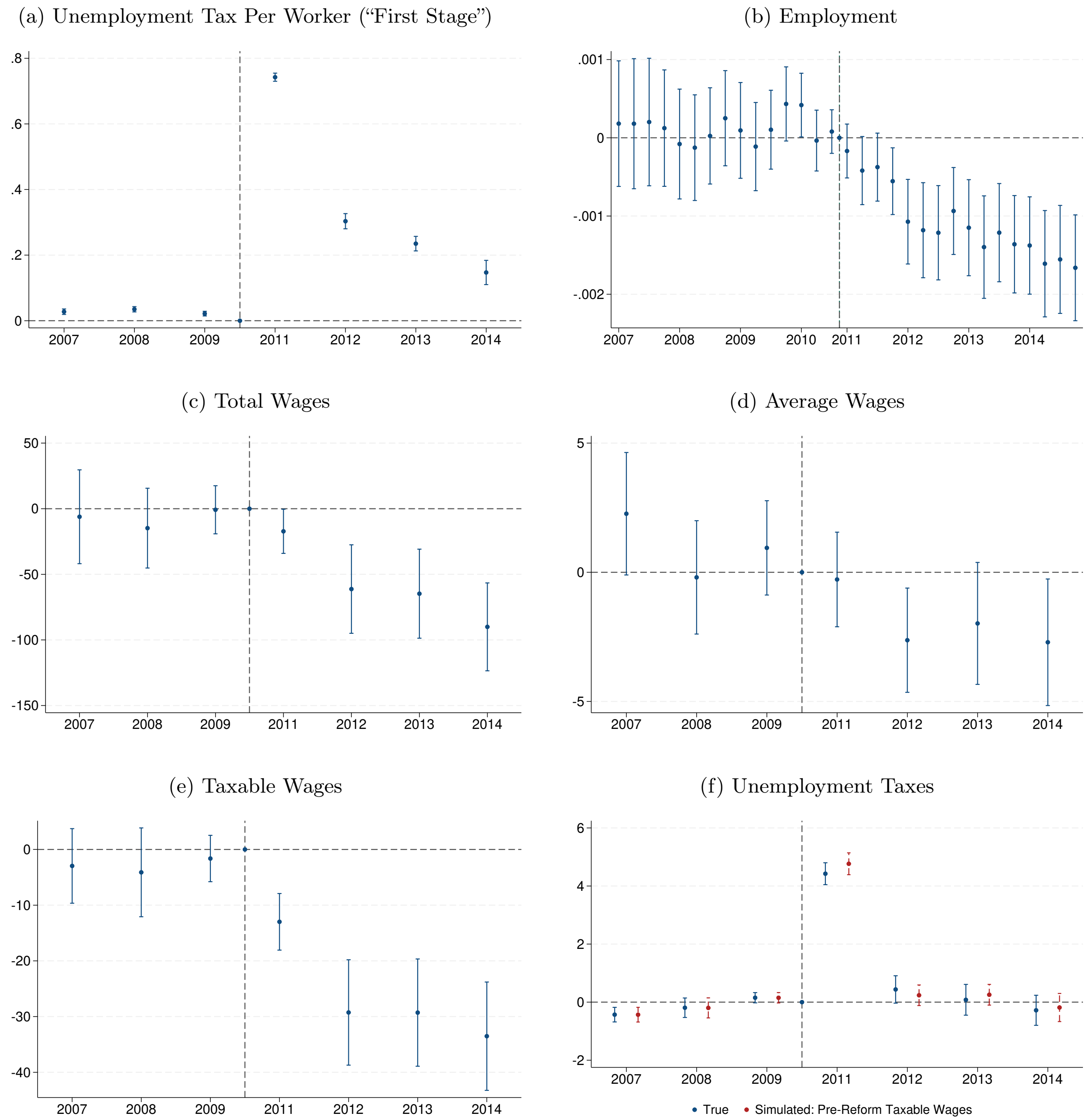


(f) Pre-Reform Employment by Benefit Ratio Class



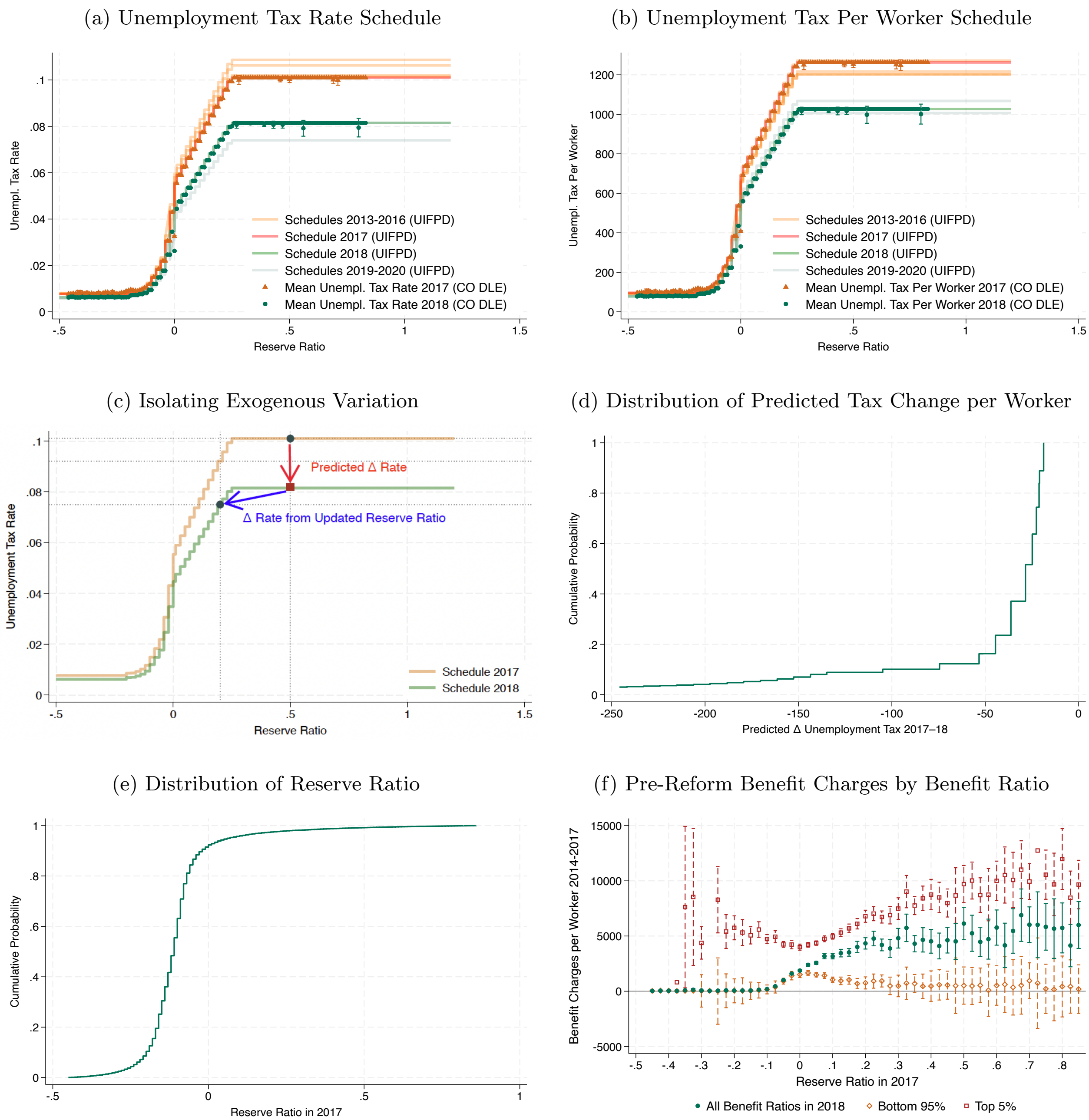
Notes: Panel (a) plots benefit ratios, trimmed at 99^{th} percentile, against predicted reserve ratios in 2011, trimmed at the 1^{st}-99^{th} percentiles, for firms in the SC DEW study sample. Equation (26) provides their formulas. Panel (b) plots frequency-weighted firms' true tax rates, based on benefit ratios and the 2011 schedule, against their predicted tax rates, based on predicted reserve ratios and the 2010 schedule, in 2011. The difference between them is the predicted rate change, which, multiplied by the taxable wage base, gives the predicted change in the unemployment tax per worker. Panel (c) shows the cumulative distribution function of the predicted change in the tax per worker, unconditional and after controlling for benefit ratio class. Panel (d) shows the average unemployment tax per worker for firms with benefit ratio equal to zero in 2011 and with a 2011 predicted tax rate equal to the minimum of the 2010 schedule (1.24%, lowest reserve ratio bin) and maximum (6.1%, highest bin). Panels (e) and (f) show the pre-reform (2007-10) average total benefit charges per worker, trimmed at 99^{th} percentile, and employment change, trimmed at the 1^{st}-99^{th} percentiles, by benefit ratio class in 2011. Table D3 provides additional details on variables construction.

Figure IV: Responses to and Incidence of Unemployment Taxes in South Carolina



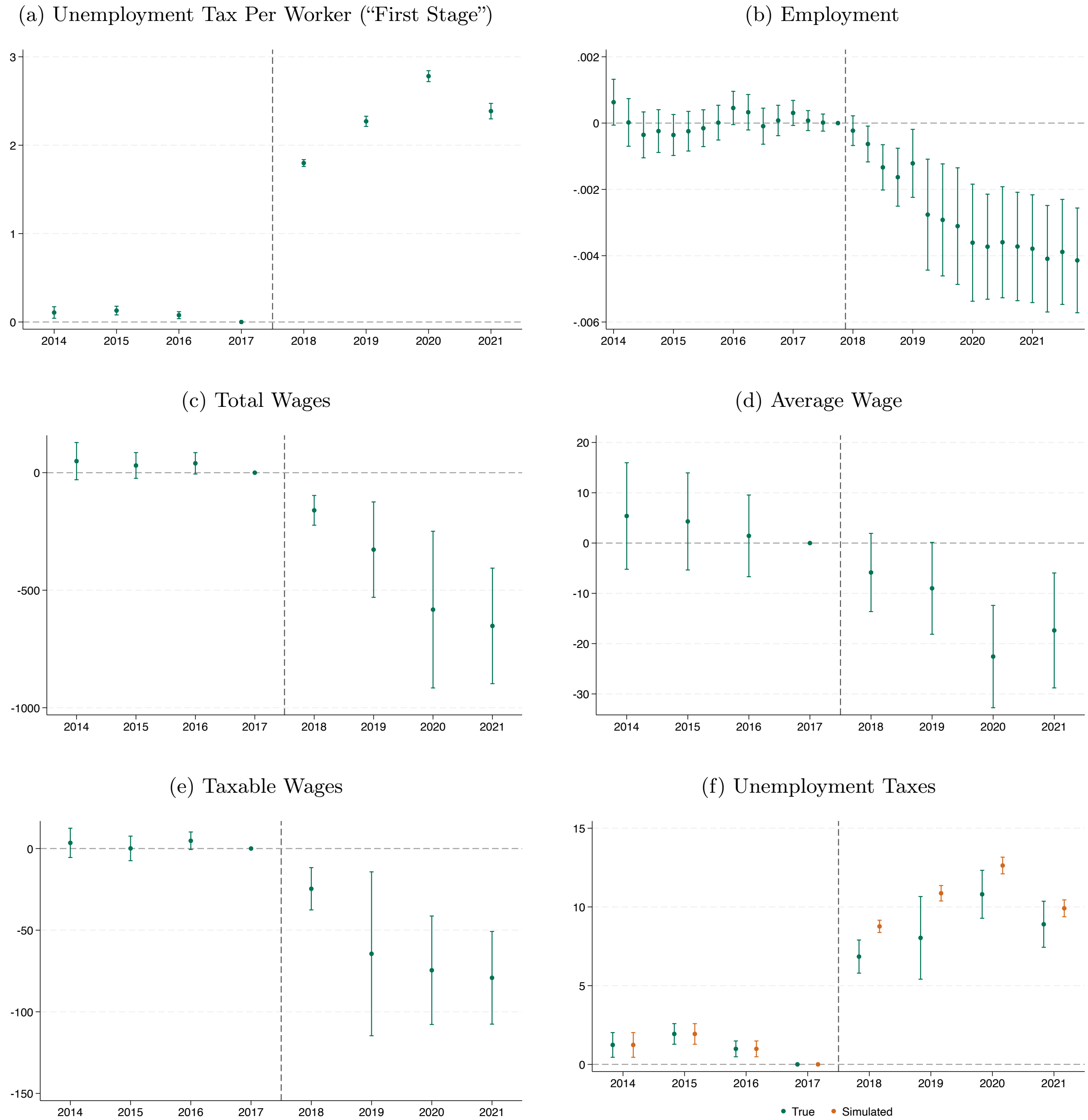
Notes: The figure shows the estimated β_y coefficients from Equation (27) for firms in the SC DEW study sample with fewer than 50 employees on average in 2010. 50 is the 95$^{\text{th}}$ percentile of the 2010 employment distribution. Appendix D.1 provides details on the construction, cleaning and winsorization of the outcomes. 95% percent confidence intervals are reported. Yearly estimates and standard errors are shown in Table E1.

Figure V: Colorado's Unemployment Tax Reform and Identifying Variation



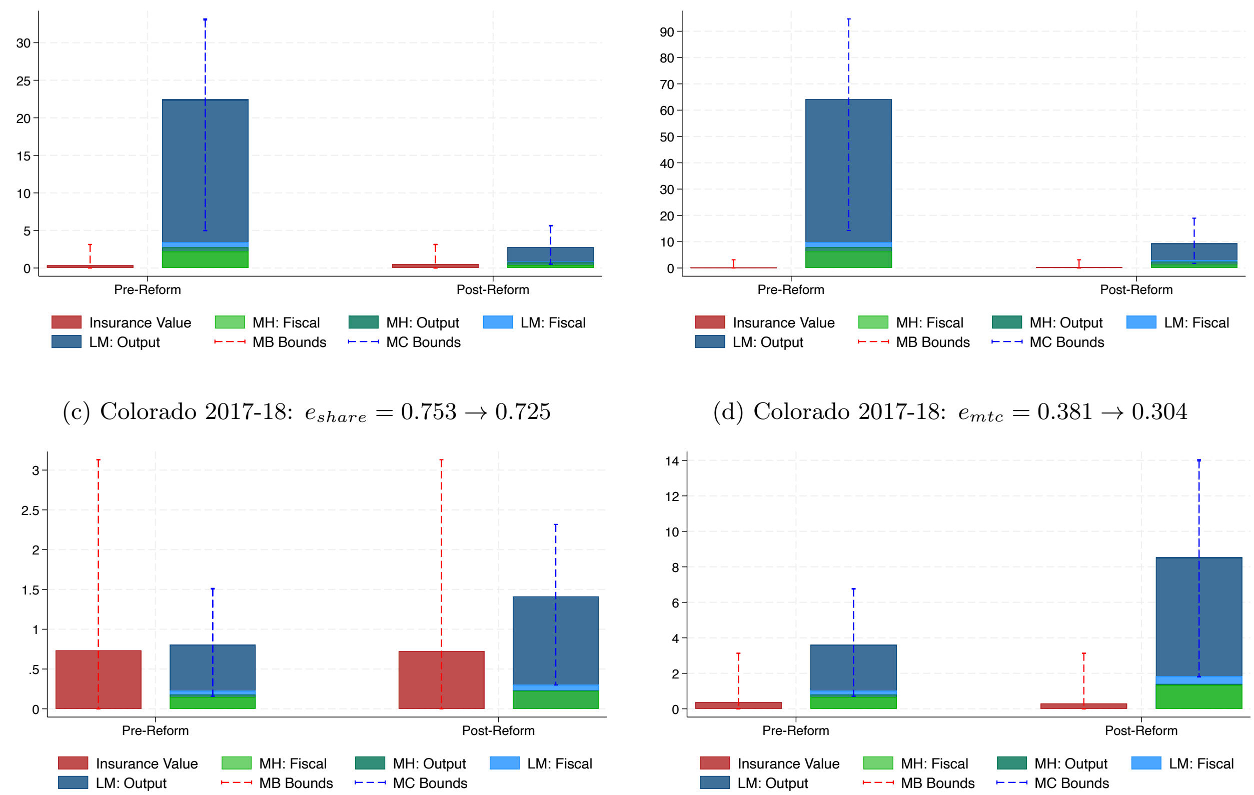
Notes: Panels (a) and (b) show the unemployment tax rate and tax per worker schedules in place in Colorado in 2013-20 from the US Unemployment Insurance Financing Policy Database and, for each level of reserve ratios, average tax rates and taxes per worker for CO DLE firms, with 95% confidence intervals. Panel (c) decomposes the total tax rate change between 2017 and 2018 into the predicted change due to the surcharge removal and the endogenous change due to the reserve ratio update (Equation 31). Panels (d) and (e) show the cumulative distribution function of the predicted change in unemployment tax per worker in 2018 and of the reserve ratio in 2017 for firms in the study sample. Panel (f) shows the average total benefit charges per worker pre-reform (2014-17) and 95% confidence intervals for 2018 reserve ratio bins in the full sample, the top 5% and the bottom 95% of the benefit ratio distribution in 2018. The 95^{th} percentile is 0.05. Bins are created by rounding the reserve ratio to 0.025 and trimming it at the 1^{st} and 99^{th} percentiles. Table D5 provides additional details on variables construction.

Figure VI: Responses to and Incidence of Unemployment Taxes in Colorado



Notes: The figure reports the estimated β_y coefficients from Equation (32) for firms in the CO DLE study sample with fewer than 12 employees on average in 2017 and with a benefit ratio in 2018 lower than 0.05. 12 is the 75^{th} percentile of the employment distribution in 2017. 0.05 is the 95^{th} percentile of the benefit ratio distribution in 2018. Appendix D.2 provides details on the construction, cleaning and winsorization of the outcomes. 95% percent confidence intervals are reported. Yearly estimates and standard errors are shown in Table F2.

Figure VII: Marginal Benefit and Cost of Coinsurance



(a) South Carolina 2010-11: $e_{share} = 0.343 \rightarrow 0.526$

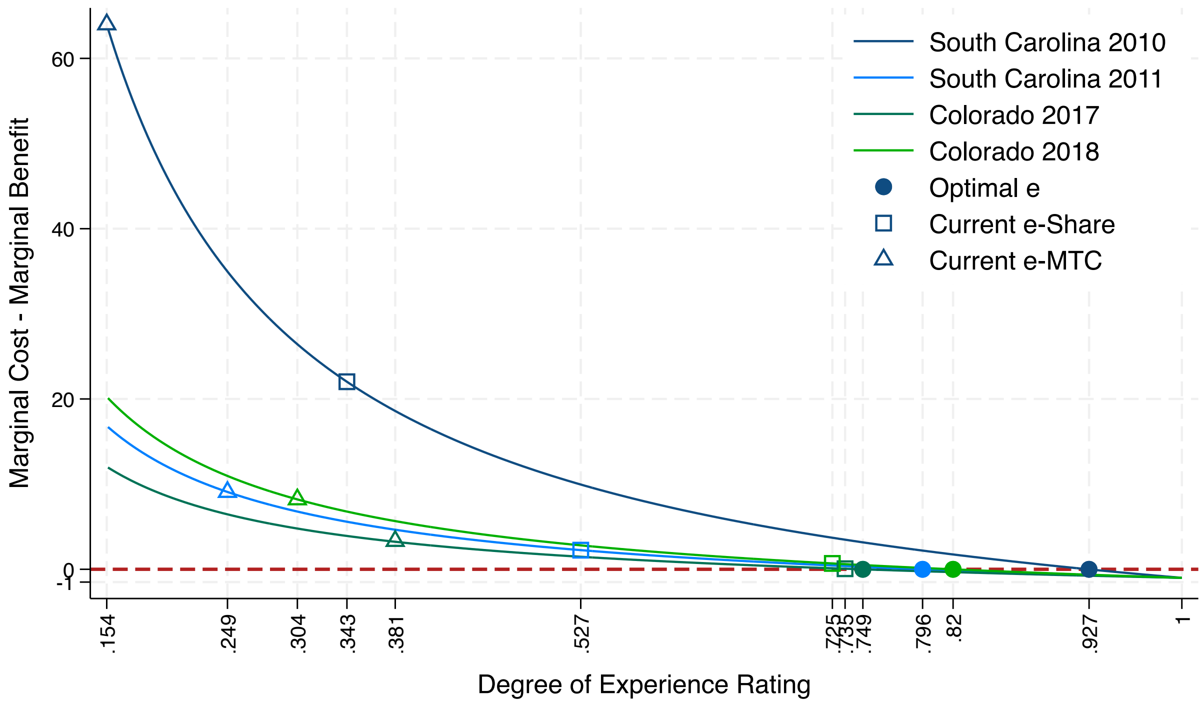
(b) South Carolina 2010-11: $e_{mtc} = 0.154 \rightarrow 0.249$

(c) Colorado 2017-18: $e_{share} = 0.753 \rightarrow 0.725$

(d) Colorado 2017-18: $e_{mtc} = 0.381 \rightarrow 0.304$

Notes: The figure shows the calibration of the formula for the optimal degree of experience rating, Equation (21), for South Carolina in 2010–2011 and Colorado in 2017–2018. Experience rating is measured either as the share of own benefit charges repaid by firms in unemployment taxes, e_{share}, or as the marginal tax cost of one dollar of benefit charges, e_{mtc}. For each state-year and experience rating measure, the figure shows the prevailing degree of experience rating, the marginal cost and its components, the marginal benefit, and upper and lower bounds for marginal benefit and cost. Bounds on the marginal cost are based on alternative elasticity estimates from the literature, while bounds on the marginal benefit reflect different assumptions about firms' risk aversion. Additional details are provided in Section 5. In the legend, LM stands for labor misallocation, MH for moral hazard, MB for marginal benefit, MC for marginal cost, Fiscal for fiscal externalities, and Output for output losses. In the graph, Pre-Reform indicates 2010 for South Carolina and 2017 for Colorado. Post-Reform indicates 2011 for South Carolina and 2018 for Colorado.

Figure VIII: Current and Optimal Degree of Experience Rating



Notes: The lines show the evolution of the difference between the marginal cost and marginal benefit in Equation (21), calibrated as summarized in Table II, when the degree of experiating changes, separately for South Carolina in 2010-2011 and Colorado in 2017-2018. All the lines converge at (x=1, y=-1): when the degree of experience rating is equal to one, the marginal cost is equal to zero and the marginal benefit is equal to one. Circles indicate the optimal degree of experience rating in a state-year, identified by a difference between marginal benefit and cost equal to zero. This optimal level is calculated assuming constant formula parameters as the degree of experience rating changes. Squares indicate the current degree of experience rating in a state-year measured as the share of own benefit charges that a firm repays in unemployment taxes, e_{share}. Triangles indicate the current degree of experience rating in a state-year measured as the marginal tax cost of a dollar of benefit charges, e_{mtc}.

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APPENDIX FOR

Optimal Unemployment Insurance Financing:
Theory and Evidence from Two U.S. States
by Sara Spaziani

A Model Derivation

A.1 Low-Risk Firm's Optimal Labor Demand

The low-risk firm chooses labor demand l_L to maximizes profits, Π_L from Equation (2), treating the degree of experience rating and the high-risk firm's layoff rate r_H and labor demand l_H as fixed. I express Π_L as a function of the degree of experience rating in Equation (36) by substituting unemployment taxes from Equation (9). I then take the derivative of profits with respect to l_L, set the derivative to zero, and rearrange terms to obtain Equation (12).

$$\Pi_L = \int_{1-l_L}^{1} \left[f_L(i)^{\smile} w_{i,L} f_L(i) \right] di - (1-e) b r_H l_H - j \tag{36}$$

$$\frac{\partial \Pi_L}{\partial l_L} = -\left[-f_L(1-l_L) - w_{1-l_L,L} f_L(1-l_L) \right](-1) - \int_{1-l_L}^{1} \frac{\partial w_{i,L}}{\partial l_L} f_L(i) d\, i = 0 \tag{37}$$

The integral is taken over $[1-l_D, 1]$ because productivity in the low-risk industry increase over the unit interval and the firm hires workers in descending order, starting from $i = 1$. The choice of labor demand can therefore be interpreted as the selection of an optimal stopping rule along the unit interval. The first-order condition in Equation (37) shows that hiring stops when the productivity of the marginal worker $i = 1 - l_L^D$ equals the marginal cost of labor. Since lower-productivity workers require higher wages to enter the industry, the wage rate offered, $w_{i,L}$, must increase as the firm keeps hiring over the unit interval (going towards $i = 0$). The marginal cost thus includes the wage paid to the marginal worker and the incremental wage to workers in the industry. However, previously hired workers $i > 1 - l_L^D$ were already attracted to the industry by their respective wage rate offers, so only the marginal worker's wage must adjust at the margin, i.e., $\frac{\partial w_{i,L}}{\partial l_L}$ is non-zero only for the marginal worker $i = 1 - l_L^D$.

A.2 First Best Solution

The first best equilibrium is the solution of the following constrained optimization problem: the government chooses the high-risk firm's layoff rate r_H, labor demand, l_H, and the degree of experience rating e to maximize social welfare (Equation 6) and output (Equation 7), subject to the budget constraint for allocating the tax burden between the firms (Equation 9, which I substitute in firms' profits shown in Equations 1 and 2), workers' optimal labor supply (Equation 11), the low-risk firm's labor demand (Equation 12), and labor market clearing in both industries.

Equation (38) shows the Lagrangian of this maximization problem, $\mathcal{L}^{FB}$. To obtain the first-best solution, I take the derivative of $\mathcal{L}^{FB}$ with respect to r_H, l_H, and e, and set them equal to zero.

$$
\begin{aligned}
\mathcal{L}^{FB} =& \underbrace{\int_{l_H}^{1} u(w_{i,L} f_L(i)) d\, i}_{U_{i,L} \text{ for workers in L}} + \underbrace{\int_0^{l_H} (1-r_H)[u(w_{i,H} f_H(i))] + r_H[u(b) + L_i]\, di}_{U_{i,H} \text{ for workers in H}} \\
&+ \underbrace{\Pi_L + \Pi_H}_{U_C} + \underbrace{\int_0^{l_H} (1-r_H) f_H(i) d\, i + \int_{l_H}^{1} f_L(i) d\, i}_{\text{Y}} \\
=& \int_{l_H}^{1} u(w_{i,L} f_L(i)) d\, i + \int_0^{l_H} (1-r_H) u(w_{i,H} f_H(i)) + r_H[u(b) + L_i]\, di \\
&+ \underbrace{\int_{l_H}^{1} [f_L(i) - w_{i,L} f_L(i)]\; di - (1-e) b l_H r_H - j}_{\Pi_L} + \int_0^{l_H} (1-r_H) f_H(i) d\, i + \int_{l_H}^{1} f_L(i) d\, i \\
&+ \underbrace{(1-r_H)\left[\int_0^{l_H} [f_H(i) - w_{i,H} f_H(i)]\; di - e b l_H r_H - j\right] + r_H[-e b l_H r_H (1+e)] - \psi(1-r_H)}_{\Pi_H}
\end{aligned} \tag{38}
$$

A.2.1 First-Best Layoff Rate

Equation (39) shows the derivative of $\mathcal{L}^{FB}$ with respect to r_H. Rearranging terms, I obtain the first-order condition for the first-best layoff rate, r_H^{FB}, shown in Equation (13).

$$
\begin{aligned}
\frac{\partial \mathcal{L}^{FB}}{\partial r_H} =& \int_0^{l_H} \overbrace{[-u(w_{i,H} f_H(i)) + u(b) + L_i]}^{=0 \text{ by Indiff. bw States (Eqn 4)}}\, di - (1-e) b l_H - \int_0^{l_H} f_H(i)\, di - r_H e b l_H (1+e) - (1-r_H) e b l_H \\
&+ \psi'(1-r_H) \underbrace{- \int_0^{l_H} [f_H(i) - w_{i,H} f_H(i)]\; di - (1-r_H) e b l_H - j}_{\text{-}\Pi_H^{good}} \underbrace{- e b l_H r_H (1+e)}_{\Pi_H^{bad}} = 0
\end{aligned} \tag{39}
$$

Notice that the difference in workers' utilities between employment in the high-risk industry and unemployment cancels out by assumption (Equation 4).

A.2.2 First-Best High-Risk Industry's Employment Share

Equation (40) shows the derivative of $\mathcal{L}^{FB}$ with respect to l_H. Rearranging terms, I obtain the first-order condition for the high-risk firm's first-best labor demand, $l_H^{D,FB}$, shown in Equation (14).

Several terms in the first-order condition cancel out due to model assumptions and an envelope condition: since workers and the low-risk firm optimize, changes in the degree of experience rating do not generate additional welfare effects. First, by Equation (10), the utility difference between industries is zero for the indifferent worker. By labor market clearing (Equation 15), the marginal worker hired by the high-risk firm at the government's chosen labor demand level, $l_H^{D,FB}$, coincides with the indifferent worker, which makes the utility difference between industries equal to zero.

$$\begin{aligned}\frac{\partial \mathcal{L}^{FB}}{\partial l_H} = & \overbrace{-u(w_{l_H,L} f_L(l_H)) + (1-r_H)u(w_{l_H,H} f_H(l_H)) + r_H[u(b) + L_{l_H}]}^{=0 \text{ for Indifferent Worker (Eqn 10)+Mkt Clearing (Eqn 15)}} \\ & + \overbrace{\int_{l_H}^{1} u'(w_{i,L} f_L(i)) f_L(i) \frac{\partial w_{i,L}}{\partial l_H}\, di + \int_0^{l_H} (1-r_H) u'(w_{i,H} f_H(i)) f_H(i) \frac{\partial w_{i,H}}{\partial l_H}\, di}^{\frac{\partial w_i}{\partial l_H} = 0 \text{ for inframarginal workers. Terms sum to 0 for } l_H \text{ by Deriving Indiff. W. (Eqn 41)+Mkt Clearing (Eqn 15)}} \\ & \overbrace{-f_L(l_H) + w_{l_H,L} f_L(l_H) - \int_{l_H}^{1} f_L(i) \frac{\partial w_{i,L}}{\partial l_H}\, di}^{=0 \text{ for L's Optimal Labor Demand (Eqn 12)}} - (1-e)br_H + (1-r_H) f_H(l_H) - f_L(l_H) \\ & + (1-r_H)\left[f_H(l_H) - w_{l_H} f_H(l_H) - \int_0^{l_H} f_H(i) \frac{\partial w_{i,H}}{\partial l_H}\, di - ebr_H \right] - r_H ebr_H(1+e) = 0 \end{aligned} \tag{40}$$

Second, to attract an additional worker to the high-risk industry, only the wage of the marginal worker needs to increase. Therefore, wage changes for all the other workers, represented by $\frac{\partial w_{i,H}}{\partial l_H}$ and $\frac{\partial w_{i,L}}{\partial l_H}$, are zero, and these workers' utilities remains unaffected. For the marginal worker, $l_H^{D,FB}$, these derivatives are non-zero. However, Equation (41), which is the derivative of the indifferent worker's condition in Equation (10), shows that the marginal utility changes across industries cancel out, as expected for an optimizing, indifferent worker. Third, by labor market clearing in both industries (Equation 15), the labor demands of the low- and high-risk firm sum to one. Substituting $l_H = 1 - l_L$ and noting that $-\frac{\partial w_{i,L}}{\partial l_H} = -\frac{\partial w_{i,L}}{\partial (1-l_L)} = \frac{\partial w_{i,L}}{\partial l_L}$ gives the first-order condition for the low-risk firm's labor demand, Equation (12), whose terms also sum to zero.

$$(1-r_H)u'(w_{i^I,H} f_H(i^I)) \frac{\partial w_{i^I,H}}{\partial i^I} = u'(w_{i^I,L} f_L(i^I)) \frac{\partial w_{i^I,L}}{\partial i^I} \tag{41}$$

A.2.3 First-Best Experience Rating

Equation (42) shows the derivative of $\mathcal{L}^{FB}$ with respect to e. Since all the terms are positive, the first-order condition implies that e^{FB} is equal to zero.

$$\frac{\partial \mathcal{L}^{FB}}{\partial e} = bl_H r_H - (1-r_H) bl_H r_H - rHbl_H r_H(1+e) - r_H ebl_H r_H = -2ebl_H r_H r_H = 0 \rightarrow e = 0 \tag{42}$$

Moreover, Equation (43) shows that e represents the gap in marginal profits between the bad and the good states of the world, leading to Equation (16).

$$\frac{\Pi_H'^{bad} - \Pi_H'^{good}}{\Pi_H'^{good}} = \frac{\frac{\partial \Pi_H^{bad}}{\partial e} - \frac{\partial \Pi_H^{good}}{\partial e}}{\frac{\partial \Pi_H^{good}}{\partial e}} = \frac{bl_H r_H(-1 - e + 1)}{-bl_H r_H} = e \tag{43}$$

A.3 Second Best Solution

The second-best equilibrium is the solution of two optimization problems. First, the high-risk firm chooses the layoff rate and labor demand to maximize profits. Second, the government sets the degree of experience rating to maximize social welfare and output, taking into account the constraints of the first-best problem listed in Appendix A.2, as well as the high-risk firm's labor demand and layoff responses to changes in experience rating.

A.3.1 Second Best Layoff Rate

The high-risk firm chooses the layoff rate r_H to maximize profits, Π_H from Equation (1), treating the degree of experience rating chosen by the government as fixed. I express Π_H as a function of experience rating in Equation (44), by substituting unemployment taxes from Equation (9).

$$\Pi_H = (1 - r_H)\left(\int_0^{l_H} [f_H(i)\, di - w_H f_H(i)]\, di - ebl_H r_H\right) + r_H[-ebl_H r_H(1+e)] - \psi(1 - r_H) \quad (44)$$

I then take the derivative of the expected profit with respect to r_H, set the derivative to zero, and rearrange terms to obtain Equation (18).

$$\begin{aligned}\frac{\partial \Pi_H}{\partial r_H} = & \underbrace{\int_0^{l_H} [-f_H(i)\, di + w_{i,H} f_H(i)]\, di + ebl_H r_H}_{\Pi_H^{good}} - (1 - r_H)ebl_H \underbrace{-ebl_H r_H(1+e)}_{\Pi_H^{bad}} \\ & - r_H ebl_H(1+e) + \psi'(1 - r_H) = 0 \end{aligned} \quad (45)$$

To learn how r_H^{SB} changes with experience rating (i.e., the sign of $\frac{\partial r_H^{SB}}{\partial e}$), I leverage the fact that this first-order condition, which I call $G(r_H^{SB}, e)$, is zero at the optimum.

$$\begin{aligned} G(r_H, e) \coloneqq \frac{\partial \Pi_H}{\partial r_H}\Big|_{r_H = r_H^{SB}} = & \int_0^{l_H} [-f_H(i)\, di + w_{i,H} f_H(i)]\, di + ebl_H r_H^{SB} - (1 - r_H^{SB})ebl_H \\ & - ebl_H r_H^{SB}(1+e) - r_H^{SB} ebl_H(1+e) + \psi'(1 - r_H^{SB}) = 0 \end{aligned} \quad (46)$$

By the Implicit Function Theorem, we obtain:

$$\frac{\partial r_H^{SB}}{\partial e} = -\frac{\frac{\partial G(r_H^{SB}, e)}{\partial e}}{\frac{\partial G(r_H^{SB}, e)}{\partial r_H}} = -\frac{-bl_H(1 + 4r_H^{SB} e)}{-2e^2 bl_H - \psi''(1 - r_H^{SB})} < 0 \quad (47)$$

Since ψ is convex by assumption, $\psi''(1 - r_H^{SB}) > 0$ and $\frac{\partial r_H^{SB}}{\partial e} < 0$.

A.3.2 Second-Best High-Risk Industry's Employment Share

The high-risk firm chooses labor demand l_H to maximize profits, Π_H from Equation (44), treating the degree of experience rating chosen by the government as fixed. I take the derivative of Π_H with respect to l_H, set the derivative to zero, and rearrange terms to obtain Equation (19).

$$\frac{\partial \Pi_H}{\partial l_H} = (1-r_H)\left(f_H(l_H) - w_{l_H,H} f_H(l_H) - \int_0^{l_H} \frac{\partial w_{i,H}}{\partial l_H} f_H(i)d\,i - ebr_H\right) - r_H[ebr_H(1+e)] = 0 \quad (48)$$

The integral is taken over $[0, l_H]$ because workers' productivities in the high-risk industry decline over the unit interval and the firm hires them in ascending order, starting from $i = 0$. The choice of labor demand can therefore be interpreted as the selection of an optimal stopping rule along the unit interval. The first-order condition in Equation (48) shows that hiring stops when the productivity of the marginal worker $i = l_H^{D,SB}$ equals the marginal cost of labor. Since by the optimal labor supply in Equation (11) lower-productivity workers require higher wages to enter the industry, the wage rate offered, $w_{i,H}$, must increase along the unit interval. The marginal cost thus includes the wage paid to the marginal worker and the incremental wage to workers in the industry. However, previously hired workers $i < l_H^{D,SB}$ were already attracted to the industry by their respective wage rate offers, so only the marginal worker's wage must adjust at the margin, i.e., $\frac{\partial w_{i,H}}{\partial l_H}$ is non-zero only for the marginal worker $i = l_H^{D,SB}$.

To learn how $l_H^{D,SB}$ changes with experience rating (i.e., the sign of $\frac{\partial l_H^{D,SB}}{\partial e}$), I leverage the fact that the first-order condition of this problem, which I call $G(l_H^{D,SB}, e)$ is equal to zero at the optimum.

$$\begin{aligned} G(l_H, e) := \frac{\partial \Pi_H}{\partial l_H}\Big|_{l_H = l_H^{D,SB}} &= (1-r_H)\left(f_H(l_H^{D,SB}) - w_{l_H^{D,SB},H} f_H(l_H) - \int_0^{l_H^{D,SB}} \frac{\partial w_{i,H}}{\partial l_H^{D,SB}} f_H(i)d\,i - ebr_H\right) \\ &\quad - r_H[ebr_H(1+e)] = 0 \end{aligned} \quad (49)$$

By the Implicit Function Theorem, we obtain the following equation:

$$\frac{\partial l_H^{D,SB}}{\partial e} = \frac{\frac{\partial G(l_H^{D,SB}, e)}{\partial e}}{\frac{\partial G(l_H^{D,SB}, e)}{\partial l_H^{D,SB}}} = \frac{-[-br_H(1+2r_H e)]}{(1-r_H)\left[f_H'(l_H^{D,SB})(1-w_{i,H}) - 2f_H(l_H^{D,SB})\frac{\partial w_{l_H^{D,SB},H}}{\partial l_H^{D,SB}} - \int_0^{l_H^{D,SB}} \frac{\partial^2 w_{i,H}}{\partial^2 l_H^{D,SB}} f_H(i)d\,i\right]} < 0 \quad (50)$$

Since productivity in the high-risk industry, $f_H(i)$, decreases over the unit interval, $f'(l_H) < 0$. Then, by the optimal labor supply in Equation (11) the wage offered in the high-risk industries increases linearly over the unit interval, implying that $w_H'(l_H) = \frac{\partial w_H}{\partial l_H} > 0$ and $w''(l_H) = 0$. Then, $\frac{\partial l_H^{D,SB}}{\partial e} < 0$.

A.3.3 Second-Best Experience Rating

The second best degree of experience rating is the solution of the following constrained optimization problem: the government chooses the degree of experience rating e to maximize social welfare (Equation 6) and output (Equation 7), subject to the budget constraint for allocating the tax burden between firms (Equation 9), workers' optimal labor supply (Equation 11), the low-risk firm's labor demand (Equation 12), labor market clearing in both industries, (Equation 20), and the high-risk firm's optimal layoff rate (Equation 18) and labor demand (Equation 19). The Lagrangian is:

$$
\begin{aligned}
\mathcal{L}^{SB} = & \underbrace{\int_{l_H}^{1} u(w_{i,L} f_L(i)) d\,i}_{U_{i,L} \text{ for workers in L}} + \underbrace{\int_{0}^{l_H} (1-r_H)[u(w_{i,H} f_H(i))] + r_H[u(b) + L_i]\, di}_{U_{i,H} \text{ for workers in H}} \\
& + \underbrace{\Pi_L + \Pi_H}_{U_C} + \underbrace{\int_{0}^{l_H} (1-r_H) f_H(i) d\,i + \int_{l_H}^{1} f_L(i) d\,i}_{\text{Y}} \qquad (51) \\
= & \int_{l_H}^{1} u(w_{i,L}) d\,i + \int_{0}^{l_H} (1-r_H) u(w_{i,H}) + r_H[u(b) + L_i]\, di \\
& + \underbrace{\int_{l_H}^{1} [f_L(i) - w_{i,L} f_L(i)]\; di - (1-e) b l_H r_H}_{\Pi_L} + \int_{0}^{l_H} (1-r_H) f_H(i) d\,i + \int_{l_H}^{1} f_L(i) d\,i \\
& + \underbrace{(1-r_H)\left[\int_{0}^{l_H} [f_H(i) - w_{i,H} f_H(i)]\; di - e b l_H r_H\right] + r_H[-e b l_H r_H (1+e)] - \psi(1-r_H)}_{\Pi_H}
\end{aligned}
$$

To obtain Equation (21), I take the derivative of $\mathcal{L}^{SB}$ with respect to e and set it to zero. I then divide the first-order condition by $2r_H b l_H r_H$ to isolate e on one side. Next, I substitute for e using the marginal profit gap from Equation (43). Last, I apply the definitions of elasticity, $\varepsilon_{x,e} = \frac{\partial x}{\partial e}\frac{e}{x}$, and of λ and ρ from Equations (22) and (23).

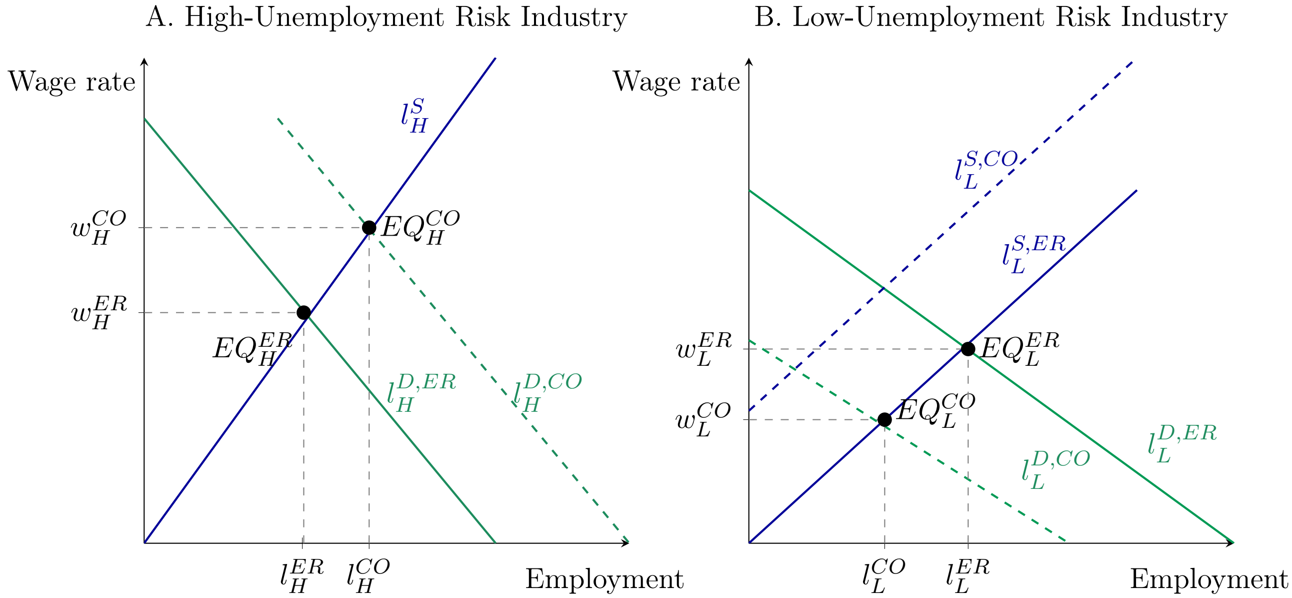
Several terms in the first-order condition cancel out due to model assumptions and an envelope condition: since workers and firms optimize, changes in the degree of experience rating do not generate additional welfare effects. In addition to the terms that cancel out in the first-order conditions for the first-best layoff rate and labor demand (Equations 39 and 40), the first-order condition for the second-best degree of experience rating also includes the first-order conditions for the second-best layoff rate and labor demand (Equations 18 and 19), which similarly cancel out. The terms corresponding to fiscal externalities and output losses appear because the budget constraint and output enter the Lagrangian.

$$
\begin{aligned}
\frac{\partial \mathcal{L}^{SB}}{\partial e} =& \overbrace{[-u(w_{l_H,L} f_L(l_H)) + (1-r_H)u(w_{l_H,H} f_H(l_H)) + r_H(u(b) + L_{l_H})]}^{=0 \text{ for Indifferent Worker (Eqn 10) + Mkt Clearing (Eqn 20)}} \frac{\partial l_H}{\partial e} \\
&+ \overbrace{\left[\int_{l_H}^{1} u'(w_{i,L} f_L(i)) \frac{\partial w_{i,L}}{\partial l_H} + \int_0^{l_H} (1-r_H) u'(w_{i,H} f_H(i)) \frac{\partial w_{i,H}}{\partial l_H}\, di\right]}^{\frac{\partial w_i}{\partial l_H} = 0 \text{ for Inframarg. W.; Terms } =0 \text{ for } l_H \text{ by Deriving Indiff. W.(Eqn 41)+Mkt Clear(Eqn 20)}} \frac{\partial l_H}{\partial e}\, di \\
&+ \int_0^{l_H} \overbrace{[-u(w_{i,H} f_H(i)) + u(b) + L_i]}^{=0 \text{ by Indiff. bw States 4}} \frac{\partial r_H}{\partial e} d\,i + \left[-(1-e) b l_H - \int_0^{l_H} f_H(i) d\,i\right] \frac{\partial r_H}{\partial e} \\
&+ \left[\overbrace{-f_L(l_H) + w_{l_H,L} f_L(l_H) - \int_{l_H}^{1} \underbrace{\frac{\partial w_{i,L}}{\partial l_H}}_{=0 \text{ except } l_H} f_L(i)\, di}^{=0 \text{ for L's Optimal Labor Demand (Eqn. 12)}} - (1-e) b r_H + (1-r_H) f_H(l_H) - f_L(l_H)\right] \frac{\partial l_H}{\partial e} \\
&+ b l_H r_H - (1-r_H) b l_H r_H - r_H b l_H r_H (1+e) - r_H e b r_H l_H \\
&+ \overbrace{[-\Pi_H^{good} + \Pi_H^{bad} - e b l_H - e b l_H r_H e + \psi'(1-r_H)]}^{=0 \text{ for H's Second-Best Layoff Rate (Eqn. 18)}} \frac{\partial r_H}{\partial e} \\
&+ \overbrace{\left[(1-r_H)\left(f_H(l_H) - w_{l_H,H} f_H(l_H) - \int_0^{l_H} \underbrace{\frac{\partial w_{i,H}}{\partial l_H}}_{=0 \text{ except } l_H} f_H(i)\, di - e b r_H\right) - r_H e b r_H (1+e)\right]}^{=0 \text{ for L's Second Best Labor Demand (Eqn 19)}} \frac{\partial l_H}{\partial e}
\end{aligned}
\tag{52}
$$

A.3.4 Labor Market Clearing from Experience Rating to Coinsurance Equilibrium

Figure A.1 shows how the labor market clears in the low- and high-unemployment risk industries when the degree of experience rating e declines. The high-risk firm's labor demand shifts outward, from $l_H^{D,ER}$ to $l_H^{D,CO}$. To attract additional workers despite their lower productivities, the firm must offer a higher marginal wage rate, $w_H^{CO} > w_H^{ER}$. At this wage, all workers $i \in [0, l_H^{D,CO}]$ prefer employment in the high-risk industry, and the market in this industry clears. This reallocation reduces labor supply to the low-risk industry, shifting it from $l_H^{S,ER}$ to $l_H^{S,CO}$. By labor market clearing (Equation 20), labor demand must correspondingly fall from $l_L^{D,ER}$ to $l_L^{D,CO}$. The resulting marginal wage rate declines from w_L^{ER} to w_L^{CO}, reflecting the higher productivity of the remaining workers and their lower requirements to work in the low-risk industry.

Figure A.1: Labor Market Clearing from Experience Rating to Coinsurance



Notes: This figure shows how labor demand and labor supply shift in the high-risk industry (Panel A) and the low-risk industry (Panel B) when moving from the second-best complete experience-rating equilibrium to the second-best coinsurance equilibrium, that is, when the degree of experience rating declines. The y-axis reports the wage rate of the marginal worker, and the x-axis the industry's employment share. Under coinsurance, the high-risk industry experiences higher wages and employment, while the low-risk industry experiences lower wages and employment compared to the complete experience rating equilibrium.

B Model Extensions

The model relies on simplifying assumptions for tractability, but its key insights remain robust when these assumptions are relaxed. I next present several model extensions.

B.1 Alternative Assumptions on Workers' Productivities

The model is derived assuming that worker productivities in the two industries are linear over the unit interval and negatively correlated. Figure B.1 illustrates alternative distributions of worker productivity to which the model is robust. Panel (a) presents the symmetric case, where f_H increases and f_L decreases over the unit interval. Panel (b) shows non-linear productivities. Panel (c) shows non-continuous productivities. Panel (d) shows non-monotonic productivities. Panel (e) shows positively correlated productivities.

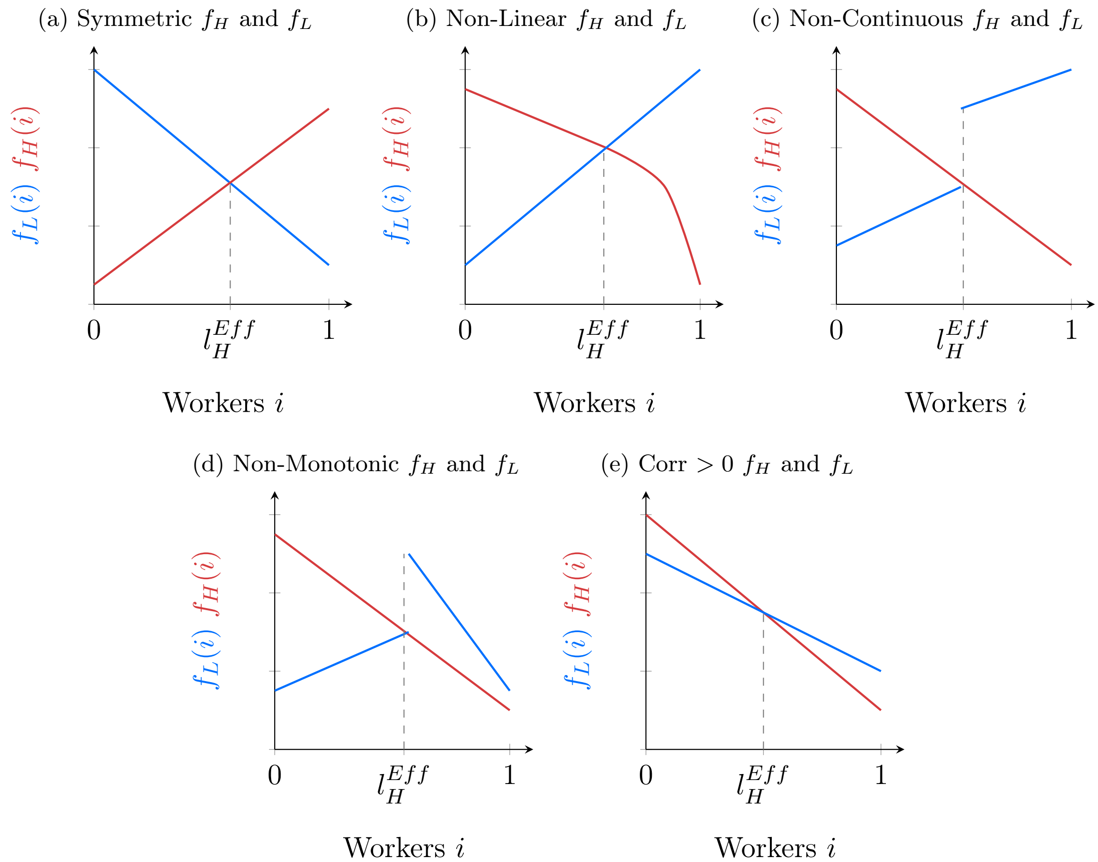
These alternative assumptions do not affect the qualitative solution of the model but may influence (i) the calibration of the formula for the optimal degree of experience rating and (ii) the prediction on equilibrium wage rates. Regarding (i), they determine whether the efficient employment level in the high-risk industry, l_H^{Eff}, is associated with a non-zero productivity gap, $(1-r_H)f_H(l^{Eff}) - f_L(l^{Eff})$. If such a gap exists, like in panels (c) and (d), it enters the formula for the second-best optimal degree of experience rating (Equation 21) as a non-zero term even when experience rating is complete (see the model solution for context), implying a residual inefficiency. In the calibration, this gap can be normalized to zero, ensuring that complete experience rating coincides with maximal efficiency. Regarding (ii), since higher productivity workers require lower wage rates, under non-monotonic productivities (case d) the wage in the high-risk industry is predicted to rise under coinsurance, opposite to the baseline model's prediction.

B.2 Payroll Unemployment Taxes

The baseline model treats unemployment taxes as head taxes, consistent with the U.S. approach to finance unemployment benefits. The framework remains equivalent if these taxes are instead modeled as payroll taxes. The high-risk firm's profits become:

$$\Pi_H \equiv \underbrace{(1-r_H)}_{\substack{\text{Prob.}\\ \text{No Shock}}} \left[\underbrace{\int_{i\in H} \overbrace{f(i)\,di}^{\text{Revenues}} - \overbrace{w_{i,H} f_H(i)}^{\text{Wages}}(1 + \overbrace{\tau_H}^{\substack{\text{Unempl.}\\ \text{Tax Rate}}})}_{\Pi_H^{good}} \right] + \underbrace{r_H}_{\substack{\text{Prob.}\\ \text{Shock}}} \underbrace{[- \overbrace{\tau_H l_H (1+e)}^{\substack{\text{Unempl. Taxes}\\ \text{with Exp. Rating}}}]}_{\Pi_H^{bad}} - \underbrace{\psi(1-r_H)}_{\text{Effort Cost}} \quad (53)$$

Figure B.1: Alternative Assumptions on Workers' Productivities



Notes: The model is solved under the assumption that workers' productivities in the high-risk industry, $f_H(i)$, decrease linearly, while those in the low-risk industry, $f_L(i)$, increase linearly over the unit interval. These productivities are illustrated in Figure I. Here, I show alternative productivity profiles to which the model is robust. Panel (a) presents the symmetric case, with $f_H(i)$ increasing and $f_L(i)$ decreasing linearly. Panel (b) shows non-linear productivity functions. Panel (c) displays non-continuous productivities. Panel (d) shows non-monotonic productivities. Panel (e) illustrates the case of positively correlated productivities across industries. For simplicity, $f_H(i)$ is not scaled by the probability of the good state, $1 - r_H$. In each panel, l_H^{Eff} denotes the efficient employment share in the high-risk industry.

The low-risk firm's profits are modified analogously. The budget constraint then becomes:

$$\underbrace{\tau_L \int_{i \in L} w_{i,L} f_L(i)\, di}_{\substack{\text{Taxes Paid by} \\ \text{Low-Risk Employer}}} + \underbrace{\tau_H \int_{i \in H} w_{i,H} f_H(i)\, di}_{\substack{\text{Taxes Paid by} \\ \text{High-Risk Employer}}} = \underbrace{b l_H r_H}_{\substack{\text{Expected Benefit} \\ \text{Expenditure}}} \tag{54}$$

Since the high-risk firm still pays a fraction e and the low-risk firm $1 - e$ of expected benefit expenditures in unemployment taxes, substituting the tax bill from the budget constraint yields the same profit function and solution as in the head-tax case.

B.3 Adjustment Costs

The baseline model assumes that experience rating affects only hiring and layoffs. However, layoff taxes may act as adjustment costs, reducing productivity by limiting firms' ability to reallocate resources efficiently (Hopenhayn et al. 1993), which can be especially detrimental for small and young firms. I introduce a layoff benefit g from layoffs, capturing productivity gains from flexibility. The high-risk firm's profits become:

$$\Pi_H \equiv \underbrace{(1-r_H)}_{\substack{\text{Prob.}\\ \text{No Shock}}} \left[\underbrace{\int_{i\in H} \overbrace{f(i)\,di}^{\text{Revenues}} - \overbrace{w_{i,H} f_H(i)}^{\text{Wages}} - \overbrace{\tau_H l_H}^{\substack{\text{Unempl.}\\ \text{Taxes}}}}_{\Pi_H^{good}} \right] + \underbrace{r_H}_{\substack{\text{Prob.}\\ \text{Shock}}} \underbrace{[- \overbrace{\tau_H l_H(1+e)}^{\substack{\text{Unempl. Taxes}\\ \text{with Exp. Rating}}} + \overbrace{g}^{\substack{\text{Productivity}\\ \text{Gain}}}]}_{\Pi_H^{bad}} - \underbrace{\psi(1-r_H)}_{\text{Effort Cost}} \tag{55}$$

In the baseline model, $g = 0$. The need for flexibility increase the marginal cost of reducing layoffs through these newly introduced productivity losses: g enters the right-hand side of Equations (13) and (18) for first- and second-best layoff rates. However, the gap between these allocations remains unchanged, as the additional term affects both equally. Consequently, the second-best degree of experience rating is unaffected relative to baseline.

B.4 Non-Wage Job Value

The baseline model assumes that workers' labor supply depend only their productivities. In practice, high-risk industries may provide additional value, or impose costs, that influence labor supply decisions, for example, by serving as entry points for young workers or by reducing mobility costs through quicker layoffs for workers accepting distant jobs (Diamond 1981). I introduce such value as an additional term V in workers' utility in the high-risk industry, $U_{i,H}$.

$$U_{i,H} \equiv \underbrace{(1-r_H)}_{\substack{\text{Prob. No}\\ \text{Shock}}} \underbrace{u(\overbrace{w_{i,H} f_H(i)}^{\text{Wage}})}_{U_{i,H}^{good}} + \underbrace{r_H}_{\substack{\text{Prob.}\\ \text{Shock}}} \underbrace{[u(\overbrace{b}^{\substack{\text{Unempl.}\\ \text{Benefit}}}) + \overbrace{L_i}^{\text{Leisure}}]}_{U_{i,H}^{bad}} + \overbrace{V}^{\text{Non-Wage Job Value}} \tag{56}$$

In the baseline model, $V = 0$. Non-wage job value lowers (or raises) the wage premium required to attract workers to the high-risk industry, reducing labor costs and increasing the first- and second-best labor demand in Equations (14) and (19) equally. Consequently, the second-best degree of experience rating is unaffected relative to baseline.

B.5 Preference for Employment

The baseline model assumes workers are indifferent between employment in the high-risk industry and unemployment (Equation 4). I relax this assumption by allowing for preferences for employment:

$$u(w_{i,H} f_H(i)) \geq u(b) + L_i, \tag{57}$$

where the baseline corresponds to equality. Since a higher layoff rate increases unemployment risk, Equation (13) for the first-best layoff rate now includes an additional benefit term, $\int_0^{l_H} [u(w_i f_H(i)) - u(b) - L_i]\, di$, capturing the increased worker utility from lower unemployment risk when the layoff rate declines. This reduces the first-best layoff rate relative to the baseline. The second-best layoff rate remains unchanged because the high-risk firm ignores workers' utilities when setting the layoff rate. Consequently, the gap between first- and second-best allocations widens, increasing the second-best degree of experience rating. The result is due to the presence of an additional term on the right-hand side of the formula for the optimal degree of experience rating (Equation 21), capturing workers' utility loss from greater unemployment risk: $\frac{\int_0^{l_H} [u(w_i f_H(i)) - u(b) - L_i]\, di}{2ebl_H r_H} \varepsilon_{r_H,e}$

B.6 Welfare Weights

The baseline model assumes equal welfare weights for workers and capitalists. I generalize it by assigning a welfare weight $\alpha_w \in [0, 1]$ to workers and $1 - \alpha_w$ to capitalists. The social welfare function becomes:

$$SWF = \alpha_w \left[\underbrace{\int_{i \in L} U_{i,L}}_{\substack{\text{Utilities of Workers} \\ \text{Employed in Industry L}}} + \underbrace{\int_{i \in H} U_{i,H}}_{\substack{\text{Utilities of Workers} \\ \text{Employed in Industry H}}} \right] + (1 - \alpha_w) \underbrace{U_C}_{\substack{\text{Capitalists'} \\ \text{Utilities}}} \tag{58}$$

In the baseline model, workers' utilities, capitalists' utilities, and output are all weighted equally. Because the terms obtained deriving workers' utilities are all equal to zero, as discussed in Appendix A, their welfare weight does not affect the solution. Conversely, the weight on capitalists' scales profits by $(1 - \alpha_w) \leq 1$, effectively inflating output by $\frac{1}{1-\alpha_w} \geq 1$ in the government's objective function. These larger output losses raise the marginal benefit of reducing the layoff rate (Equation 13) and the marginal cost of expanding labor demand for the high-risk firm (Equation 14), resulting in lower first-best layoff rate and labor demand. However, the second-best layoff rate (Equation 18) and labor demand (Equation 19) are unchanged, because the high-risk firm disregards output losses when choosing them. Consequently, the gap between first- and second-best allocations widens: the output losses λ_o (Equation 22) and ρ_f (Equation 23) are inflated by $\frac{1}{1-\alpha_w} \geq 1$ and increase the second-best experience rating in Equation (21) relative to baseline. Intuitively, the greater the welfare weight assigned to workers, the lower the value the government places on protecting firms from risk, thereby increasing the emphasis on minimizing social losses.

Relaxing worker's indifference condition between employment in the high-risk industry and unemployment in Equation (4) strenghtens the result. As discussed in Appendix B.5, in this case Equation (21) includes an additional cost term representing workers' utility loss from the increased unemployment risk. This term is non-zero and, with welfare weights, it is scaled by α_w:

$\alpha_w \frac{\int_0^{l_H}[-u(w_i f_H(i))+u(b)+L_i]\,di,}{2ebl_H r_H}\varepsilon_{r_H,e}$. A higher welfare weight on workers increases the weight placed on workers' losses upon unemployment and therefore raises the second-best degree of experience rating.

B.7 Incomplete Eligibility and Benefit Take-Up

The baseline model assumes full eligibility and take-up of unemployment benefits. However, only 43–57% of the unemployed meet eligibility criteria, and among them, 61–77% actually claim benefits (Blank et al. 1991; Auray et al. 2019; Birinci et al. 2023), implying that roughly one-third of the unemployed receive benefits. I generalize the model by assuming a take-up rate of $\alpha \in [0, 1]$. The budget contraint becomes:

$$\underbrace{\tau_L l_L}_{\substack{\text{Taxes Paid by}\\ \text{Low-Risk Employer}}} + \underbrace{\tau_H l_H}_{\substack{\text{Taxes Paid by}\\ \text{High-Risk Employer}}} = \underbrace{\overbrace{\alpha}^{\text{Take-Up}} b l_H r_H}_{\substack{\text{Expected Benefit}\\ \text{Expenditure}}} \tag{59}$$

Moreover, leisure must now compensates for the lower consumption during unemployment, making worker indifference condition in Equation (4) become:

$$u(w_{i,H} f_H(i)) = u(\overbrace{\alpha}^{\text{Take-Up}} b) + L_i \tag{60}$$

In the baseline model, $\alpha = 1$. Lower unemployment taxes reduce both the marginal benefit of lowering the layoff rate and the marginal cost of increasing labor demand for the high-risk firm, leading to higher first- and second-best layoff rates and labor demand levels (Equations 13, 18, 14, and 19). Because the gap between first- and second-best allocations remains unchanged, the second-best degree of experience rating is unaffected relative to the baseline. This occurs as α cancels out in λ_f (Equation 22) and ρ_f (Equation 23).

Benefit take-up, previously treated as exogenous, may depend on the degree of experience rating. For example, higher unemployment taxes can induce firms to reduce take-up by contesting claims more aggressively (Lachowska et al. 2020). Moreover, in contexts with informal or contract labor, higher layoff costs may discourage hiring formal workers (Bertrand et al. 2021), thereby reducing benefit eligibility. Allowing benefit take-up to decrease with the degree of experience rating ($\frac{\partial \alpha}{\partial e} < 0$) adds a new cost term to the formula for the optimal degree of experience rating (Equation 21): $-\frac{(1+r_H e^2)}{2r_H e}\varepsilon_{\alpha,e} > 0$: reducing experience rating generates a novel fiscal externality by increasing benefit take-up and eligibility, expected benefit expenditures, and taxes on firms. This additional cost term raises the second-best degree of experience rating relative to the baseline. Further relaxing workers' indifference condition between employment in the high-risk industry and unemployment in Equation (4) makes workers prefer employment to unemployment and introduces two new terms: a cost term representing workers' utility loss from higher unemployment risk (as detailed in Appendix B.5) and a new benefit term, $-\frac{u'(\alpha b)}{2r_H e}\varepsilon_{\alpha,e} > 0$, capturing the welfare gain from increased insurance when reduced experience rating leads to higher benefit eligibility and take-up. In this case, the net effect on the second-best degree of experience rating becomes ambiguous.

B.8 Corporate Taxes

The baseline model assumes that firms are only subject to payroll taxes. I introduce corporate taxation to finance other public goods. I model corporate taxes by assuming that profits generate a positive externality $t \geq 0$. The social welfare function becomes:

$$SWF = \underbrace{\int_{i \in L} U_{i,L}}_{\substack{\text{Utilities of Workers} \\ \text{Employed in Industry L}}} + \underbrace{\int_{i \in H} U_{i,H}}_{\substack{\text{Utilities of Workers} \\ \text{Employed in Industry H}}} + \overbrace{(1+t)}^{\text{Corporate Tax}} \underbrace{U_C}_{\substack{\text{Capitalists'} \\ \text{Utilities}}} \tag{61}$$

In the baseline model, $t = 0$. The weight on profits scales output down by $\frac{1}{1+t} \leq 1$ in the government's objective function. These lower output losses reduce the marginal benefit of a lower layoff rate (Equation 13) and the marginal cost of expanding labor demand for the high-risk firm (Equation 14), resulting in higher first-best layoff rate and labor demand. However, the second-best layoff rate (Equation 18) and labor demand (Equation 19) are unchanged, because the high-risk firm disregards output losses when choosing them. Consequently, the gap between first- and second-best allocations declines: the output losses λ_o (Equation 22) and ρ_o (Equation 23) are divided by $1 + t$ and reduce the second-best experience rating in Equation (21) relative to baseline. Intuitively, the greater the externality generated by profits, the higher the value the government places on bad state profits, thereby reducing the emphasis on minimizing social losses.

B.9 Externalities of the High-Risk Industry

In the baseline model, there is no reason to expand employment in the high-risk industry beyond the efficient level. In practice, however, several factors may justify expansion. High risk may deter entrepreneurship (Van Doornik et al. 2022) or induce workers to demand compensating wages (Abowd et al. 1981) that firms struggle to sustain, potentially making high-risk industries inefficiently small and justifying subsidies for their expansion (see Van Doornik et al. (2022) for unemployment insurance and Michaud et al. (2018) for disability insurance). Similarly, it may be efficient to reduce the size of the low-risk industry if it coincides with the public sector, often viewed as oversized, and if the high-risk industry coincides with the productive private sector. High-risk, high-return industries may also generate broader societal benefits: for example, technology shocks can induce layoffs but increase productivity by replacing labor with more efficient technologies. Higher complementarity between the goods produced in the two industries raise efficiency gains in the low-risk sector from expanding the high-risk sector (Michaud et al. 2018).

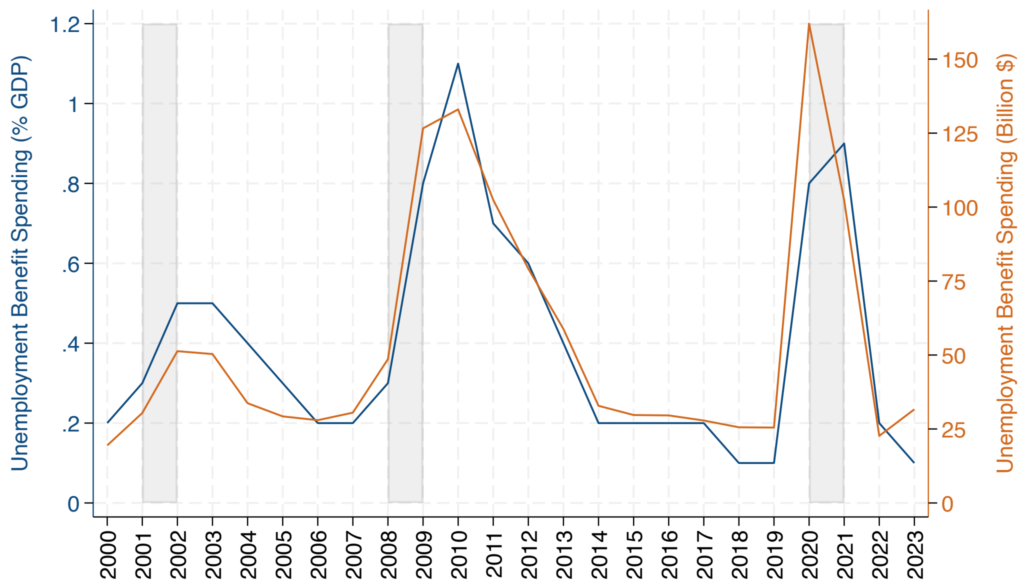
A simple way to model these scenarios is to assume that output in the high-risk industry generates a positive externality of size γ:

$$Y = \underbrace{(1 - r_H)y_H}_{\substack{\text{Output of Workers} \\ \text{in Industry H}}} \cdot \underbrace{\gamma}_{\text{Externality}} + \underbrace{y_L}_{\substack{\text{Output of Workers} \\ \text{in Industry L}}} \tag{62}$$

In the baseline model, $\gamma = 1$. Because output in the high-risk industry generates additional value, the marginal benefit of reducing the layoff rate in Equation (13) increases, leading to a lower layoff rate in the first-best allocation. At the same time, the marginal cost of increasing employment in the high-risk industry in Equation (14) declines, as hiring workers from the low-risk industry entails smaller productivity losses, leading to a higher first-best employment share in the high-risk industry. Since the high-risk firm disregards output when making decisions, the second-best layoff rate and labor demand remain unchanged. Consequently, the gap between the first- and second-best allocations widens for the layoff rate but narrows for labor demand, leading to ambiguous effects on the optimal second-best degree of experience rating. This reflects the fact that ρ_o increases in absolute value (multiplied by γ) amplifying moral hazard, while λ_o decreases, reducing labor misallocation.

C Auxiliary Figures

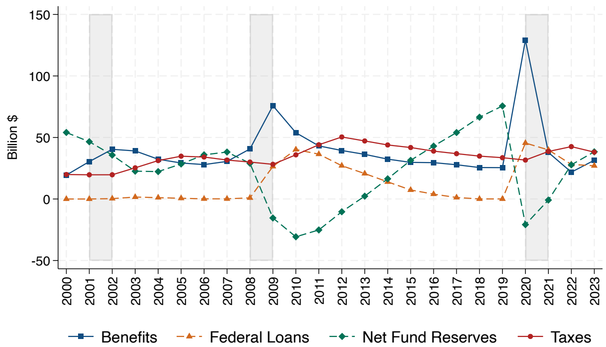
Figure C.1: Unemployment Benefit Spending Increases During Downturns



Notes: The figure shows the evolution of spending in unemployment benefits in absolute value and as a share of GDP for the United States. Benefit spending includes regular benefits, extended benefits, and emergency federal benefits. Gray areas correspond to economic recessions. Data sources: ET Financial Handbook 394, OECD, and US Business Cycle Expansions and Contractions from the NBER.

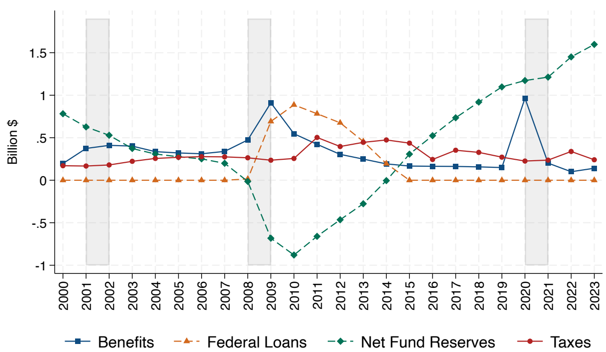
Figure C.2: Unemployment Taxes Increase During and After Downturns

(a) Benefits, Taxes and Solvency, United States

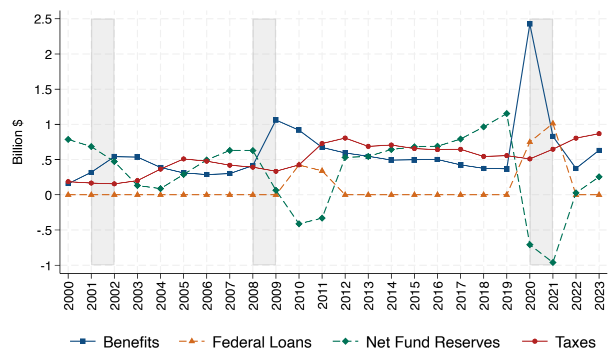


(b) Taxes as % of Total Wages

(c) Benefits, Taxes and Solvency, South Carolina

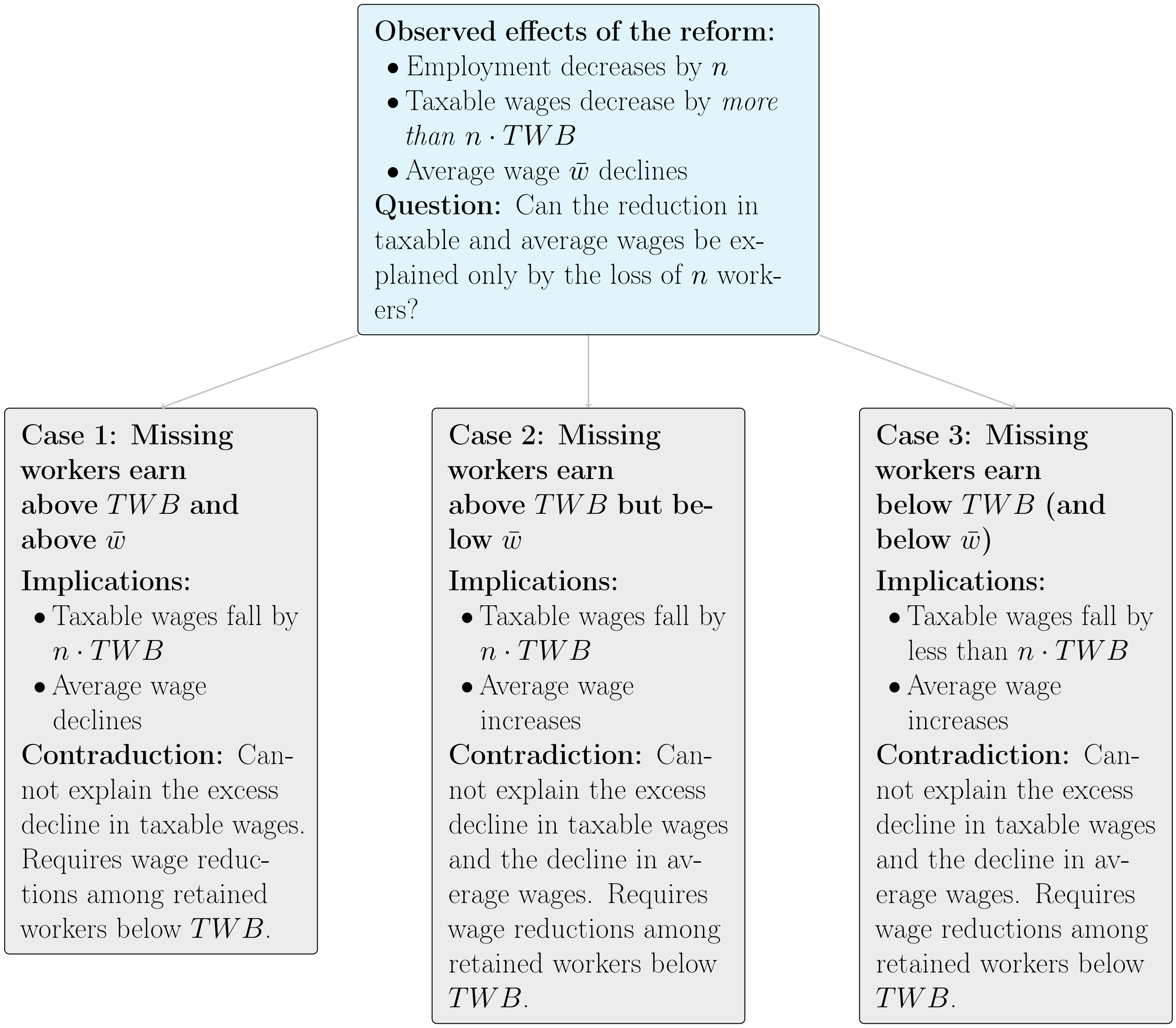


(d) Benefits, Taxes and Solvency, Colorado



Notes: Panels (a), (c), and (d) show the evolution of the total amount of unemployment benefits paid out to workers (including only state-provided benefits and excluding extended and emergency benefits), federal government loans (requested by states whenever their reserves in the Unemployment Trust Fund are insufficient to pay benefits), reserves net of loans, and unemployment taxes collected in the United States (sum of all states), South Carolina, and Colorado, respectively. Panel (b) shows the evolution of unemployment taxes, total wages, and unemployment taxes as a share of total wages. Taxes are expressed in billions of dollars, while wages are expressed in billions of dollars and divided by a factor of 100 to facilitate comparison. Gray areas correspond to economic recessions. Data sources: ET Financial Handbook 394 and US Business Cycle Expansions and Contractions from the NBER.

Figure C.3: Why Reduced Wages of Retained Workers Are Necessary



D Data Appendix

I describe the data used in the analysis, including the data provided by the South Carolina Department of Employment and Workforce, the data provided by the Colorado Department of Labor, and several aggregate public data sources.

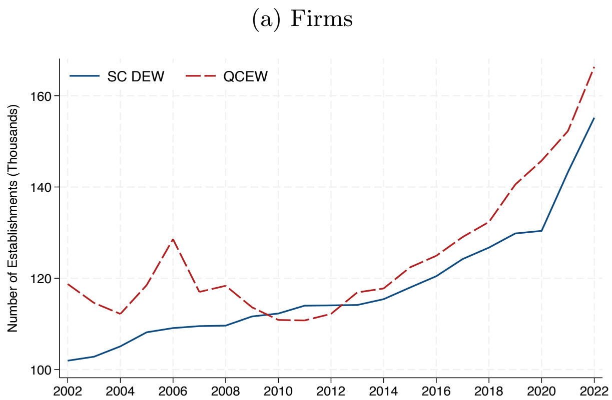
D.1 South Carolina Department of Employment and Workforce

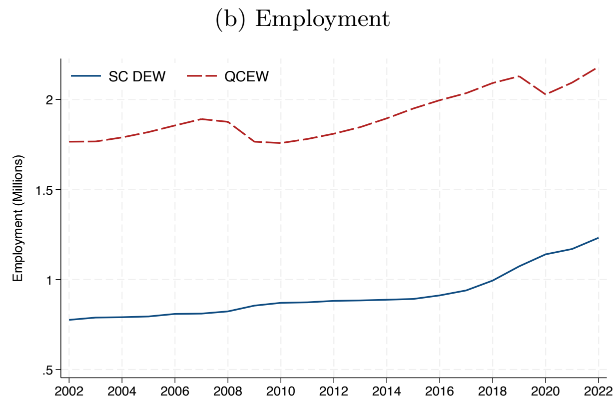
Data Description The data provided by the South Carolina Department of Employment and Workforce (SC DEW) cover the near universe of firms (=employers) operating in the state between 2002 and 2022, including 358,758 employers and 9,903,144 employer-quarter observations. Data are unavailable before 2002 and largely incomplete before 2007. The series ends in 2022 due to the timing of the data request in 2023.

South Carolina employers submit quarterly reports including monthly employment counts and individual gross wages paid to each worker. SC DEW uses this information and benefit charges from workers' claims to calculate annual reserve ratios (until 2010) or benefit ratios (from 2011), and unemployment tax rates, which are multiplied by quarterly taxable wages to calculate quarterly unemployment taxes. The data thus include quarterly employment, total, minimum, average, and maximum wages, taxable wages, and benefit charges; annual reserve ratios (until 2010), benefit ratios (from 2011) and unemployment tax rates; NAICS four-digit sector codes and date of establishment.

Data Representativeness To ensure confidentiality and prevent identification, the largest firms were excluded from the data. Following SC DEW's disclosure procedure, all the firms with more than 211 employees in any given year were removed, so that the excluded firms represent the top 1% largest employers in 2002, leaving a dataset covering approximately 99% of firms in the state. To assess the implications of this sample restriction for state-level representativeness, Figure D1 compares the number of firms and employment between the SC DEW data and the Quarterly Census of Employment and Wages (QCEW): SC DEW covers on average 95% of the state firms (panel [a]), representing on average 48% of state employment (panel [b]). SC DEW has confirmed that the top 1% largest firms account for 47% of employment in the state. Table D1 shows that the distribution of firms and workers across industries is similar between SC DEW and QCEW. SC DEW slightly over-represents primary sector and construction firms but underrepresents public administrations. Average wages are higher in the QCEW data for construction, manufacturing, FIRE, and public administration, but lower for trade and transport. These differences are possibly driven by a small number of large firms in these sectors. Overall, excluding the largest firms does not appear to substantially affect the representativeness of the SC DEW data.

Figure D1: Employment Coverage of the SC DEW Dataset





Notes: The figure compares the number of firms (panel [a]) and workers (panel [b]) in South Carolina between the data provided by the South Carolina Department of Employment and Workforce (SC DEW) and the Quarterly Census of Employment and Wages (QCEW).

Table D1: Sectoral Representativeness of the SC DEW Dataset

	% Firms		*% Workers*		*Average yearly wage*	
	SC DEW	QCEW	SC DEW	QCEW	SC DEW	QCEW
Primary	.011	.010	.014	.007	\$30,836	\$30,516
Construction	.132	.101	.086	.046	\$31,526	\$41,030
Manufacturing	.043	.043	.094	.118	\$40,715	\$49,568
Trade and Transport	.224	.252	.207	.194	\$41,177	\$32,460
FIRE	.086	.096	.059	.053	\$43,789	\$47,185
Services	.502	.482	.536	.519	\$34,921	\$35,290
Public Administration	.002	.016	.004	.063	\$25,475	\$39,221

Notes: The table compares the share of firms, the share of workers, and average wages across economic sectors between South Carolina Department of Employment and Workforce (SC DEW) data and the Quarterly Census of Employment and Wages (QCEW) in 2010. Average wages are calculated as the ratio of total annual wages and average annual employment. Sectors are defined based on their NAICS 2-digit industry codes: primary (11, 21), construction (23), manufacturing (31–33), trade and transport (42, 44, 48, 49), FIRE (finance, insurance, and real estate, 52, 53), services (22, 51, 54, 55, 56, 61, 62, 71, 72, 81), and public administration (92).

Data Cleaning As noted in Anderson et al. (1997), unemployment tax filing data contain errors. I clean such potential errors arising from manual data entry by firms, such as adding or omitting a digit, or entering wages instead of employment counts. The procedures target extreme outliers, remove noise, limit researcher discretion, and are validated through systematic checks.

Taxable wages: Quarterly taxable wages are missing for 17% of observations (firm–quarters). I impute missing values using a variable measuring annual taxable wages, or by reconstructing annual taxable wages as the ratio of unemployment taxes to the unemployment tax rate. The latter method is feasible only for 2006–2009, as unemployment taxes for these years were included in the files used to calculate reserve ratios in 2007–2010. For example, the file containing 2010 reserve ratios reports unemployment taxes paid between 2008 Q2 and 2009 Q2. Since the taxable wage base—the portion of a worker's earnings subject to taxation—was at most $12,000 and thus much lower than average wages during the study period (see panel [b] of Figure E1), most wages of workers hired in the first quarter become non-taxable by the second quarter, and most unemployment taxes are paid in the first quarter of the year. I thus use taxes paid between 2008 Q2 and 2009 Q2 as proxy for taxes in 2009, and divide these taxes by the 2009 tax rate to approximate taxable wages for that year. Imputed annual taxable wages are assigned to the first quarter of each year; quarterly taxable wages are then set to zero in the other quarters. After these replacements, missing observations fall to 9%, of which 58% belong to the last year in which an employer is observed. Additionally, 101 extreme cases where taxable wages were at least ten times larger than adjacent quarters and more than twice as large as the second-highest observed value for a firm, were replaced with the average of taxable wages in the same quarter of adjacent years.

Employment: Quarterly employment is missing in 24% of observations. I impute missing values using a variable containing the average of monthly employment in the fourth quarter and a variable measuring the number of workers for whom wages were paid each quarter, both of which closely match employment when non-missing. After these replacements, the share of missing observations falls to 19.5%, of which 45% belong to the last year in which a firm is observed.

I then correct implausible employment values, including: cases where employment is at least ten times larger or smaller than in adjacent quarters and the same quarter of adjacent years, or at least ten times larger or smaller in one quarter and seven times in the other; cases where employment is at least 200 and the implied taxable wages per worker are less than $100, corresponding to an annual wage below $100 per employee; and cases where employment is substantially higher than typical for firms that historically reported only one or two employees and appear to have maintained the same workforce, identified by taxable wages equal to the taxable wage base in the first quarter and zero in the remaining quarters. Remaining implausible cases are manually corrected. Errors are replaced with either the average employment in adjacent quarters or the firm's modal employment. In total, 82,900 replacements are made (0.8% of the data).

Below, I report examples of corrections of quaterly employment values:

..., 132, 138, 129, 127, 136, 133, 1332, 140, 161, 166, 165, ...	→ Replace with 136.5
..., 1, 1, 1, 1, 1, 1, 3120, 1, 1, 1, 1, 1, ...	→ Replace with 1
..., 10, 10, 11, 12, 12, 12, 12, 112, 12, 10, 10, 10, 10, ...	→ Replace with 12
..., 210, 207, 190, 169, 167, 0, 179, 160, 177, 177, 178, ...	→ Replace with 173
..., 57, 60, 57, 59, 62, 0, 61, 60, 62, 63, 69, ...	→ Replace with 61.5

Wages: Quarterly wage variables, that is, total, minimum, average, and maximum wages, are missing in 32% of the observations. Because these variables are non-missing only when employment is positive, I make them unconditional by setting them to zero when employment is zero. When annual wage data are available from a separate variable, I impute missing quarterly wages as the difference between annual wages and the sum of observed quarterly wages within the year, distributing the residual equally across quarters with missing wages and positive employment. Missing average wages are then replaced with the ratio of total quarterly wages to quarterly employment. After these adjustments, the share of missing observations falls to 27% for minimum and maximum wages, 20% for average wages, and 9.9% for total quarterly wages.

I then correct implausible wage values. Single occurrences for a firm in which at least three employees are reported in a quarter but total wages are equal to the minimum, average, and maximum wages are replaced with the average of these wages in adjacent quarters. I also correct extreme deviations without corresponding employment changes, including cases in which wages are at least twenty times higher than in adjacent quarters and ten times higher than the second-largest value for a firm, or twenty times higher than in adjacent quarters and two quarters apart, and cases in which wages are positive but at least fifty times smaller than in adjacent quarters and in the same quarter of adjacent years. This latter adjustment is not applied to minimum quarterly wages, where the number of replacements was substantially higher than for other wage variables, suggesting that genuine low wage values were mistakenly treated as errors. In total, 1,872, 2,544, 1,761, and 2,228 replacements are made for total, minimum, average, and maximum quarterly wages, respectively, each representing less than 0.05% of the data.

Winsorization Following common practice (Lachowska et al. 2018, Johnston 2021), I address outliers and any remaining residual errors by winsorizing the key variables used in the analysis, which would otherwise introduce substantial noise into the estimates. Total wages, taxable wages, and unemployment taxes, variables considered by SC DEW reliable and subject to few errors, are winsorized at the 99.9^{th} percentile. Quarterly employment, considered the least accurate measure, is winsorized at the 99^{th} percentile, while average annual employment, which exhibits less extreme variation, is winsorized at the 99.9^{th} percentile. Annual average wages, calculated as total wages divided by average annual employment, are winsorized at the 99^{th} percentile. For wage distribution variables, the maximum wage is winsorized at the 99.9^{th} percentile, and the minimum and mean wages at the 99^{th} percentile. When outcomes are scaled by their 2010 level, winsorization is applied at the 99.9^{th} percentile, except for the average wage and taxable wages, which are winsorized at the 99^{th} percentile. Results remain qualitatively similar with less strict or no winsorization, although estimates are noticeably noisier.

Sample The study sample includes 40,272 private sector experience-rated firms continuously observed between 2007 and 2014. The sample excludes three categories of firms, based on information provided by SC DEW and the South Carolina statute, SC Code of Laws 41-31: (i) public-sector firms, identified by NAICS 2-digit code 92; (ii) new firms, assigned a fixed unemployment tax rate (3.4% until 2010, or class 12 rate from 2011) until they have operated for at least 12 months; and (iii) firms missing in any quarter between 2007 and 2014 or present throughout but with missing employment values in their final year in the data.

Table D2 compares firms in the study sample with excluded firms in 2010. Consistent with the exclusion of new firms, study sample firms have been active for 11 additional years on average. Because firms going out of business between 2007 and 2014 are excluded, the sample is positively selected: included firms have, on average, 5 more employees, pay \$5,000 more in average annual wages, and were charged \$178 less in unemployment benefits per worker. Sectoral representation is broadly similar. Data quality increases substantially within the sample. Missing values for employment, total wages, and taxable wages are nearly nonexistent, and the rate of identified and cleaned errors remains minimal, reaching at most 2.2% of observations for employment.

Table D2: SC DEW Study Sample Description and Representativeness

	Study Sample			***Excluded Firms***			***Comparison***	
	N	Mean	Std. dev	N	Mean	Std. dev.	Diff.	p-value
Panel A: Firm Characteristics								
Employment	40,201	11.224	21.054	54,645	5.971	32.501	-5.253	0.000
Average Wage	39,607	39,343.614	56,999.720	52390	34,621.641	56264.835	-4,721.973	0.000
Benefit Charges per Worker	39,607	358.762	1,731.787	52,390	536.499	4,256.556	177.737	0.000
Reserve Ratio	40,219	-0.088	1.132	69,694	-0.017	1.177	0.070	0.000
Unemployment Tax Rate	40,272	0.020	0.013	72,002	0.028	0.013	0.007	0.000
Year of Establishment	40,272	1993.311	10.948	71,890	2004.059	7.893	10.748	0.000
Primary	40,257	0.015	0.123	71,636	0.009	0.094	-0.007	0.000
Construction	40,257	0.125	0.331	71,636	0.136	0.343	0.011	0.000
Manufacturing	40,257	0.056	0.230	71,636	0.036	0.187	-0.020	0.000
Trade and Transport	40,257	0.212	0.409	71,636	0.188	0.391	-0.025	0.000
FIRE	40,257	0.025	0.156	71,636	0.029	0.167	0.004	0.000
Services	40,257	0.566	0.496	71,636	0.600	0.490	0.033	0.000
Panel B: Data Quality								
1(Employment Missing)	40,272	0.002	0.042	72,002	0.241	0.428	0.239	0.000
1(Total Wages Missing)	40,272	0.000	0.000	72,002	0.141	0.348	0.141	0.000
1(Taxable Wages Missing)	40,272	0.000	0.000	72,002	0.141	0.348	0.141	0.000
1(Employment Error)	40,272	0.022	0.147	72,002	0.016	0.126	-0.006	0.000
1(Total Wages Error)	40,272	0.000	0.011	72,002	0.000	0.016	0.000	0.158
1(Taxable Wages Error)	40,272	0.000	0.005	72,002	0.000	0.000	-0.000	0.181

Notes: The table reports summary statistics and tests for differences in key characteristics measured in 2010 between firms in the study sample and excluded firms from the SC DEW data. Employment and the average wage are defined Table D3. The reserve ratio is defined in Equation (26) and is winsorized at the 0.1 and 99.9 percentiles. Sectors are defined based on their NAICS 2-digit industry codes: primary (11, 21), construction (23), manufacturing (31–33), trade and transport (42, 44, 48, 49), FIRE (finance, insurance, and real estate, 52, 53), services (22, 51, 54, 55, 56, 61, 62, 71, 72, 81), and public administration (92). The indicators for missing values and corrected errors of key outcomes refer to 2010Q3. The table excluded 246,274 firms who do not appear in 2010.

Variables Creation Table D3 describes the construction of key variables employed in the analysis based on the SC DEW data.

Table D3: Construction of Employer Outcomes and Treatment in the SC DEW Dataset

Variable	**Frequency**	**Construction**
Panel A: Outcomes		
Unemployment Tax per Worker	Annual	Product of the unemployment tax rate and the taxable wage base.
Employment	Quarterly	Provided.
Total Wages	Annual	Sum of quarterly total wages within each year.
Average Wage	Annual	Ratio of total annual wages to average annual employment.
Taxable Wages	Annual	Sum of quarterly taxable wages within each year.
Unemployment Taxes	Annual	Sum of quarterly unemployment taxes within each year. Quarterly unemployment taxes are calculated as the product of the annual unemployment tax rate and quarterly taxable wages.
Simulated Unemployment Taxes	Annual	Until 2010, equal to true unemployment taxes. From 2011 on, calculated as product of the simulated unemployment tax rate and simulated taxable wages. Simulated tax rates are determined from the schedule in place and employers' simulated benefit ratios each year. Simulated benefit ratios in each year are calculated following Equation (26) as the ratio of true benefit charges during the lookback period and simulated taxable wages. Until 2010, simulated taxable wages are equal to true taxable wages. From 2011 on, they are equal to taxable wages in 2010, scaled by the growth rate of the taxable wage base in each year relative to 2010. Firm specific trends are unnecessary as they would be absorbed by employer fixed effects.
Panel B: Treatment and Identification		
Benefit Ratio Class 2011	Constant	Values 1-20 assigned based on employers' 2011 benefit ratios. The range of benefit ratios corresponding to each class is reported on the SC DEW website (accessible only from within the US).
Predicted Reserve Ratio 2011	Constant	Calculated following Equation (26) as the ratio of firms' reserves as of 2010 Q2 to taxable wages between 2009 Q3 and 2010 Q2. Reserves as of 2010 Q2 are calculated as the sum of reserves as of 2009 Q2 (provided) plus unemployment taxes and minus benefit charges between 2009 Q3 and 2010 Q2.
Predicted Unemployment Tax Rate 2011:	Constant	Rate that would have been assigned based on firms' predicted 2011 reserve ratio under the 2010 tax schedule.
Predicted Δ Unemployment Tax Rate 2011	Constant	Difference between the 2011 true unemployment tax rates (based on firms' benefit ratios) and the 2011 predicted rates (based on firms' predicted reserve ratios).
Predicted Δ Unemployment Tax per Worker 2011	Constant	Product of predicted Δ unemployment tax rate and taxable wage base in 2011 ($10,000).

D.2 Colorado Department of Labor and Employment

Data Description The data provided by the Colorado Department of Labor and Employment (CO DLE) covers the universe of 366,125 firms (=employers) operating in the state between 2013 and 2022. Data are unavailable before 2013 because earlier records were deleted. The dataset ends in 2022 due to the timing of the data request in 2023.

Colorado employers submit quarterly reports including monthly employment counts and individual gross wages paid to each worker. CO DLE uses this information and benefit charges from workers' claims to calculate annual reserve ratios and unemployment tax rates, which are multiplied by quarterly taxable wages to determine quarterly unemployment taxes. Consequently, the data provided includes monthly employment; quarterly total wages, taxable wages, unemployment taxes, and benefit charges; annual unemployment tax rates and reserve ratios; the NAICS 6-digit sector code and date of estalishment.

Data Cleaning As noted in Anderson et al. (1997), unemployment tax filing data contain errors. I clean such potential errors arising from manual data entry by firms, such as adding or omitting a digit, or entering wages instead of employment counts. The procedures target extreme outliers, remove noise, limit researcher discretion, and are validated through systematic checks.

Employment: I correct implausible monthly employment values, including: cases in which employment is at least ten times larger or smaller than in the two preceding months, in the two subsequent months, and than in the same month of adjacent years; cases in which the implied taxable wages per worker are below \$200, corresponding to an annual wage of less than \$200 per employee; cases in which employment is ten times smaller in all months of a quarter than in adjacent quarters, with employment in adjacent quarters exceeding 100 employees and wages no more than twenty percent lower than in adjacent quarters; and cases in which employment is reported as zero but exceeds ten employees in adjacent months and gross wages are no more than twenty percent lower than in adjacent quarters. Remaining implausible cases are corrected manually. Errors are replaced with either the average employment in adjacent months or quarters, or with modal employment. In total, 9,143 firm–month observations are corrected (0.04% of the data). Below, I report examples of corrections of monthly employment values:

$\ldots, 3, 3, 3, 3, 3, 33, 3, 3, 3, 3, \ldots$ → Replace with 3

$\ldots, 216, 216, 210, 214, 211, 215, 2163, 213, 217, 215, 215, 215, \ldots$ → Replace with 214

$\ldots, 1, 1, 1, 1, 1, 0, 0, 25000, 1, 1, 1, 1, 1, 1, 1, 1, \ldots$ → Replace with 0

$\ldots\ 92, 90, 89, 90, 90, 91, 90, 0, 91, 93, 94, 90, 90, 90, 89, \ldots$ → Replace with 90.5

$\ldots, 403, 397, 393, 395, 401, 400, 405, 3, 3, 3, 421, 434, 431, 434, \ldots$ → Replace with 415.33

Total wages: I correct implausible values for total quarterly wages, including single cases in which firms display: wages at least ten times larger than in adjacent quarters and more than twice as large as the second-highest wage value, and employment no more than twenty percent higher; wages at least ten times smaller than in adjacent quarter, while employment has decreased by at most twenty percent; and employment exceeding two employees but wages equal to zero. Errors are replaced with the average from adjacent quarters. In total, 1,017 firm–quarter observations are corrected (0.015% of the data).

Taxable wages and Unemployment Taxes: I correct cases in which annual taxable wages divided by the sum of quarterly employment in the year (typically higher than the number of distinct employees in the year) is higher than the taxable wage base, and quarterly taxable wages are at least five times larger than in the same quarter of adjacent years. Errors are replaced with the average of taxable wages in the same quarter of adjacent years. 271 firm-quarter observations are corrected (0.004 percent of the data). Unemployment taxes are recalculated as the product of the corrected taxable wages and the unemployment tax rate. I do not recalculate reserve ratios, tax rates and unemployment taxes in following years.

Winsorization Following common practice (Lachowska et al. 2018, Johnston 2021), I address outliers and any remaining residual errors by winsorizing the key variables used in the analysis, which would otherwise introduce substantial noise into the estimates. Total wages, taxable wages, and unemployment taxes, variables generally considered reliable and subject to few errors are winsorized at the 99.9^{th} percentile. Employment, which unemployment agencies consider the least accurate measure, and average wages, constructed as the ratio of total wages to employment, are winsorized at the 99^{th} percentile. The same thresholds apply for variables scaled by their pre-reform (2017) level and when quarterly employment is averaged over the year. Results are qualitatively similar with less strict or without winsorization, but they are noticeably noisier.

Sample The study sample includes 69,776 private-sector experience-rated firms observed continuously between 2014 and 2021. The sample excludes three categories of firms, based on information provided by CO DLE and in the Colorado statute CO Rev Stat § 8-76-102.5: (i) public-sector firms (NAICS two-digit code 92); (ii) non–experience-rated firms, including reimbursable firms (tax rate of zero), group-rated firms such as political subdivisions (tax rates of 0.003% or 0.002%), and new firms, who are assigned a fixed, unrated tax rate during their initial months of operation; for new firms, both the rate and duration of the non–experience-rated period vary by sector and are higher for construction; and (iii) firms missing in any quarter between 2014 and 2021.

Table D4 compares firms in the study sample with excluded firms in 2017. Consistent with the exclusion of new firms, study sample firms have been active for ten more years. Because firms going out of business between 2014 and 2021 are excluded, the sample is positively selected: included firms have, on average, ten more employees, pay $4,000 more in average annual wages, and were charged

$59 less in benefit charges. Sectoral representation is similar. Data quality is high.

Table D4: CO DLE Study Sample Description and Representativeness

	Study Sample			***Excluded Employers***			***Comparison***	
	N	Mean	Std. dev	N	Mean	Std. dev.	Diff.	p-value
Panel A: Employer Characteristics								
Employment	69776	21.003	176.280	97189	10.796	142.167	-10.207	0.000
Average Annual Wage	69415	65826.169	96276.960	94395	61827.165	96753.758	-3999.004	0.000
Benefit Charges per Worker	69415	175.533	1761.989	94395	234.088	3320.400	58.555	0.000
Reserve Ratio	69776	-0.082	0.921	97189	-0.034	0.891	0.048	0.000
Unemployment Tax Rate	69776	0.019	0.021	97189	0.025	0.018	0.006	0.000
Year of Establishment	69776	2001.754	10.843	97189	2012.190	8.319	10.436	0.000
Primary	69775	0.019	0.138	96872	0.016	0.124	-0.004	0.000
Construction	69775	0.111	0.314	96872	0.111	0.315	0.001	0.694
Manufacturing	69775	0.042	0.202	96872	0.026	0.158	-0.017	0.000
Trade and Transport	69775	0.175	0.380	96872	0.146	0.353	-0.029	0.000
FIRE	69775	0.116	0.321	96872	0.098	0.297	-0.019	0.000
Services	69775	0.536	0.499	96872	0.596	0.491	0.060	0.000
Public Administration	69775	0.000	0.000	96872	0.007	0.086	0.007	0.000
Panel B: Data Quality								
1(Employment Missing)	69776	0.000	0.000	97189	0.000	0.000	0.000	.
1(Total Wages Missing)	69776	0.000	0.000	97189	0.000	0.000	0.000	.
1(Taxable Wages Missing)	69776	0.000	0.000	97189	0.000	0.000	0.000	.
1(Employment Error)	69776	0.002	0.040	97189	0.001	0.032	-0.001	0.001
1(Total Wages Error)	69776	0.000	0.005	97189	0.000	0.012	0.000	0.025
1(Taxable Wages Error)	69776	0.000	0.005	97189	0.000	0.008	0.000	0.336

Notes: This table reports summary statistics and tests for differences in relevant characteristics measured in 2017 between firms in the study sample and excluded employers from the CO DLE data. Average annual employment and wage are defined in Table D5. The average annual wage is winsorized to the 99.9^{th} percentile to limit the influence of outliers. Benefits per worker are the ratio of total benefit charges and average annual employment. The reserve ratio is defined in Equation 26. Sector classifications are as follows: Primary sector—NAICS 11 and 21; Construction—23; Manufacturing—31; Trade and Transport—42, 44, 48; FIRE—52 and 53; Services—22, 51, 54, 55, 56, 61, 62, 71, 72, and 81; Public Administration–92. The indicators for missing values and corrected errors of key outcomes refer to the third quarter of the year. The table excludes 186,607 firms who do not appear in 2017.

Variables Creation Table D5 details the construction of key variables employed in the analysis based on the CO DLE data.

Table D5: Costruction of Employer Outcomes and Treatment in the CO DLE Dataset

Variable	Frequency	Construction
Panel A: Outcomes		
Unemployment Tax per Worker	Annual	Product of the unemployment tax rate and the taxable wage base.
Employment	Quarterly	Average of monthly employment within each quarter.
Total Wages	Annual	Sum of quarterly total wages within each year.
Average Wage	Annual	Ratio of total annual wages to average annual employment.
Taxable Wages	Annual	Sum of quarterly taxable wages within each year.
Unemployment Taxes	Annual	Sum of quarterly unemployment taxes within each year.
Simulated Unemployment Taxes	Annual	Until 2017, equal to true unemployment taxes. From 2018 onward, calculated as the product of the simulated unemployment tax rate and simulated taxable wages. Simulated tax rates are determined from the schedule in place and firms' simulated reserve ratios each year. Simulated reserve ratios are calculated following Equation (26) as the ratio of simulated reserves and simulated taxable wages. Simulated reserves are calculated using true benefit charges and a combination of true (until 2017) and simulated (from 2018 on) unemployment taxes. Until 2017, simulated taxable wages are equal to true taxable wages. From 2018 onward, they are equal to taxable wages in 2017 scaled by the growth rate of the taxable wage base. Firm-specific trends are unnecessary as they would be absorbed by firm fixed effects.
Panel B: Treatment and Identification		
Benefit Ratio	Annual	For year t, ratio of benefit charges to taxable wages cumulated between years $t-4$ and $t-1$
Predicted Unemployment Tax Rate 2018	Constant	Rate that would have been assigned to firms in 2018 had their reserve ratios remained constant to their 2017 levels. Defined in Equation (29).
Predicted Δ Unemployment Tax Rate 2018	Constant	Difference between the predicted unemployment tax rate in 2018, based on the 2017 reserve ratio and the 2018 schedule without the surcharge, and the true unemployment tax rate in 2017, based on the 2017 reserve ratio and the 2017 schedule with the surcharge. Defined in Equation (30).
Predicted Δ Unemployment Tax per Worker 2018	Constant	Product of the Predicted Δ Unemployment Tax Rate and the taxable wage base in 2018 ($12,600).

D.3 Public Use Data

The analysis relies on several public data sources:

QCEW The Quarterly Census of Employment and Wages (QCEW) data are available here. I extract state-year and state-year–industry (NAICS 2-digit code) data on employment and employer counts, state-quarter–industry (NAICS 6-digit code) data on employment, average weekly wages, taxable wages, and unemployment taxes.

ETA 204 Report The Experience Rating Report 204 from the U.S. Department of Labor is available here. I extract state-year–level information on non-charged, inactively charged, and ineffectively charged unemployment benefit payments.

ETA 394 Handbook The Annual Program and Financial Data Handbook 394 from the U.S. Department of Labor is available here. I extract state-year–level data on unemployment taxes, unemployment benefit payments, federal government loans, trust fund net reserves, total wages paid in covered employment, taxable wage bases, average weekly wages, average weekly benefit amounts, and average actual benefit duration in weeks.

DOL UI Data Summary The Unemployment Insurance Data Summary is available here. I extract state-quarter level data on benefit recipiency rate.

U.S. Unemployment Insurance Financing Policy Database This dataset is currently under development and will be made publicly accessible in the near future. It includes harmonized historical information on the unemployment tax rate schedules in place in each U.S. state over recent years.

E Additional Tables and Figures: South Carolina

Table E1: Responses to and Incidence of Unemployment Taxation in South Carolina

	(1) Unempl. Tax Per Worker	(2) Employ-ment	(3) Total Wages	(4) Average Wage	(5) Taxable Wages	(6) Unempl. Taxes	(7) Sim. Unempl. Taxes
2007 × Pred Δ UTPW	0.0275***	-5.78e-05	-6.123	2.266*	-2.958	-0.431***	-0.432***
	(0.00426)	(0.000374)	(18.24)	(1.209)	(3.415)	(0.128)	(0.129)
2008 × Pred Δ UTPW	0.0349***	-7.05e-05	-14.80	-0.197	-4.109	-0.192	-0.199
	(0.00411)	(0.000284)	(15.51)	(1.119)	(4.067)	(0.171)	(0.176)
2009 × Pred Δ UTPW	0.0217***	7.09e-07	-0.808	0.945	-1.630	0.152*	0.150*
	(0.00364)	(0.000201)	(9.351)	(0.931)	(2.122)	(0.0899)	(0.0911)
2011 × Pred Δ UTPW	0.742***	-0.000445**	-17.23**	-0.278	-12.99***	4.423***	4.765***
	(0.00632)	(0.000196)	(8.583)	(0.934)	(2.592)	(0.193)	(0.192)
2012 × Pred Δ UTPW	0.303***	-0.00130***	-61.23***	-2.629**	-29.26***	0.437*	0.237
	(0.0118)	(0.000328)	(17.21)	(1.029)	(4.825)	(0.241)	(0.181)
2013 × Pred Δ UTPW	0.235***	-0.00148***	-64.76***	-1.979	-29.29***	0.0808	0.254
	(0.0113)	(0.000344)	(17.31)	(1.204)	(4.912)	(0.270)	(0.182)
2014 × Pred Δ UTPW	0.147***	-0.00167***	-90.04***	-2.709**	-33.52***	-0.281	-0.185
	(0.0188)	(0.000324)	(17.09)	(1.250)	(4.960)	(0.264)	(0.247)
Observations	306,904	306,902	306,904	301,192	306,904	306,904	306,904
R-squared	0.772	0.875	0.880	0.861	0.831	0.719	0.767
P-val Test 2007-09=0	0.000	0.990	0.467	0.0928	0.747	0.000	0.000
Avg Effect 2011-14	0.357	-0.001	-58.32	-1.899	-26.26	1.165	1.268
Avg Effect Std. Err	0.008	0.000	13.03	0.906	3.866	0.196	0.167
P-val Test Avg Effect=0	0.000	0.000	0.000	0.036	0.000	0.000	0.000
Full Sample Mean 2007-10	134.9	11.71	424,716	37,732	95,820	1,973	1,973

Notes: The table presents the estimated β_y coefficients from Equation (27) for firms in the SC DEW study sample with less than 50 employees on average in 2010. 50 is the 95^{th} percentile of the employment distribution in 2010. Appendix D.1 provides details on the construction, cleaning and winsorization of the outcomes. Corresponding estimates and 95% confidence intervals are shown in Figure IV. *** p<0.01, ** p<0.05, * p<0.1.

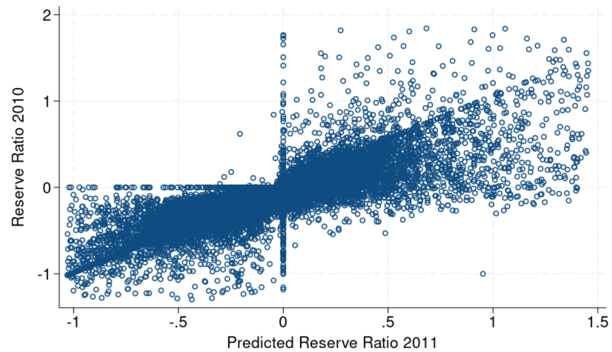
Table E2: Addressing Dose-Related Treatment Effect Heterogeneity in South Carolina

	(1)	(2)	(3)	(4)	(5)	(6)
	Δ Unempl. Tax Per Worker		Δ Employment		Δ Average Wage	
Treated	-130.6***	-47.07***	0.657***	0.121**	-756.3	552.6**
	(3.044)	(1.482)	(0.102)	(0.0498)	(528.6)	(257.1)
Observations	306,904	306,904	306,904	306,904	306,120	306,120
R-squared	0.534	0.533	0.013	0.013	0.004	0.004
Predicted Δ UTPW Control	[-10, 10]	[-100,100]	[-10,10]	[-100,100]	[-10,10]	[-100,100]
Share Control	2.36	24.96	2.36	24.96	2.36	24.96
Mean Outcome 2007-10	134.6	134.6	8.173	8.173	39716	39716

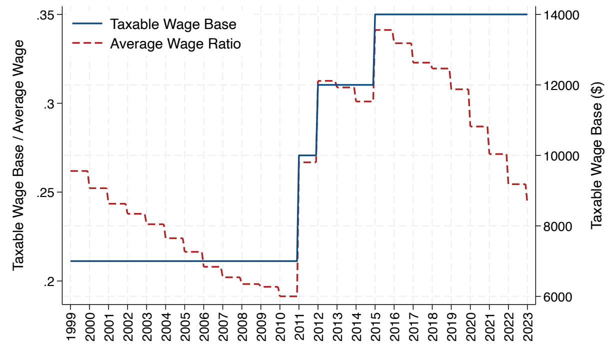
Notes: Following Callaway et al. (2024) (Equation 4.13 on page 26 of the current version of the manuscript), I regress the change in key outcomes between the pre-reform (2007–2010) and post-reform (2011–2014) periods on an indicator for having a non-zero predicted change in the unemployment tax per worker, controlling for benefit-ratio class fixed effects. To avoid pooling firms with positive and negative treatment, I take the negative of all outcomes for positive-treatment units, so that untreated units are compared with units experiencing a negative treatment. Because no firm has a predicted change in the tax per worker equal to zero, I define untreated units as those with an absolute predicted change below 10 (2.36% of the sample, columns 1-3-5) or below 100 (25% of the sample, columns 2-4-6). Outcomes are winsorized at the 0.1 and 99.9 percentiles. *** p<0.01, ** p<0.05, * p<0.1.

Figure E1: South Carolina's 2011 Unemployment Tax Reform

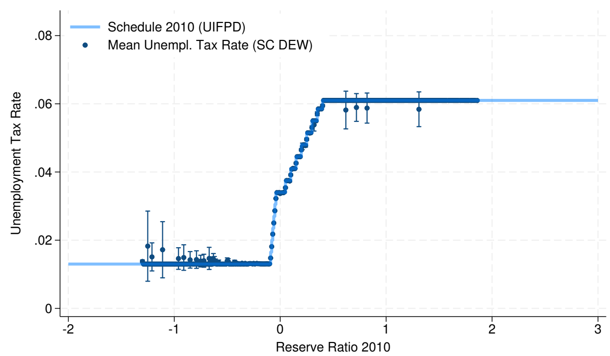
(a) True (2010) and Predicted (2011) Reserve Ratios



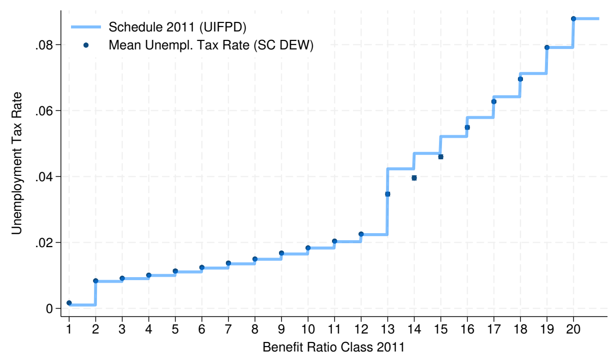
(b) Taxable Wage Base



(c) Reserve Ratio Schedule 2004-2010

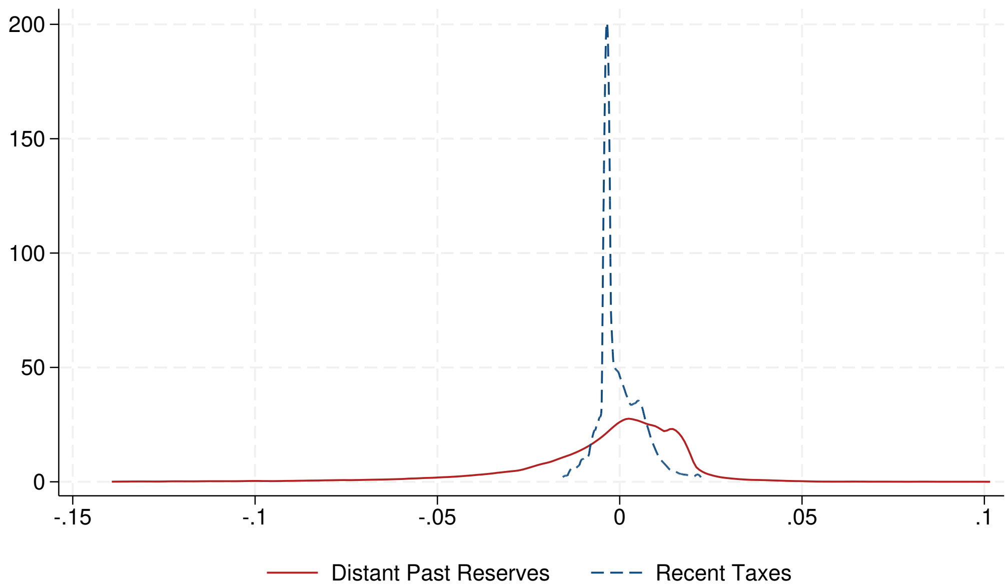


(d) Benefit Ratio Schedule 2011



Notes: Panel (a) plots reserve ratios in 2010 against predicted reserve ratios in 2011 for firms in the SC DEW study sample, both, trimmed at the 1^{st} and 99^{th} percentiles. Equation (26) gives the formula of the reserve ratio. The 2010 reserve ratio is provided. Table D3 describes the construction of the predicted reserve ratio. Panel (b) shows the evolution of the taxable wage base in South Carolina in absolute term and as percentage of the average annual wage each year, using data from the ETA 394 Handbook. The average annual wage is calculated as the ratio of total annual wages paid in covered employment and average monthly employment. Panels (c) and (d) show the unemployment tax rate schedules in place in South Carolina in place between 2004 and 2010 (based on the reserve ratio) and 2011 (based on the benefit ratio) respectively, from the US Unemployment Insurance Financing Policies Database. The figure also shows the average unemployment tax rates of SC DEW firms for each level of reserve ratio, trimmed at the 1^{st} and 99^{th} percentiles at rounded at the second decimal position, and for each benefit ratio class. Markers lie on the schedules, confirming their implementation.

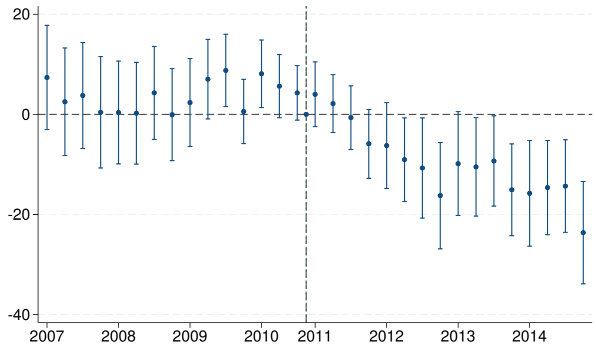
Figure E2: Distant Past Determinant of Treatment in South Carolina



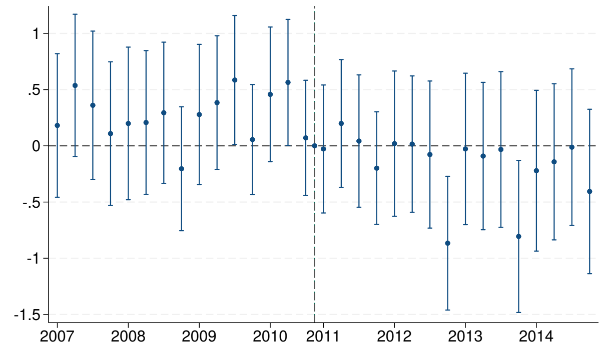
Notes: The figure shows the distribution of distant past reserves and recent taxes from Equation (28), scaled by recent taxable wages, for firms in the SC DEW study sample in 2010. Distant past reserves are calculated as the difference between total reserves in 2009 (provided) and recent reserves. Recent reserves are calculated as the difference between benefit charges and unemployment taxes during the lookback period (July 2003–July 2010). Recent taxes are the unemployment tax payments made during the lookback period. Both variables are trimmed at the 1^{st} and 99^{th} percentiles.

Figure E3: Responses to Unemployment Tax in South Carolina: Wage Distribution

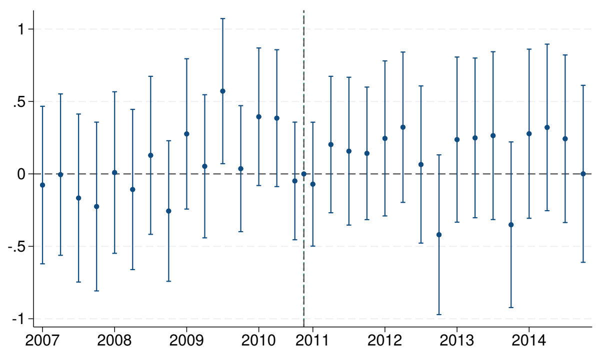
(a) Total Quarterly Wages



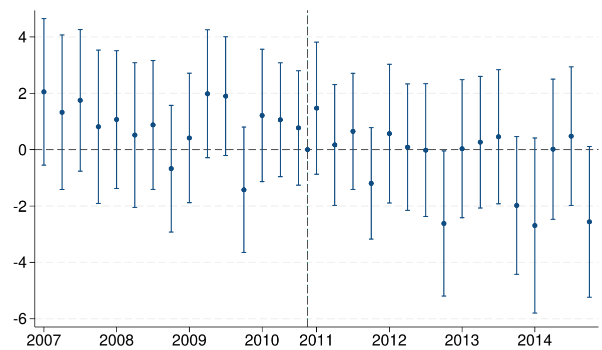
(b) Average Quarterly Wage



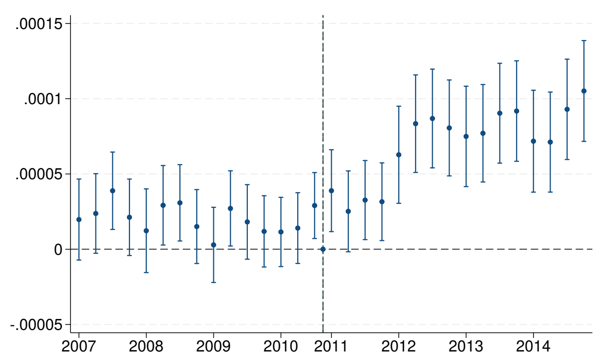
(c) Minimum Quarterly Wage



(d) Maximum Quarterly Wage



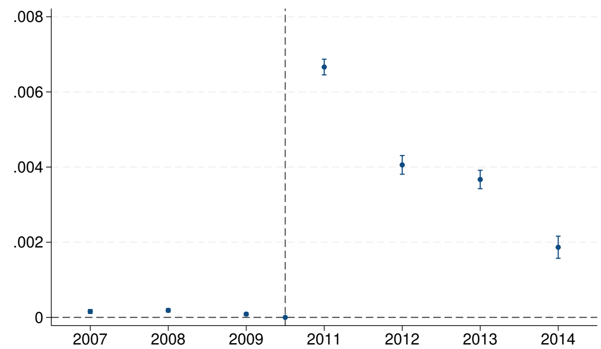
(e) Share of Workers Earning Below Taxable Wage Base



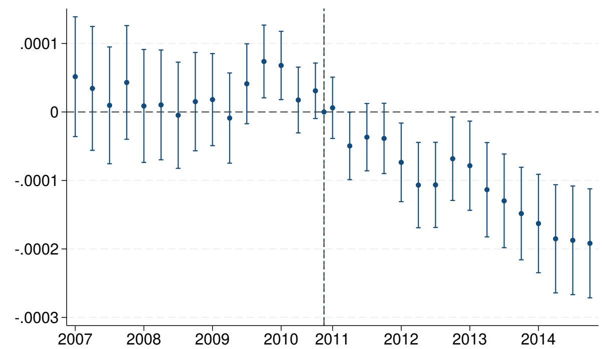
Notes: The figure shows the estimated β_y coefficients from Equation (27) for firms in the SC DEW study sample with fewer than 50 employees on average in 2010. 50 is the 95$^{\text{th}}$ percentile of the 2010 employment distribution. The share of workers earning below the taxable wage base is defined as the number of workers with quarterly earnings below the taxable wage base divided by average quarterly employment. For each firm, I observe the number of workers in specific quarterly wage brackets. Until 2010, workers earning less than \$7,000 per quarter are correctly classified as below the taxable wage base; in 2011, the threshold is \$10,000 per quarter; and in 2012–2014, due to the available wage brackets, I use \$15,000 as the threshold, but it exceeds the taxable wage base of \$12,000 (see Panel (b) of Figure E1). Appendix D.1 provides details on the construction, cleaning and winsorization of the outcomes. 95% percent confidence intervals are reported.

Figure E4: Responses to Unemployment Tax in South Carolina: Large Employers

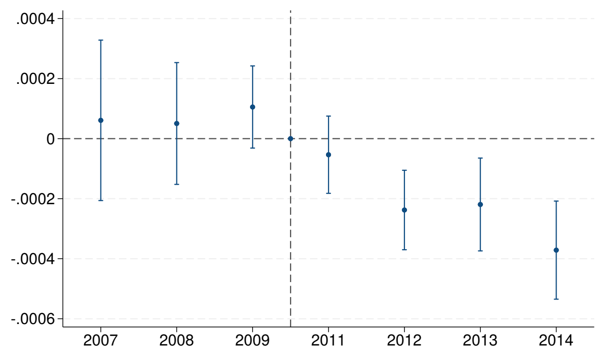
(a) Unemployment Tax Per Worker (Scaled by 2010 Level)



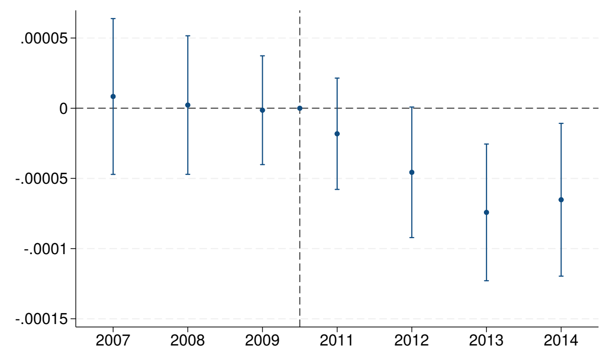
(b) Employment (Scaled by 2010 level)



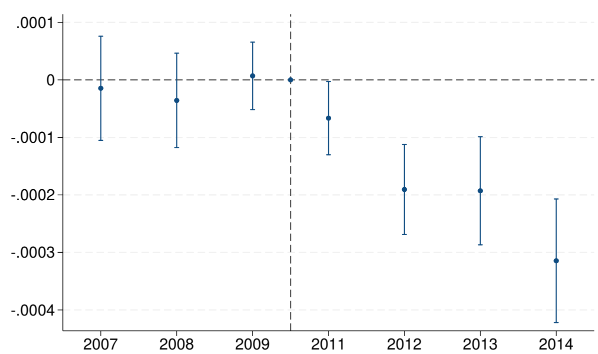
(c) Total Wages (Scaled by 2010 level)



(d) Average Wage (Scaled by 2010 level)



(e) Taxable Wages (Scaled by 2010 level)

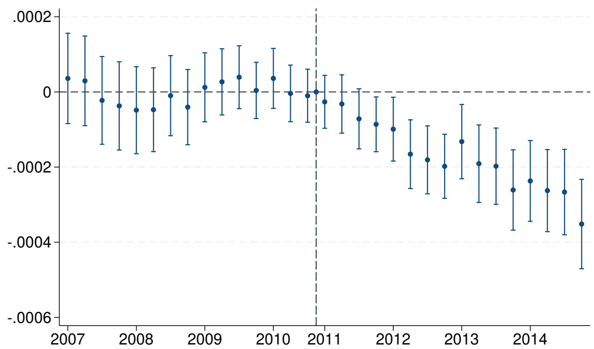


(f) Unemployment Taxes (Scaled by 2010 level)

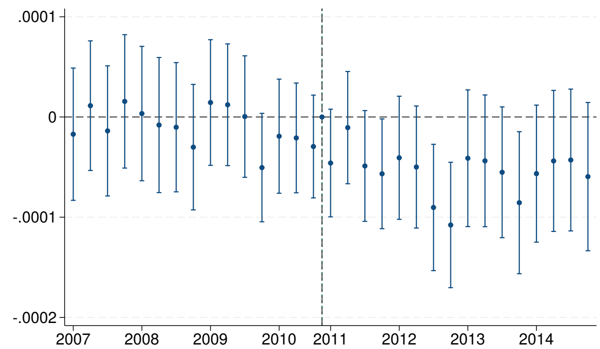
Notes: The figure shows the estimated β_y coefficients from Equation (27) for firms in the SC DEW study sample with one or more employees in 2010. Outcomes are scaled by their 2010 level. Appendix D.1 provides details on the construction, cleaning and winsorization of the outcomes. 95% percent confidence intervals are reported.

Figure E5: Responses to Unemployment Tax in South Carolina: Large Employers' Wage Distributions

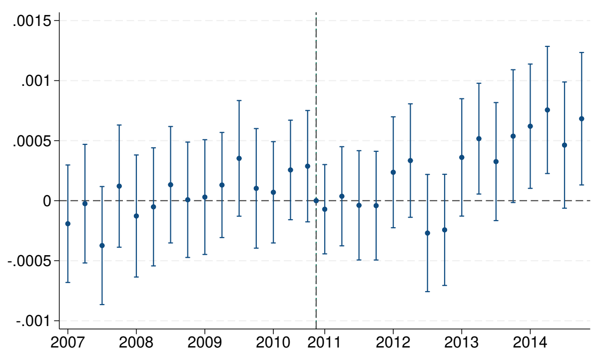
(a) Total Quarterly Wages (Scaled by 2010 level)



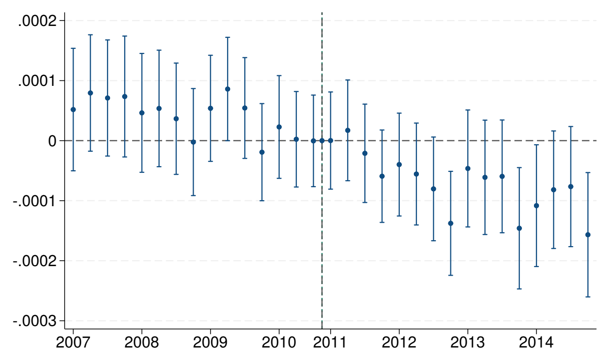
(b) Average Quarterly Wage (Scaled by 2010 level)



(c) Minimum Quarterly Wage (Scaled by 2010 level)



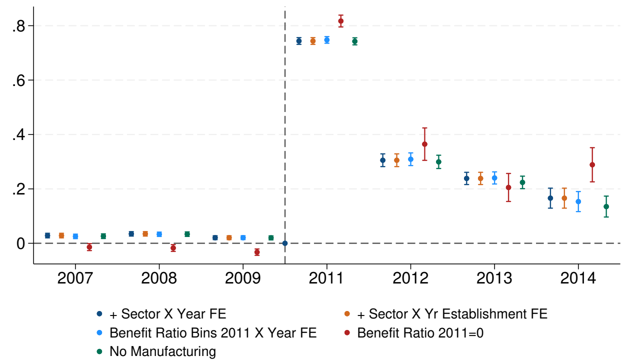
(d) Maximum Quarterly Wage (Scaled by 2010 level)



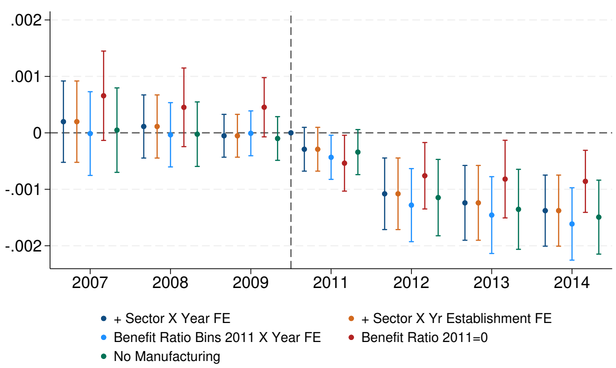
Notes: The figure shows the estimated β_y coefficients from Equation (27) for firms in the SC DEW study sample with one or more employees in 2010. Outcomes are scaled by their 2010 level. Appendix D.1 provides details on the construction, cleaning and winsorization of the outcomes. 95% percent confidence intervals are reported.

Figure E6: Robustness of Firms' Responses to Unemployment Tax in South Carolina

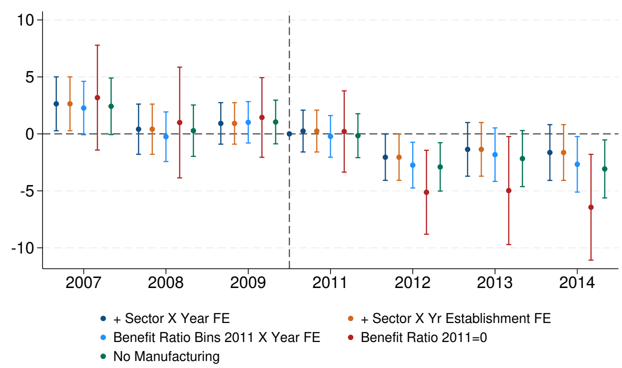
(a) Unemployment Tax Per Worker ("First Stage")



(b) Employment



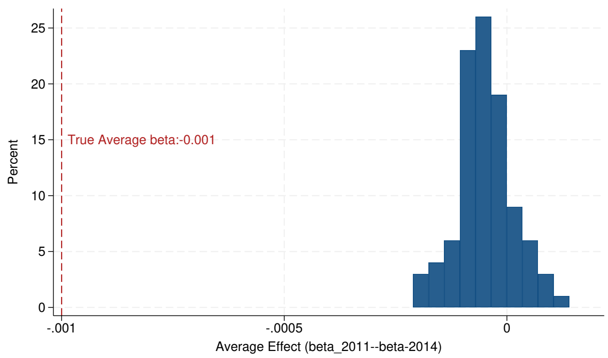
(c) Average Wage



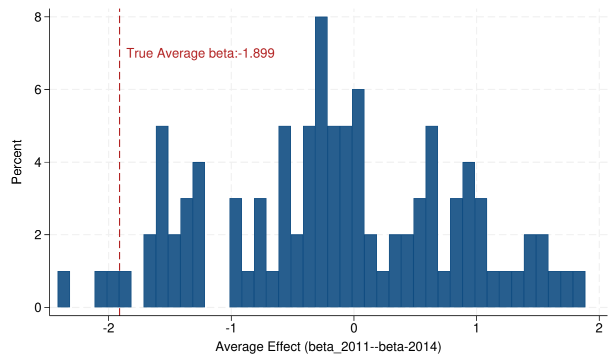
Notes: The figure shows the estimated β_y coefficients from modifications of Equation (27) for firms in the SC DEW study sample with fewer than 50 employees on average in 2010. 50 is the 95$^{\text{th}}$ percentile of the 2010 employment distribution. Appendix D.1 provides details on the construction, cleaning and winsorization of the outcomes. 95% percent confidence intervals are reported. Modifications: I include NAICS-four-digit-sector-code-by-year fixed effects or year-of-establishment-by-year fixed effects; I substitute the twenty benefit ratio classes with 330 bins of benefit ratios in 2011 rounded to the third decimal position; I restrict the sample to firms with benefit ratio in 2011 equal to zero; I exclude manufacturing firms.

Figure E7: Responses to Random Treatment Assignment in South Carolina

(a) Employment



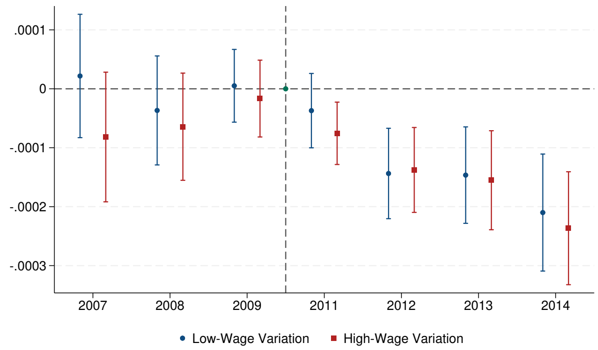
(b) Average Wage



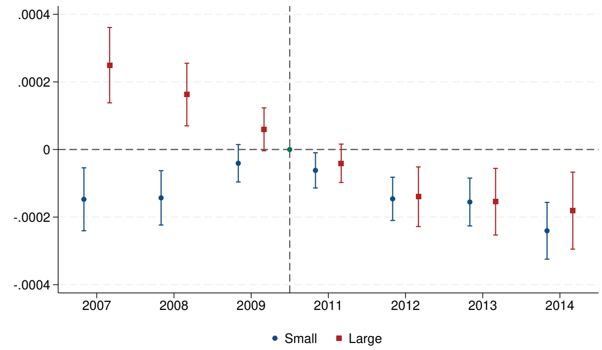
Notes: The figure shows the distribution of the average post-reform coefficients (β_{2011}–β_{2014}) from Equation (27), estimated using a random treatment for firms in the SC DEW study sample with fewer than 50 employees on average in 2010. 50 is the 95$^{\text{th}}$ percentile of the 2010 employment distribution. Appendix D.1 provides details on the construction, cleaning and winsorization of the outcomes. In each of one hundred iterations, firms are randomly assigned a predicted reserve ratio in 2011 drawn from a uniform distribution between −1.5 and 1.5, and their predicted change in the unemployment tax per worker (treatment) is calculated based on this simulated predicted reserve ratio. The dashed line indicates the average effects based on the true treatment, reported in panel A of Table I.

Figure E8: Heterogeneous Employment Responses to Unemployment Tax in South Carolina

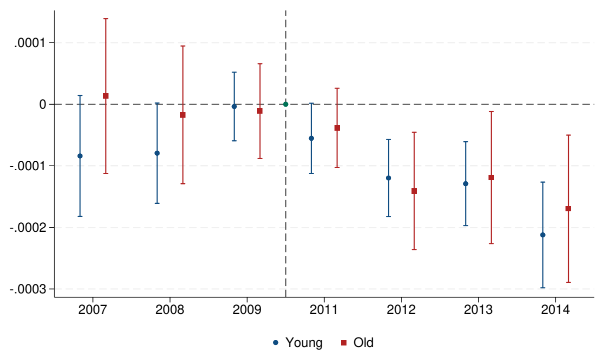
(a) Test Competition: Het. by Industry Wage Variation



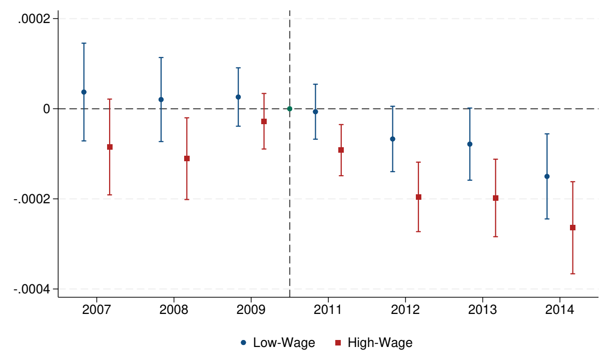
(b) Test Cash Constraint: Het. By Firm Size



(c) Test Cash Constraint: Het. By Firm Age



(d) Test Head Tax: Het. By Average Wage



Notes: The figure shows the estimated β_y coefficients from Equation (27) for firms in the SC DEW study sample with at least one worker in 2010. The outcome is average annual employment scaled by its 2010 level. Estimates are shown separately by industry wage heterogeneity (within-industry standard deviation of firm-level average wages between 2007 and 2010 below/above median), with industries classified by four-digit NAICS codes, in panel (a); by firm size (below/above ten employees in 2010) in panel (b); by firm age (below/above the 2010 median of fourteen years) in panel (c); by average wages (below/above the 2010 median of $29,000) in panel (d).

F Additional Tables and Figures: Colorado

Table F1: Surcharge Applied to Unemployment Tax Rates in Colorado, 2012-2018

Year	2012	2013	2014	2015	2016	2017	2018
Surcharge (%)	0.00	19.39	22.19	25.20	24.47	23.94	0.00

Notes: The surcharge is the percentage by which statutory unemployment tax rates in the Colorado unemployment tax rate schedule were increased to obtain total rates. For instance, in 2017 all the schedule rates were multiplied by 1.2394, reflecting a 23.94% surcharge.

Table F2: Responses to and Incidence of Unemployment Taxation in Colorado

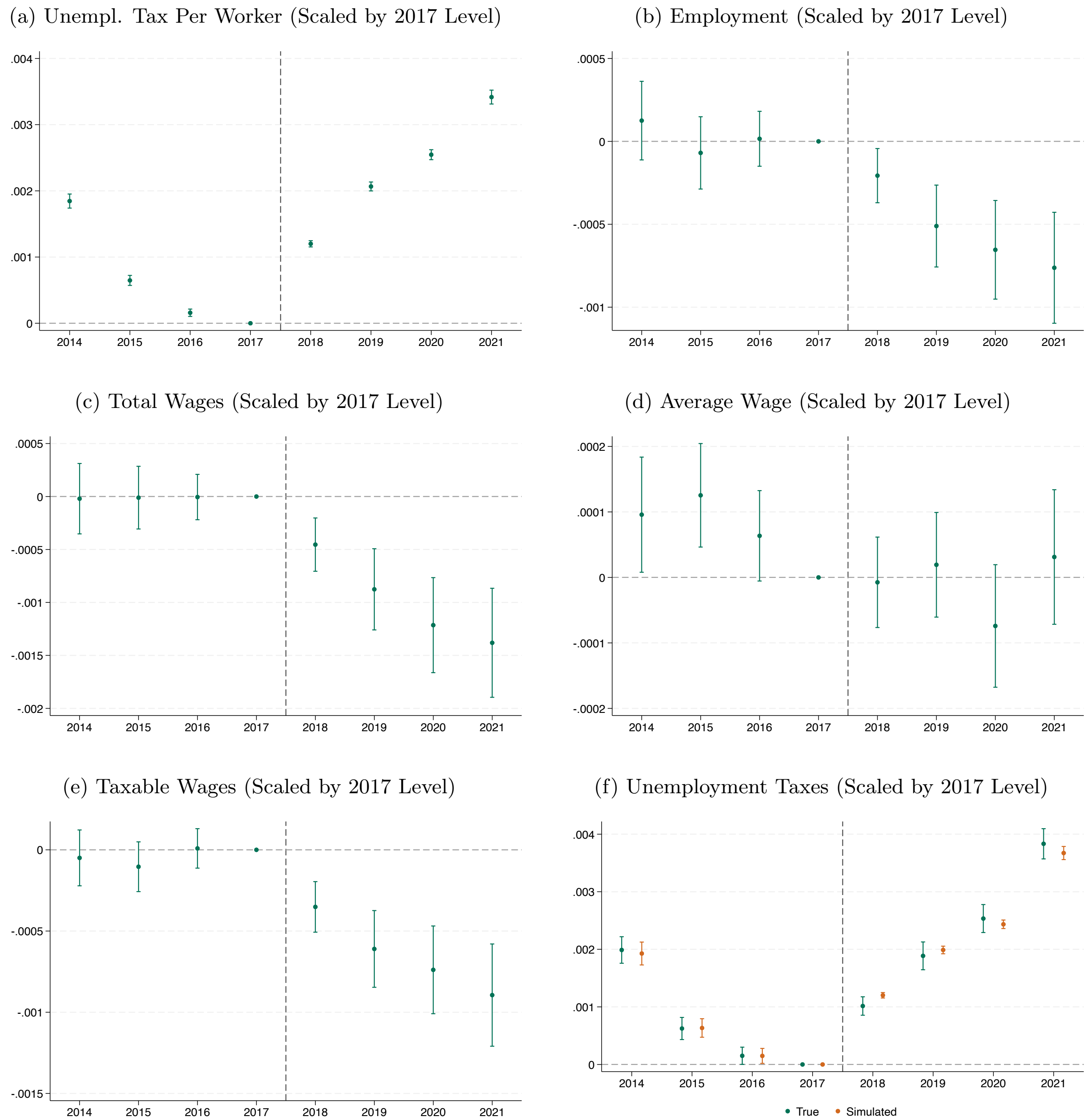
	(1) Unempl. Tax Per Worker	(2) Employ-ment	(3) Total Wages	(4) Average Wage	(5) Taxable Wages	(6) Unempl. Taxes	(7) Sim. Unempl. Taxes
2014 × Pred Δ UTPW	0.107***	3.45e-05	49.21	5.387	3.476	1.234***	1.234***
	(0.0333)	(0.000290)	(40.40)	(5.403)	(4.574)	(0.401)	(0.401)
2015 × Pred Δ UTPW	0.129***	-9.11e-05	30.62	4.305	0.0873	1.933***	1.933***
	(0.0251)	(0.000231)	(27.96)	(4.926)	(3.834)	(0.334)	(0.334)
2016 × Pred Δ UTPW	0.0773***	0.000345**	39.94*	1.438	4.764*	0.985***	0.985***
	(0.0201)	(0.000164)	(23.22)	(4.143)	(2.742)	(0.256)	(0.256)
2018 × Pred Δ UTPW	1.798***	-0.00125***	-160.3***	-5.848	-24.65***	6.846***	8.767***
	(0.0197)	(0.000290)	(32.30)	(3.966)	(6.610)	(0.537)	(0.198)
2019 × Pred Δ UTPW	2.270***	-0.00294***	-327.4***	-8.998*	-64.46**	8.036***	10.87***
	(0.0293)	(0.000815)	(103.6)	(4.661)	(25.61)	(1.340)	(0.250)
2020 × Pred Δ UTPW	2.781***	-0.00382***	-582.6***	-22.57***	-74.56***	10.80***	12.63***
	(0.0316)	(0.000818)	(170.1)	(5.190)	(16.94)	(0.776)	(0.270)
2021 × Pred Δ UTPW	2.385***	-0.00410***	-651.9***	-17.37***	-79.18***	8.900***	9.913***
	(0.0446)	(0.000794)	(125.4)	(5.830)	(14.48)	(0.746)	(0.273)
Observations	400,224	400,224	400,224	397,321	400,224	400,224	400,224
R-squared	0.750	0.672	0.630	0.830	0.614	0.488	0.665
p-val Test 2014–16=0	0.000	0.012	0.309	0.758	0.109	0.000	0.000
Avg. Effect 2018–21	2.309	-0.003	-430.5	-13.70	-60.71	8.646	10.55
Avg. Effect Std. Err	0.0290	0.001	95.16	4.039	12.81	0.609	0.235
p-val Test Avg.Eff=0	0.000	0.000	0.000	0.001	0.000	0.000	0.000
Full Sample Mean 2014-17	254	20.31	1187863	59913	282889	6204	6204

Notes: The table presents the estimated β_y coefficients from Equation (32) for firms in the CO DLE study sample with fewer than 12 employees on average in 2017 and with a benefit ratio in 2018 lower than 0.05. 12 is the 75^{th} percentile of the employment distribution in 2017. 0.05 is the 95^{th} percentile of the benefit ratio distribution in 2018. Appendix D.2 provides details on the construction, cleaning and winsorization of the outcomes. Corresponding estimates and 95% confidence intervals are shown in Figure VI. *** p<0.01, ** p<0.05, * p<0.1.

Figure F1: Taxable Wage Base in Colorado

Notes: The figure shows the evolution of the taxable wage base in Colorado in absolute term and as percentage of the average annual wage each year, using data from the ETA 394 Handbook. The average annual wage is calculated as the ratio of total annual wages paid in covered employment and average monthly employment.

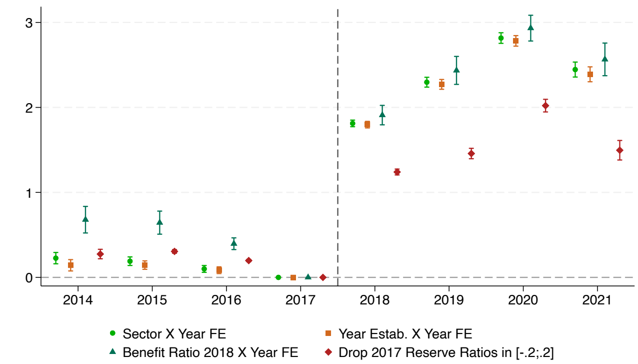
Figure F2: Responses to Unemployment Tax in Colorado: Large Employers



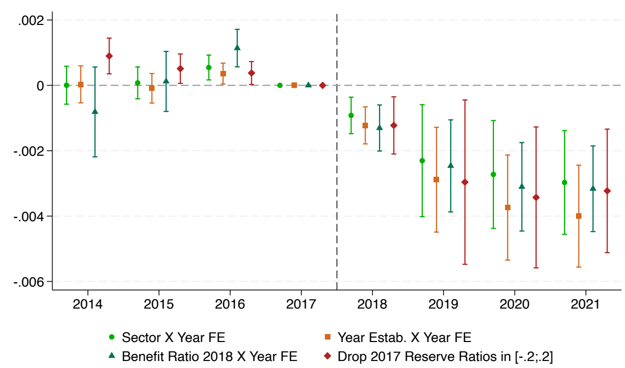
Notes: The figure shows the estimated β_y coefficients from Equation (32) for firms in the CO DLE study sample with a benefit ratio in 2018 lower than 0.05. 0.05 is the 95^{th} percentile of the benefit ratio distribution in 2018. Appendix D.2 provides details on the construction, cleaning and winsorization of the outcomes. 95% percent confidence intervals are reported.

Figure F3: Robustness of Firms' Responses to Unemployment Tax in Colorado

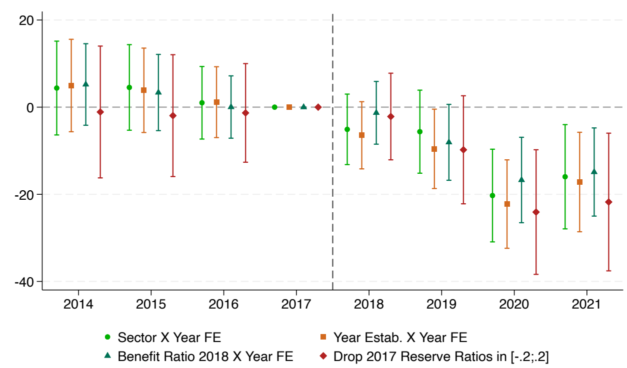
(a) Unemployment Tax Per Worker



(b) Employment



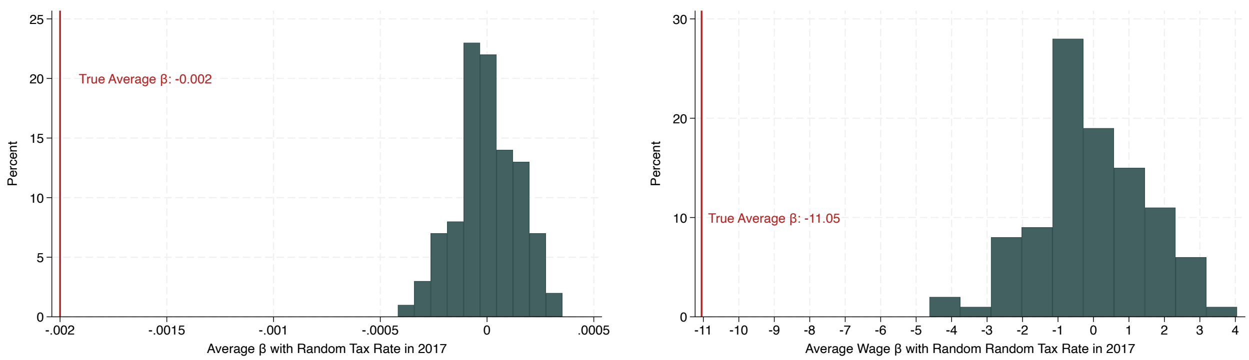
(c) Average Wage



Notes: The figure shows estimated β_y coefficients from modifications of Equation (32) for firms in the CO DLE study sample with fewer than 12 employees in 2017 and with a benefit ratio in 2018 lower than 0.05. 12 is the 75^{th} percentile of the employment distribution in 2017. 0.05 is the 95^{th} percentile of the benefit ratio distribution in 2018. Appendix D.2 provides details on the construction, cleaning and winsorization of the outcomes. 95% percent confidence intervals are reported. Modifications: I include NAICS-six-digit-sector-code-by-year fixed effects or year-of-establishment-by-year fixed effects; I include firms with any benefit ratio level in 2018 but control for 2018-benefit-ratio-by-year fixed effects; I drop firms with a reserve ratio in 2017 between -0.2 and 0.2, with positive pre-reform average benefit charges in panel (f) of Figure V.

Figure F4: Responses to Random Treatment Assignment in Colorado

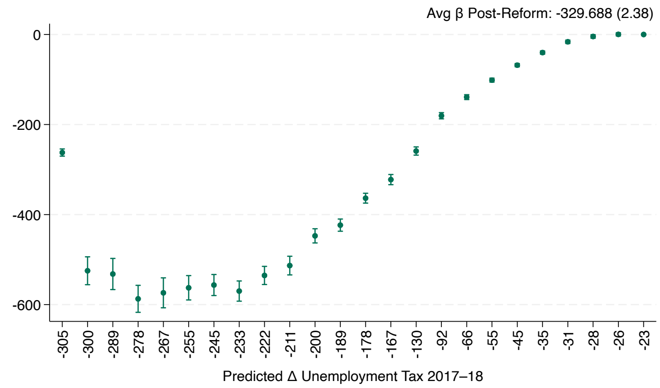
(a) Employment (b) Average Wage



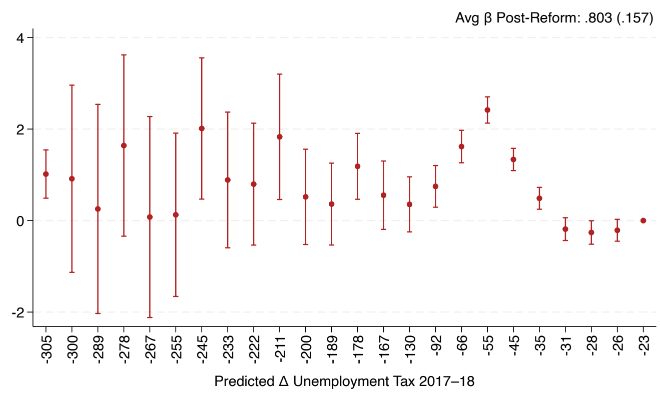
Notes: The figure shows the distribution of the average post-reform coefficients (β_{2018}–β_{2021}) from Equation (32), estimated using a random treatment for firms in the CO DLE study sample with fewer than 12 employees in 2017 and with a benefit ratio in 2018 lower than 0.05. 12 is the 75^{th} percentile of the employment distribution in 2017. 0.05 is the 95^{th} percentile of the benefit ratio distribution in 2018. Appendix D.2 provides details on the construction, cleaning and winsorization of the outcomes. In each of one hundred iterations, firms are randomly assigned an unemployment tax rate from the 2017 schedule, which is used to calculate their predicted change in their unemployment tax per worker (treatment). The dashed line indicates the average effects based on the true treatment, reported in panel A of Table I.

Figure F5: Addressing Dose-Related Treatment Effect Heterogeneity in Colorado

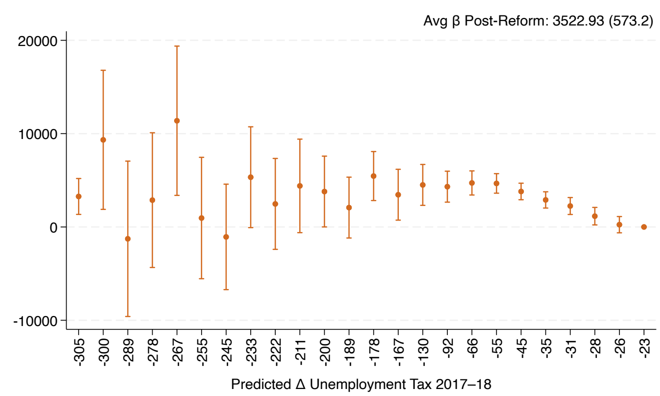
(a) Δ Unemployment Tax Per Worker ("First Stage")



(b) Δ Employent



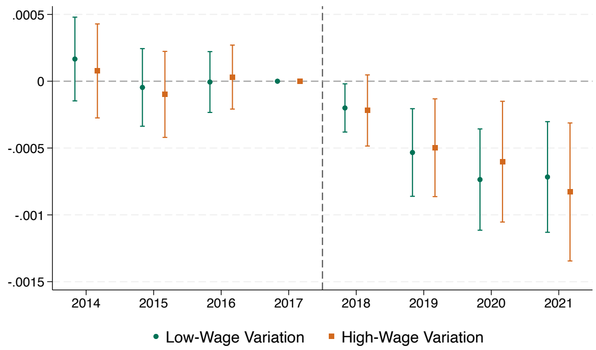
(c) Δ Average Wage



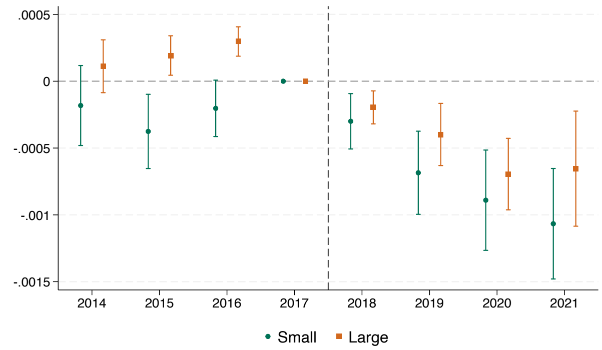
Notes: Following Callaway et al. (2024) (Equation 4.1 on page 21), I regress the change in outcomes between the pre-reform (2014–17) and post-reform (2018-21) periods on indicators for each level of predicted change in the unemployment tax per worker in 2018, for firms in the CO DLE study sample with a benefit ratio in 2018 lower than 0.05. 0,05 is the 95^{th} percentile of the distribution. I drop firms with predicted tax change equal to -51 and -82 because estimates are noisy due to the small number of observations with that treatment level. Because no firm has a predicted change equal to zero, I normalize to zero the coefficient for the smallest treatment dose ($\beta_{d=-19}$). Outcomes are winsorized at the 1^{st} and 99^{th} percentiles.

Figure F6: Heterogeneous Employment Responses to Unemployment Tax in Colorado

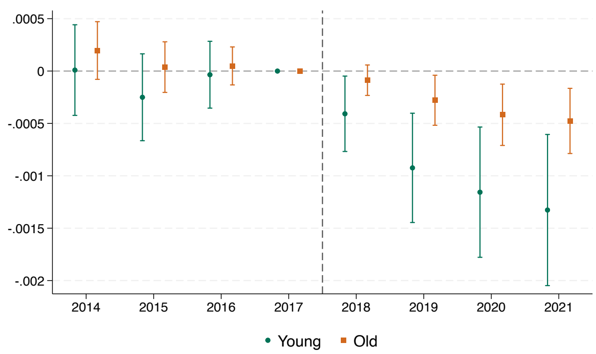
(a) Test Competition: Het. by Industry Wage Variation



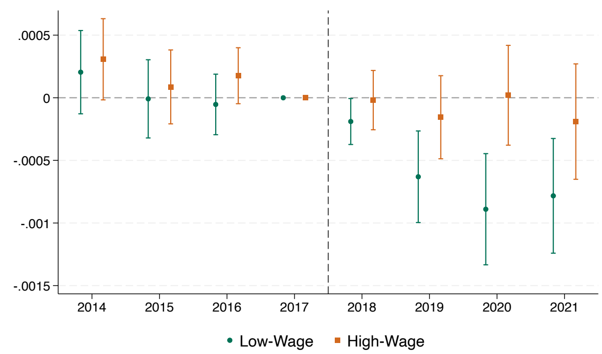
(b) Test Cash Constraint: Het. By Firm Size



(c) Test Cash Constraint: Het. By Firm Age



(d) Test Head Tax: Het. By Average Wage



Notes: The figure shows the estimated β_y coefficients from Equation (32) for firms in the CO DLE study sample with a benefit ratio in 2018 lower than 0.05. 0.05 is the 95^{th} percentile of the distribution. The outcome is average annual employment scaled by its 2017 level. Estimates are shown separately by industry wage heterogeneity (within-industry standard deviation of firm-level average wages between 2014 and 2017 below/above median), with industries classified by six-digit NAICS codes, in panel (a); by firm size (below/above ten employees in 2017) in panel (b); by firm age (below/above the 2017 median of twelve years) in panel (c); by average wages (below/above the 2017 median of $44,000) in panel (d).

G Cash Constraints and Labor Demand Elasticity

In the United States, unemployment taxes rise during economic downturns, precisely when firms are most cash-constrained. First, layoffs increase firms' experience factors, raising the tax per worker. Second, states often increase tax rates and the taxable wage base to fund higher unemployment benefits. I show with a simple example that firms with cash constraints display weakly larger employment responses to unemployment taxes than firms without such constraints.

Consider a firm with three workers with productivities $f_1 > f_2 > f_3$ and rigid wages $w_i f_i$, subject to the unemployment tax per worker τ. Define each worker's net productivity as: $Net_i = f_i - w_i f_i - \tau$. The third worker is the marginal worker, defined by the condition $Net_3 = 0$, so that total profits are: $\Pi = \sum_{i=1}^{3} Net_i = Net_1 + Net_2$.

Case 1: No cash constraint. Suppose the unemployment tax increases to τ', but total profits remain positive, $\Pi' = \sum_{i=1}^{3} Net'_i > 0$. After the tax increase, the net productivity of the marginal (third) worker becomes negative, $Net'_3 < 0$, so the firm does not hire worker 3. Togeher, $Net'_1 > Net'_2$ and $\Pi' > 0$ imply that $Net'_1 > 0$. Whether the firm also hires worker 2 depends on the sign of Net'_2. If $Net'_2 > 0$, not hiring worker 2 would reduce profits, and the firm retains them. Thus, in the absence of cash constraints, a tax increase leads the firm to not hire the marginal worker, while hiring worker 2 is ambiguous.

The minimum elasticity of labor demand with respect to the unemployment tax per worker is $\varepsilon^{N}_{l,\tau} = \frac{-1}{3} \frac{\tau}{\Delta\tau^{+}}$, where -1 is the number of workers no longer hired, 3 is the original firm size, τ is the initial tax, and $\Delta\tau^{+}$ is the maximum tax increase that keeps profits positive, $\Pi' > 0$.

Case 2: Cash-constrained. Suppose instead that the unemployment tax increases to τ' and total profits become negative, $\Pi' = \sum_{i=1}^{3} \mathrm{Net}'_i < 0$. As in Case 1, after the tax increase the net productivity of the marginal (third) worker becomes negative, $\mathrm{Net}'_3 < 0$, so the firm does not hire worker 3. However, because profits are now negative, the firm must further reduce employment to restore profitability. Togeher, $Net'_1 > Net'_2$ and $\Pi' < 0$ imply that $Net'_2 < 0$. As a result, not hiring worker 2 increases profit. When the firm is cash-constrained, a tax increase leads to a larger employment reduction than in the unconstrained case.

The minimum elasticity of labor demand with respect to the unemployment tax per worker is $\varepsilon^{C}_{l,\tau} = \frac{-2}{3} \frac{\tau}{\Delta\tau^{-}}$, where -2 is the number of workers no longer hired, 3 is the original firm size, τ is the initial tax, and $\Delta\tau^{-}$ is the minimum tax increase that makes profits negative, $\Pi' < 0$.

Comparison: I obtain that $\varepsilon^{C}_{l,\tau} = 2\varepsilon^{N}_{l,\tau} \frac{\Delta\tau^{+}}{\Delta\tau^{-}} \approx 2\varepsilon^{N}_{l,\tau}$.

H Calibration Details

I describe in detail the calibration of each parameter entering the formula for the optimal degree of experience rating (Equation 21). The parameters are calibrated separately for South Carolina in 2010 and 2011 and for Colorado in 2017 and 2018.

H.1 Experience Rating e

I calibrate the degree of experience rating using two approaches corresponding to the two interpretations of experience rating in the model.

Share of Own Benefits Repaid in Taxes First, e represents the share of a firm's benefit expenditure that the firm repays in unemployment taxes. Following Vroman et al. (2017), I calculate the share of benefits repaid in taxes in state s and year t, as the fraction of non-socialized benefits in that state-year, as shown in Equation (63). Socialized benefits include:

i *Non-charged benefits*: Benefits not assigned to any firm, such as dependent allowances or extended emergency benefits;

ii *Inactive charges*: Benefits charged to firms that have gone out of business;

iii *Ineffective charges*: Benefits charged to firms at the maximum tax rate, for whom the unemployment tax rate cannot increase further.

$$e_{share,s,t} = 1 - \frac{(\text{Non-Charged}_{s,t} + \text{Inactive}_{s,t} + \text{Ineffective}_{s,t})}{\text{Total Benefit Payments to Taxable Employers}_{s,t}} \tag{63}$$

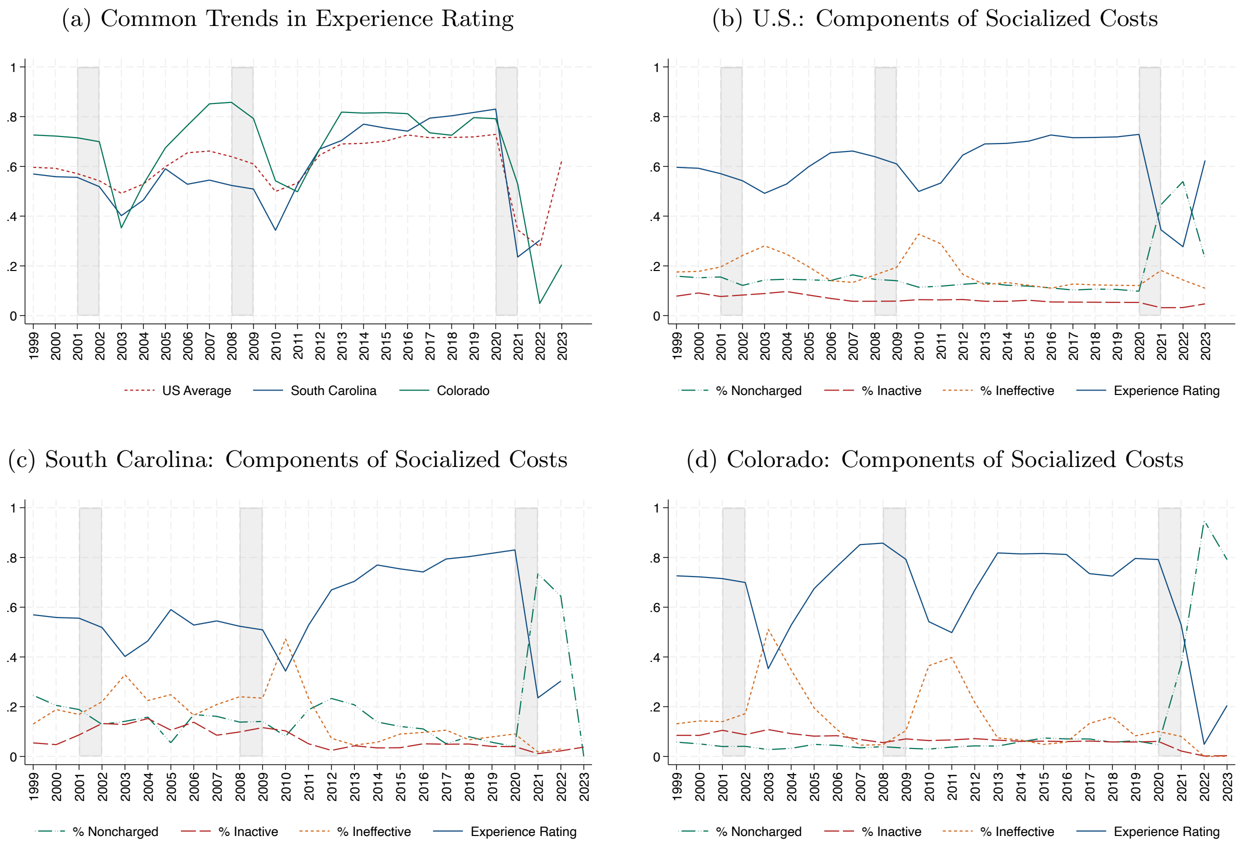
Non-charged and inactive benefits are reported for each state and year in the ETA 204 Report, Sections A and B. Ineffective charges are derived from Section C as the sum of benefit payments exceeding unemployment taxes across reserve or benefit ratios.

Figure H1 shows the evolution of socialized costs and the implied degree of experience rating in South Carolina, Colorado, and in the United States, averaging across states. The degree of experience rating is 34.3% in 2010 and 52.7% in 2011 in South Carolina, and 73.5% in 2017 and 72.5% in 2018 in Colorado. These values indicate that 34.3–52.7% and 72.5–73.5% of benefit charges were repaid by the firms who generated them in South Carolina and Colorado, respectively.

For comparison, Vroman et al. (2017) report an average degree of experience rating of 49.2% in South Carolina (2005–2009) and 65.4% in Colorado (2010–2014). When I replicate my measure for the same periods as Vroman et al. (2017), I obtain 53.9% for South Carolina and 66.8% for Colorado, closely matching their estimates.

Figure H1: Degree of Experience Rating: Share of Own Benefits Repaid in Taxes

(a) Common Trends in Experience Rating; (b) U.S.: Components of Socialized Costs; (c) South Carolina: Components of Socialized Costs; (d) Colorado: Components of Socialized Costs



Notes: Panel (a) shows the degree of experience rating, calculated in Equation (63) as the share of unemployment benefits repaid in unemployment taxes by firms who generated them, in South Carolina, Colorado, and the United States. U.S. values represent averages across states. Panels (b)–(d) show the evolution of the share of non-charged, inactive, and ineffective benefits, and the resulting degree of experience rating for the United States, South Carolina, and Colorado. Thirty-five state-year observations in which the share of socialized benefits exceeds one are dropped. Data source: ETA 204 Report.

Marginal Tax Cost of Benefit Charge Second, e represents the marginal tax cost of a dollar of benefit charge. Following Pavosevich (2020) and Duggan et al. (2023), I calculate the average marginal tax cost in state s and year t in Equation (64) as the product of four components:

(i) The change in the experience factor induced by the one-dollar increase in benefit charges, $\Delta ER_{s,t}$, which depends on the formula used in the state-year;

(ii) The average change in the unemployment tax rate induced by the change in the experience factor, captured by the average slope of the unemployment tax rate schedule in the state-year, $slope_{s,t}$;

(iii) The taxable wage base in place in the state-year, $TWB_{s,t}$, to convert the rate change into the dollar change in the unemployment tax per worker;

(iv) The average firm size in the state-year, $Emp_{s,t}$, to obtain the total tax change due to the benefit charge for the average employer in the state.

$$e_{mtc,s,t} = \Delta ER_{s,t} \cdot slope_{s,t} \cdot TWB_{s,t} \cdot Emp_{s,t} \tag{64}$$

Calculating $\Delta ER_{s,t}$ requires knowing the experience-rating formula used in each state. In 2010, South Carolina employed the reserve ratio defined in Equation (26). The change in the reserve ratio from an additional dollar of benefit charges is equal to the benefit charge (one dollar) divided by taxable wages in the previous year, 2009. I assume that all the workers earn more than the taxable wage base and calculate taxable wages for the average firm in the state as the product of the taxable wage base in 2009, $TWB_{SC,2009}$, and the average number of employees in the state in 2009, $Emp_{SC,2009}$.

$$\Delta ER_{SC,2010} = \frac{1}{TWB_{SC,2009} \cdot Emp_{SC,2009}}$$

In 2011, South Carolina used the benefit ratio defined in Equation (26). The change in the benefit ratio from an additional dollar of benefit charges is equal to the benefit charge (one dollar) divided by the sum of taxable wages in the seven years prior to the reform.

$$\Delta ER_{SC,2011} = \frac{1}{\sum_{y=2004}^{y=2010} TWB_{SC,y} \cdot Emp_{SC,y}}$$

In 2017 and 2018, Colorado used a reserve ratio equal to that in South Carolina, except that the denominator is the average of the taxable wages from the previous three years. Therefore, the change in the reserve ratio from an additional dollar of benefit charges is equal to the benefit charge (one dollar) divided by the average taxable wages over the three preceding years.

$$\Delta ER_{CO,2017} = \frac{1}{\frac{TWB_{CO,2014} \cdot Emp_{CO,2014} + TWB_{CO,2015} \cdot Emp_{CO,2015} + TWB_{CO,2016} \cdot Emp_{CO,2016}}{3}}$$

$$\Delta ER_{CO,2018} = \frac{1}{\frac{TWB_{CO,2015} \cdot Emp_{CO,2015} + TWB_{CO,2016} \cdot Emp_{CO,2016} + TWB_{CO,2017} \cdot Emp_{CO,2017}}{3}}$$

I describe the calibration of each parameter in Equation (64) in turn. Taxable wage bases, $TWB_{s,t}$, are taken from the ETA 394 Handbook. In South Carolina, the taxable wage base was \$7,000 between 2004 and 2010. In Colorado, it was \$11,700 in 2014, \$11,800 in 2015, \$12,200 in 2016, and \$12,500 in 2017.

The average firm size, $Emp_{s,t}$, is calculated as the ratio of average annual employment in private firms to the average annual establishment count from the QCEW. In South Carolina, average firm size was 13.56 in 2004, 13.03 in 2005, 12.26 in 2006, 13.80 in 2007, 13.42 in 2008, 12.99 in 2009, 13.29 in 2010, and 13.56 in 2011. In Colorado, it was 11.53 in 2014, 11.48 in 2015, 11.39 in 2016, 11.25 in 2017, and 11.18 in 2018.

I calculate the average slope of the unemployment tax rate schedule, $slope_{s,t}$, using information on the statutory schedules in each state-year from the U.S. Unemployment Insurance Financing Policies Database. Both in South Carolina and Colorado, the schedules take the form of step functions. For each step, corresponding to a reserve-ratio or benefit-ratio interval, I calculate the slope as the ratio of the change in the tax rate between adjacent steps, $\Delta\tau$, to the corresponding reserve ratio change, representing the length of the interval, ΔEF (where EF stands for experience factor). The flat tails of the schedule, representing the minimum and maximum tax rates and of effectively infinite length, are treated as single steps with slope zero. Panels (a), (b), and (c) of Figure H2 show the tax rate schedules and their slopes in South Carolina in 2010 and 2011 and in Colorado in 2017 and 2018.

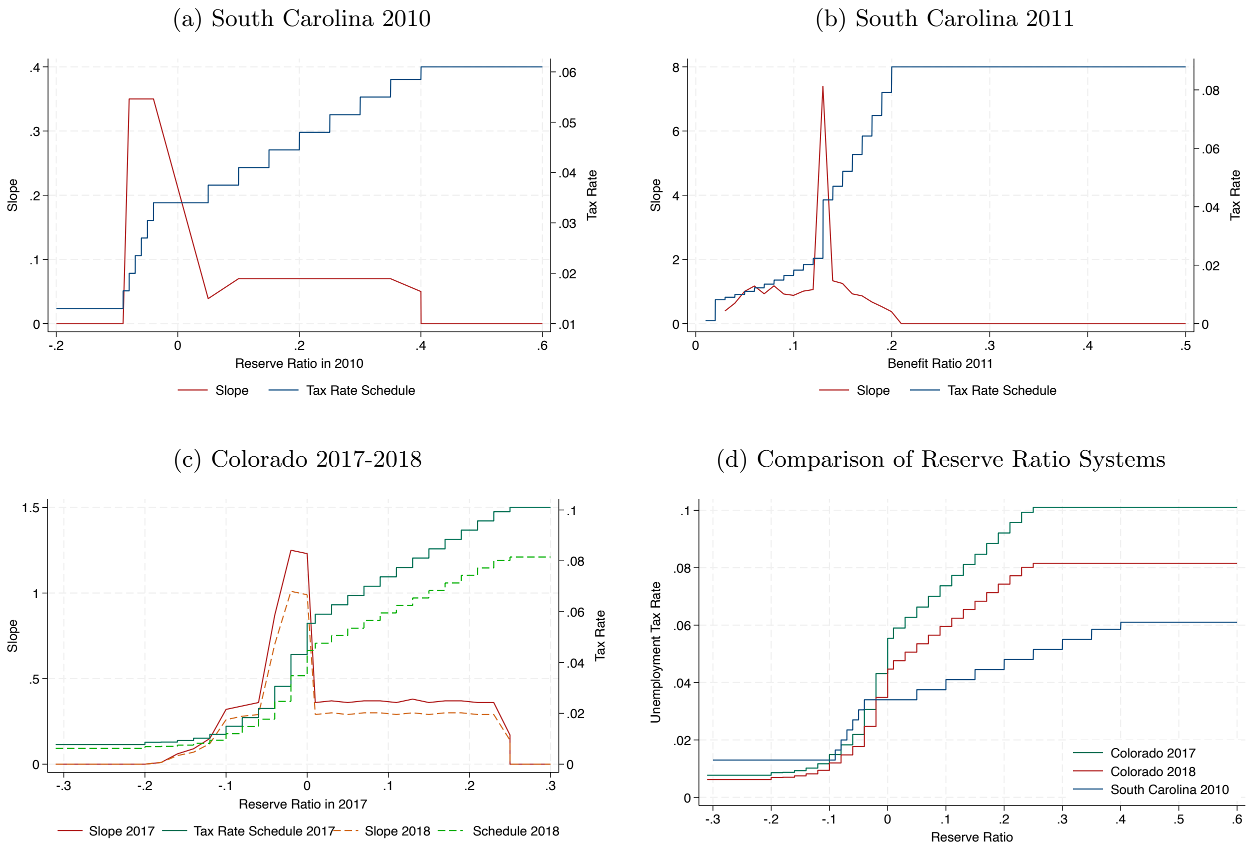
The average slope is 0.15 in South Carolina in 2010, 1.186 in 2011, 0.37 in Colorado in 2017, and 0.30 in 2018. Panels (a) and (b) of Figure H2 show that the sharp increase in the average slope occurred in South Carolina between 2010 and 2011 is driven by the transition to the benefit ratio system and the introduction of higher tax rates, with the maximum rate increasing from 6.1% to 8.79%. Panel (c) compares the 2010 South Carolina schedule with Colorado's schedules, as they are all based on reserve ratios. The higher average slope in Colorado reflects Colorado's higher maximum tax rate (10.1% in 2017, compared with 6.1% in South Carolina in 2010) and its shorter flat segments. The reduction in Colorado's maximum tax rate in 2018, while maintaining the same flat segments, reduced the average slope of the schedule.

These values are consistent with Duggan et al. (2023), who calculate slopes of the unemployment tax rate schedules in place in the various U.S. states using the empirical distribution of average tax rates from the ETA 204 Report. They find that the slopes of benefit-ratio schedules range between 0.3 and 3, while the slopes of reserve reserve-ratio schedules rarely exceed one. The key advantage of my approach is its precision, since the slope is calculated from the statutory shape of the system rather than from the endogenous distribution of firms along the schedule. When employers are concentrated in specific parts of the schedule, the empirical distribution may fail to capture its full range. Moreover, the average tax rates reported for different reserve-ratio bins in the ETA 204 Report may differ from the true tax rates assigned by the schedule, as they may average statutory rates with rates for new and delinquent firms. While in some states the resulting empirical distribution closely matches the statutory schedule, in others is very noisy.

Table H1 reports the calibrated values of each parameter in Equation (64) for the marginal tax cost of benefit charges in the two states. The resulting degree of experience rating is $e_{mtc} = 0.154$ for South Carolina in 2010, $e_{mtc} = 0.249$ in 2011, $e_{mtc} = 0.381$ for Colorado in 2017, and $e_{mtc} = 0.304$ in 2018, implying that firms repay, on average, 15.4-24.9 and 38.1-30.4 cents in unemployment taxes, respectively, for each dollar of benefit charges.

For comparison, Pavosevich (2020) report very similar values of 0.33 for South Carolina and 0.27 for Colorado, both in 2018. Conversely, Duggan et al. (2023) estimate the marginal per worker tax cost

Figure H2: Slope of the Unemployment Tax Rate Schedule



Notes: Panels (a) and (b) shows the unemployment tax rate schedules in place in South Carolina in 2010 and 2011 and their slopes. They are not reported in the same graph because the variable on the x-axis is different: it is the reserve ratio in 2010 and the benefit ratio in 2011. Panel (c) shows the unemployment tax rate schedules in place in Colorado in 2017 and 2018 and their slopes. The slope is calculated as the difference in the tax rate between adjacent steps of the schedule and the length of the step, $\Delta\tau/\Delta EF$. Panel (c) compares the South Carolina and Colorado schedules based on reserve ratios. Data source: Unemployment Insurance Financing Policies Database.

of laying off 10% of the workforce. I thus replicate their approach in one year per state to further validate my findings. Such layoffs generate benefit charges equal to $0.1 \cdot Emp \cdot \text{take-up}_{s,t} \cdot b_{s,t}$, where Emp is the average firm size, $\text{take-up}_{s,t}$ is the benefit take-up rate, and $b_{s,t}$ is the benefit level. The average firm size, Emp cancels out because it appears in both the numerator (benefit charges) and the denominator (taxable wages) of the formula. I measure benefit take-up rate as the average of quarterly recipiency rates reported in the Unemployment Insurance Data Summary. In South Carolina in 2010, the average take-up rate is 27.2%, and in Colorado in 2017 it is 31.9%. I measure the benefit level as the product of the average weekly benefit amount and the average potential duration in weeks from the ETA 394 Handbook. This yields $b_{SC,2010} = \$5{,}201.20$ and $b_{CO,2017} = \$10{,}065.96$. Using these values, I obtain a marginal tax cost of laying off 10% of the workforce equal to \$21.32

in South Carolina in 2010 and \$124.45 in Colorado in 2017. While Duggan et al. (2023) do not report state-level estimates, they report a 10^{th} percentile of \$39 and a 90^{th} percentile of \$135 across states, and employment weighted average of \$89. My estimates have consistent magnitude and their employment weighted average is \$84, indicating substantial alignment.

Table H1: Degree of Experience Rating: Marginal Tax Cost

Parameter	**Source**	**South Carolina**	**Colorado**
Taxable wage base, $TWB_{s,t}$	ETA 394 Handbook	\$7,000 (2004-10)	\$11,700 (2014) \$11,800 (2015) \$12,200 (2016) \$12,500 (2017)
Average Number of Employees, $Emp_{s,t}$	QCEW	13.56 (2004) 13.03(2005) 12.26 (2006) 13.80 (2007) 13.42 (2008) 12.99 (2009) 13.29 (2010) 13.56 (2011)	11.53 (2014) 11.48 (2015) 11.39 (2016) 11.25 (2017) 11.18 (2018)
Average slope of tax rate schedule, $slope_{s,t}$	UIFP	0.15 (2010) 1.186 (2011)	0.37 (2010) 0.30 (2018)
Marginal Tax Cost	Equation (64)	0.154 (2010) 0.249 (2011)	0.381 (2017) 0.304 (2018)

Notes: The table summarizes the calibration of the parameters derermining the marginal tax cost of a dollar of benefit charge in Equation (64) for South Carolina in 2010-2011 and Colorado in 2017-2018.

H.2 Labor Market

I calibrate key labor market attributes of low- and high-unemployment-risk industries using data from the Quarterly Census of Employment and Wages (QCEW), which provides employment and wages data at the state–industry (NAICS 6-digits)–quarter level.

Industries' Classification as Low- or High- Risk I classify industries within each state as low- or high-unemployment risk based on the average within-year standard deviation of quarterly employment between 1990–2020. The correlation between the average and the median standard deviation is 97.8% in South Carolina and 94.5% in Colorado, indicating that the averages are not driven by transitory, year-specific shocks but instead capture persistent, long-run industry patterns. Figure H3 shows the distribution of industries' average within-year quarterly employment standard deviation for South Carolina (panel [a]) and Colorado (panel [b]). Most industries exhibit limited within-year employment variation, but the distributions are right-skewed.

I classify as high-risk those industries in the top 5% of the within-year standard deviation distribution. The 95^{th} percentile, which serves as the cutoff, is shown in the figures with dashed lines. The figures also display examples of industries classified as high-risk. Consistent with intuition, these industries have pronounced seasonal employment patterns, such as ski facilities, restaurants, hotels, tax preparation services, and highway, street, and bridge construction.

Employment Share in High-Risk Industries l_H: It is defined as the fraction of total state employment accounted for by industries classified as high-unemployment-risk: 40.4% in South Carolina in 2010; 40.3% in 2011, 32.7% in Colorado in 2017; and 32.6 % in 2018.

Layoff Rate in High Risk Industries, r_H: It is defined in Equation (65) as the differential average percentage change between the minimum and maximum quarterly employment values over the year between low- and high-unemployment risk industries. Consistent with the model, I aim to measure the "excess" layoff probability in the high-risk industry. I obtain 4.4% in South Carolina in 2010; 11.3% in 2011; 12.4% in Colorado in 2017; and 8.6% in 2018. Two factors may account for differences over time and across states. First, layoff rates may have been unusually high even among low-risk industries during the Great Recession. Second, the smaller employment base of low-risk industries in South Carolina may mechanically inflate their layoff rates. In both cases, we would observe a smaller gap in South Carolina in 2010.

$$r_H = \underbrace{\frac{Emp_{max,H} - Emp_{min,H}}{Emp_{max,H}}}_{\text{Layoff Prob. (High-Risk Industry)}} - \underbrace{\frac{Emp_{max,L} - Emp_{min,L}}{Emp_{max,L}}}_{\text{Layoff Prob. (Low-Risk Industry)}} \tag{65}$$

Marginal Productivities $f_H(l_H^{SB})$ and $f_L(l_H^{SB})$: The second-best marginal productivity in the high-risk industry, $f_H(l_H^{SB})$, is defined in Equation (19) as the sum of the wage of the marginal worker hired in the industry, the unemployment tax per worker, and borrowing costs. I calibrate the marginal wage using the average annual wage in high-unemployment risk industries from the QCEW. The tax per worker and borrowing costs are functions of e, r_H and b, whose calibration approach has already been discussed. Since there are two values of e for each state, the share of benefit repaid, e_{share}, and the marginal tax cost, e_{mtc}, I obtain two corresponding values for the marginal productivity in high-risk industries for each state-year. In South Carolina in 2010, the value implied by e_{share} is \$34,839, while the value implied by e_{mtc} is \$34,795. In 2011, these values are \$37,649 and \$37,487 respectively. In Colorado in 2017, the value implied by e_{share} is \$50,209, and the value implied by e_{mtc} is \$49,700. In 2018, these values are \$52,324 and \$51,915, respectively.

The second-best marginal productivity in the low-risk industry, $f_L(l_H^{SB})$, is defined in Equation (12) as the wage of the marginal worker in the industry, which I calibrate with the average annual wage in low-unemployment-risk industries from the QCEW. I obtain \$39,362 in South Carolina in 2010,

\$41,114 in 2011, \$58,393 in Colorado in 2017, and \$61,047 in 2018.

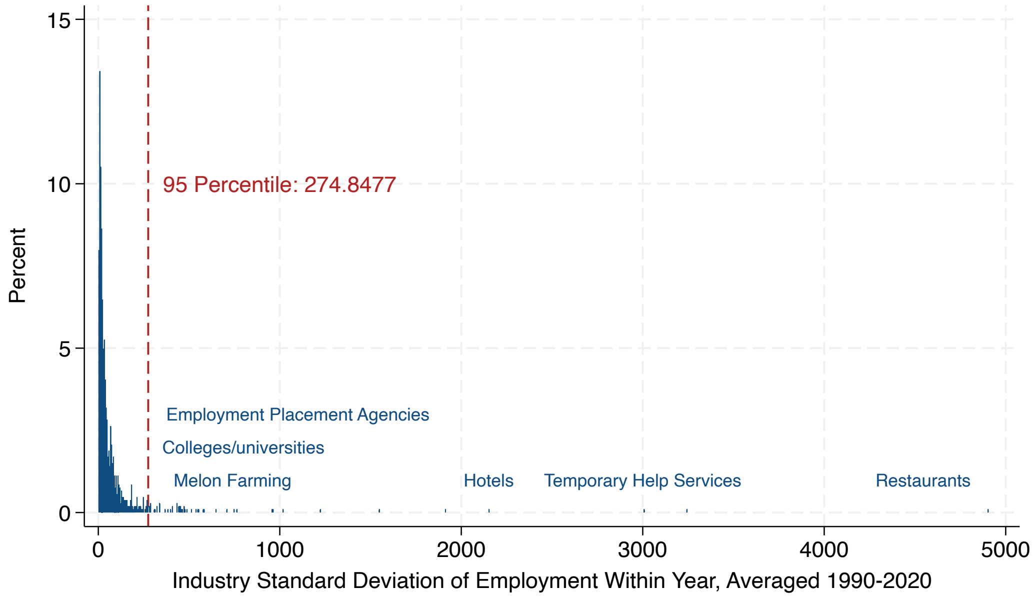
These numbers indicate that, in each state-year, productivity is higher in low-risk industries than in high-risk industries. Although this conclusion is based on average wages rather than the marginal wages required by the model, marginal wages by industry are not available, and the conclusion would change only if the relationship between marginal and average wages differed systematically between low- and high-risk industries.

Output in the High-Risk Industry $\int_0^{l_H^{SB}} f_(i)\, di$: It is calibrated as the area of the gray and green parallelograms in Figure I. The area is the product of the base, which is the second-best employment share in the high-risk industrtìy, l_H, and the height, which is equal to the change in the output of the high-risk industry between the first- and second-best, or the extra output at risk of loss in the second-best, $(r_H - r_H^{FB}) f_H(l_H^{SB})$. I calibrate r_H^{FB}, the first-best layoff rate, as the lowest non-negative layoff rate observed across 1990-2020 in the QCEW. For South Carolina, the layoff rate is negative and close to zero in only one year, and the next highest value is 0.7%; for Colorado, the layoff rate is always positive, and the lowest value is 7.6%.

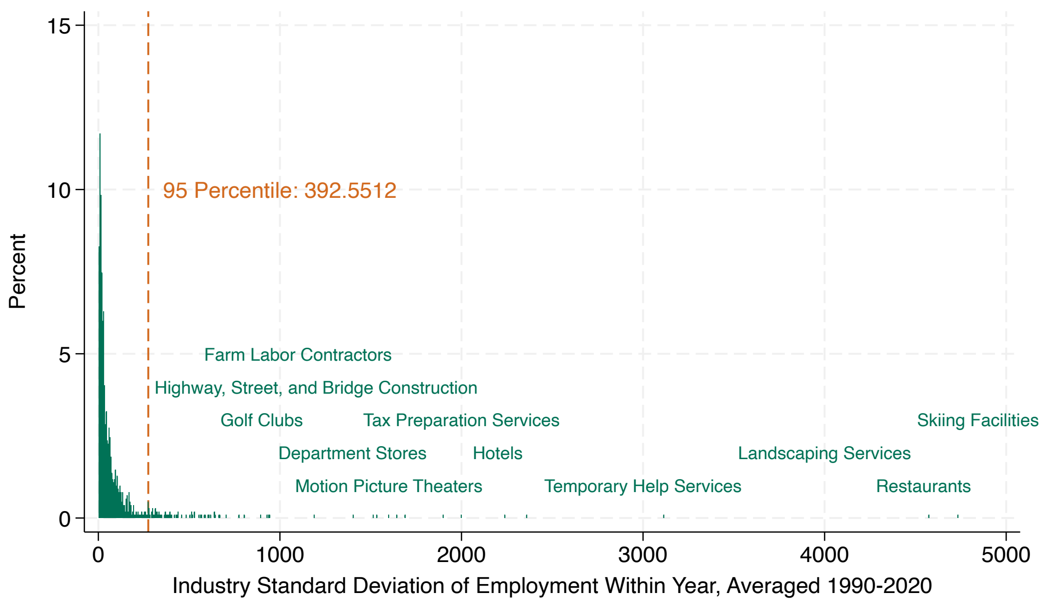
Since there are two values of e for each state, the share of benefit repaid, e_{share}, and the marginal tax cost, e_{mtc}, I obtain two corresponding values for productivity in the high-risk industry and hence for output at risk of loss for each state-year. In South Carolina in 2010, the value implied by e_{share} is \$516 while the value implied by e_{mtc} is \$515. In 2011, these values are \$1,614 and \$1,607, respectively. In Colorado in 2017, the value implied by e_{share} is \$780 and the value implied by e_{mtc} is \$772. In 2018, these values are \$218 and \$214, respectively. The differences between states and over time are explained primarily by differences in the gap between the realized and the first-best layoff rates.

Figure H3: Distribution of Average Industry Within-Year Employment Standard Deviation

(a) South Carolina



(b) Colorado



Notes: The figure shows the distribution of industries' average within year standard deviation of quarterly employment between 1990 and 2020 from the QCEW. Industries are defined by their six-digit NAICS codes.

H.3 Bounds on the Marginal Benefit and Cost

Table H2: Estimated Insurance Value for Workers from the Literature

Study	Estimated value	Approach
Gruber (1997)	0.89	Consumption drop
Hendren (2017)	1.32	Consumption drop
Hendren (2017)	1.87	Ex-ante consumption drop
Landais et al. (2021)	1.52	Consumption drop
Landais et al. (2021)	1.59	Marginal propensity to consume
Landais et al. (2021)	3.13	Revealed preference

Notes: The table lists various estimates for the value of insurance for workers. The value of 0.89 for Gruber (1997) is obtained by multiplying the percentage drop in consumption at layoff (22.2% in Table 1) by the highest value of risk aversion considered, 4. The value of 1.32 for Hendren (2017) comes from column 1 of Table 5, and 1.87 comes from column 1 of Table 6. The value of 1.52 for Landais et al. (2021) is the highest estimate in their Figure 1; 1.59 comes from column 1 of Table 2; and 3.13 comes from column 1 of Table 3.

Table H3: Estimated Elasticity of Labor Demand w.r.t Labor Costs from the Unemployment Insurance Financing Literature

Study	Estimated value
Anderson et al. (1997)	-2.3 [-1.4, -3.2] (Washington)
Johnston (2021)	[-4, -12] (Florida)
Guo (2024)	Lower bound of -2.24 (Multiple U.S. States)
This paper	-3.9 (Colorado); -9.1 (South Carolina)

Notes: The table reports estimates of the elasticity of labor demand with respect to labor costs from the literature on the financing of unemployment insurance in the United States. The values for Anderson et al. (1997) are taken from the online Appendix of Johnston (2021). The values for Johnston (2021) come from Table 2 and Online Appendix Figure 8. The estimate from Guo (2024), along with its interpretation as a lower bound, is taken from the introduction.

I convert the lower-bound estimate of the labor demand elasticity with respect to labor costs across U.S. states, -2.24 from Guo (2024), into an elasticity with respect to the unemployment tax per worker using Equation (66) and pre-reform estimates of wages and unemployment taxes from Table I. I obtain $\varepsilon_{l,\tau} = -0.0081$ for South Carolina and $\varepsilon_{l,\tau} = -0.0094$ for Colorado.

$$\varepsilon_{l,\tau} = \frac{\bar{\beta}_l}{\bar{\beta}_\tau} \cdot \frac{\overline{\tau_{Pre}}}{\overline{l_{Pre}}} = \epsilon_{l,(\tau+w)} \frac{\overline{l_{Pre}}}{\overline{\tau_{Pre}} + \overline{w_{Pre}}} \cdot \frac{\overline{\tau_{Pre}}}{\overline{l_{Pre}}} = \epsilon_{l,(\tau+w)} \frac{\overline{\tau_{Pre}}}{\overline{\tau_{Pre}} + \overline{w_{Pre}}} \tag{66}$$

Table H4: Estimated Elasticity of Layoffs w.r.t Experience Rating from the Literature

Study	**Estimated value**	**Parameter**
Topel (1984)	-0.27	Elasticity of temporary layoffs w.r.t. experience rating
Card et al. (1994)	-0.43	Elasticity of temporary layoffs w.r.t. experience rating
Card et al. (1994)	-0.1	Elasticity of permanent layoffs w.r.t. experience rating
Anderson et al. (1994)	[-0.33, -0.15]	Elasticity of temporary layoffs w.r.t. experience rating
Johnston (2021)	0	Elasticity of layoffs w.r.t. unemployment tax rate

Notes: The table lists various estimates for the elasticity of layoffs with respect to the degree of experience rating. The value of -0.27 for Topel (1984) is backed up from the conclusions of the paper. The values of -0.43 and -0.1 for Card et al. (1994) are calculated from Table 2 (20 and 5% reductions in layoffs relative to 47% increase in experience rating relative to mean of 0.68). Anderson et al. (1994) reports various estimates ranging between -0.33 and -0.15 in Section 5. The value of zero for Johnston (2021) is taken from Table 2.